MA4G6 Lecture Notes

Introduction to the Modern Calculus of Variations

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Preface

These lecture notes, written for the *MA4G6 Calculus of Variations* course at the University of Warwick, intend to give a modern introduction to the Calculus of Variations. I have tried to cover different aspects of the field and to explain how they fit into the "big picture". This is not an encyclopedic work; many important results are omitted and sometimes I only present a special case of a more general theorem. I have, however, tried to strike a balance between a pure introduction and a text that can be used for later revision of forgotten material.

The presentation is based around a few principles:

- The presentation is quite "modern" in that I use several techniques which are perhaps not usually found in an introductory text or that have only recently been developed.
- For most results, I try to use "reasonable" assumptions, not necessarily minimal ones.
- When presented with a choice of how to prove a result, I have usually preferred the (in my opinion) most conceptually clear approach over more "elementary" ones. For example, I use Young measures in many instances, even though this comes at the expense of a higher initial burden of abstract theory.
- Wherever possible, I first present an abstract result for general functionals defined on Banach spaces to illustrate the general structure of a certain result. Only then do I specialize to the case of integral functionals on Sobolev spaces.
- I do not attempt to trace every result to its original source as many results have evolved over generations of mathematicians. Some pointers to further literature are given in the last section of every chapter.

Most of what I present here is not original, I learned it from my teachers and colleagues, both through formal lectures and through informal discussions. Published works that influenced this course were the lecture notes on microstructure by S. Müller [73], the encyclopedic work on the Calculus of Variations by B. Dacorogna [25], the book on Young measures by P. Pedregal [81], Giusti's more regularity theory-focused introduction to the Calculus of Variations [44], as well as lecture notes on several related courses by J. Ball, J. Kristensen, A. Mielke. Further texts on the Calculus of Variations are the elementary introductions by B. van Brunt [96] and B. Dacorogna [26], the more classical two-part treatise [39, 40] by M. Giaquinta & S. Hildebrandt, as well as the comprehensive treatment of the geometric measure theory aspects of the Calculus of Variations by Giaquinta, Modica & Souček [41,42].

In particular, I would like to thank Alexander Mielke and Jan Kristensen, who are responsible for my choice of research area. Through their generosity and enthusiasm in sharing their knowledge they have provided me with the foundation for my study and research. Furthermore, I am grateful to Richard Gratwick and the participants of the MA4G6 course for many helpful comments, corrections, and remarks.

These lecture notes are a living document and I would appreciate comments, corrections and remarks – however small. They can be sent back to me via e-mail at

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or anonymously via

`http://tinyurl.com/MA4G6-2015`

or by scanning the following QR-Code:

Moreover, every page contains its individual QR-code for immediate feedback. I encourage all readers to make use of them.

F. R.
January 2015

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Chapter 1

Introduction

Physical modeling is a delicate business – trying to balance the *validity* of the resulting model, that is, its *agreement* with nature and its *predictive* capabilities, with the *feasibility* of its mathematical treatment is often highly non-straightforward. In this quest to formulate a useful picture of an aspect of the world, it turns out on surprisingly many occasions that the formulation of the model becomes clearer, more compact, or more descriptive, if one introduces some form of a *variational principle*, that is, if one can define a quantity, such as energy or entropy, which obeys a minimization, maximization or saddle-point law.

How "fundamental" such a scalar quantity is perceived depends on the situation: For example, in classical mechanics, one calls forces *conservative* if they can be expressed as the derivative of an "energy potential" along a curve. It turns out that many, if not most, forces in physics are conservative, which seems to imply that the concept of energy is fundamental. Furthermore, Einstein's famous formula $E = mc^2$ expresses the equivalence of mass and energy, positioning energy as the "most" fundamental quantity, with mass becoming "condensed" energy. On the other hand, the other ubiquitous scalar physical quantity, the entropy, has a more "artificial" flavor as a measure of *missing information* in a model, as is now well understood in *Statistical Mechanics*, where entropy becomes a *derived* quantity. We leave it to the field of *Philosophy of Science* to understand the nature of variational concepts. In the words of William Thomson, the Lord Kelvin[1]:

> *The fact that mathematics does such a good job of describing the Universe is a mystery that we don't understand. And a debt that we will probably never be able to repay.*

Our approach to variational quantities here is a pragmatic one: We see them as providing *structure* to a problem and to enable different mathematical, so-called *variational*, methods. For instance, in elasticity theory it is usually unrealistic that a system will tend to a *global* minimizer by itself, but this does not mean that – as an approximation – such a principle cannot be useful in practice. On a fundamental level, there probably is no minimization as such in physics, only quantum-mechanical interactions (or whatever lies "below" quantum

[1] Quotes prove nothing, however. Thomson also said "*X-rays will prove to be a hoax*".

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mechanics). However, if we wait long enough, the inherent noise in a realistic system will move our system around and it will settle *with a high probability* in a low-energy state.

In this course, we will focus solely on minimization problems for integral functionals of the form ($\Omega \subset \mathbb{R}^d$ open and bounded)

$$\int_\Omega f(x, u(x), \nabla u(x))\,\mathrm{d}x \quad \to \quad \min, \qquad u\colon \Omega \to \mathbb{R}^m,$$

possibly under conditions on the boundary values of u and further side constraints. These problems form the original core of the Calculus of Variations and are as relevant today as they have always been. The systematic understanding of these integral functionals starts in Euler's and Bernoulli's times in the late 1600s and the early 1700s, and their study was boosted into the modern age by Hilbert's 19th, 20th, 23rd problems, formulated in 1900 [47]. The excitement has never stopped since and new and important discoveries are made every year.

We start this course by looking at a parade of examples, which we treat at varying levels of detail. All these problems will be investigated further along the course once we have developed the necessary mathematical tools.

1.1 The brachistochrone problem

In June 1696 Johann Bernoulli published a problem in the journal *Acta Eruditorum*, which can be seen as the time of birth of the Calculus of Variations (the name, however, is from Leonhard Euler's 1766 treatise *Elementa calculi variationum*). Additionally, Bernoulli sent a letter containing the question to Gottfried Wilhelm Leibniz on 9 June 1696, who returned his solution only a few days later on 16 June, and commented that the problem tempted him "like the apple tempted Eve" because of its beauty. Isaac Newton published a solution (after the problem had reached him) without giving his identity, but Bernoulli identified him "ex ungue leonem" (from Latin, "by the lion's claw"). The problem was formulated as follows[2]:

> *Given two points A and B in a vertical* [meaning "not horizontal"] *plane, one shall find a curve AMB for a movable point M, on which it travels from A to B in the shortest time, only driven by its own weight.*

The resulting curve is called the **brachistochrone** (from Ancient Greek, "shortest time") curve.

We formulate the problem as follows: We look for the curve y connecting the origin $(0,0)$ to the point $(\bar{x}, \bar{y})$, where $\bar{x} > 0$, $\bar{y} < 0$, such that a point mass $m > 0$ slides from rest at $(0,0)$ to $(\bar{x}, \bar{y})$ quickest among all such curves. See Figure 1.1 for several possible slide paths. We parametrize a point (x, y) on the curve by the time $t \geq 0$. The point mass has

[2] The German original reads "Wenn in einer verticalen Ebene zwei Punkte A und B gegeben sind, soll man dem beweglichen Punkte M eine Bahn AMB anweisen, auf welcher er von A ausgehend vermöge seiner eigenen Schwere in kürzester Zeit nach B gelangt."

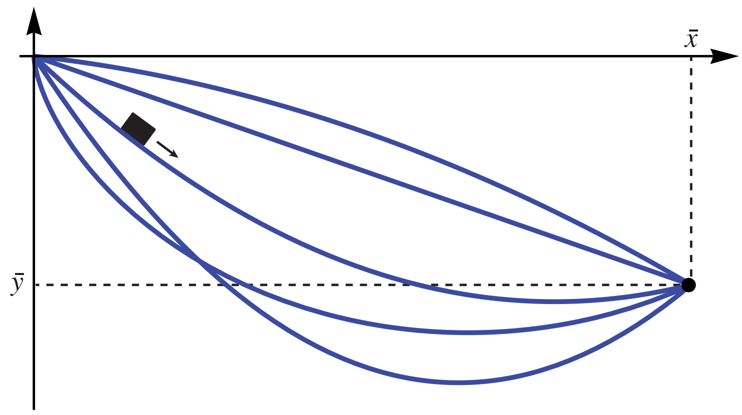

Figure 1.1: Several slide curves from the origin to $(\bar{x}, \bar{y})$.

kinetic and potential energies

$$E_{\mathrm{kin}} = \frac{m}{2}\left\{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2\right\} = \frac{m}{2}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2\left\{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2\right\},$$
$$E_{\mathrm{pot}} = mgy,$$

where $g \approx 9.81\,\mathrm{m/s}^2$ is the (constant) gravitational acceleration on Earth. The total energy $E_{\mathrm{kin}} + E_{\mathrm{pot}}$ is zero at the beginning and conserved along the path. Hence, for all y we have

$$\frac{m}{2}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2\left\{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2\right\} = -mgy.$$

We can solve this for $\mathrm{d}t/\mathrm{d}x$ (where $t = t(x)$ is the inverse of the x-parametrization) to get

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \sqrt{\frac{1 + (y')^2}{-2gy}} \qquad \left(\frac{\mathrm{d}t}{\mathrm{d}x} \geq 0\right),$$

where we wrote $y' = \frac{\mathrm{d}y}{\mathrm{d}x}$. Integrating over the whole x-length along the curve from 0 to $\bar{x}$, we get for the total time duration $T[y]$ that

$$T[y] = \frac{1}{\sqrt{2g}} \int_0^{\bar{x}} \sqrt{\frac{1 + (y')^2}{-y}}\,\mathrm{d}x.$$

We may drop the constant in front of the integral, which does not influence the minimization problem, and set $\bar{x} = 1$ by reparametrization, to arrive at the problem

$$\begin{cases} \mathscr{F}[y] := \displaystyle\int_0^1 \sqrt{\frac{1 + (y')^2}{-y}}\,\mathrm{d}x \quad \to \quad \min, \\ y(0) = 0,\ y(1) = \bar{y} < 0. \end{cases}$$

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Clearly, the integrand is *convex* in y', which will be important for the solution theory. We will come back to this problem in Example 2.34.

A related problem concerns *Fermat's Principle* expressing that light (in vacuum or in a medium) always takes the fastest path between two points, from which many optical laws, e.g. about reflection and refraction, can be derived.

1.2 Electrostatics

Consider an electric charge density $\rho\colon \mathbb{R}^3 \to \mathbb{R}$ (in units of $\mathrm{C/m^3}$) in three-dimensional vacuum. Let $E\colon \mathbb{R}^3 \to \mathbb{R}^3$ (in $\mathrm{V/m}$) and $B\colon \mathbb{R}^3 \to \mathbb{R}^3$ (in $\mathrm{T} = \mathrm{Vs/m^2}$) be the electric and magnetic fields, respectively, which we assume to be constant in time (hence electro*statics*). Assuming that $\mathbb{R}^3$ is a linear, homogeneous, isotropic electric material, the **Gauss law** for electricity reads

$$\nabla \cdot E = \operatorname{div} E = \frac{\rho}{\varepsilon_0},$$

where $\varepsilon_0 \approx 8.854 \cdot 10^{-12}\,\mathrm{C/(Vm)}$. Moreover, we have the **Faraday law of induction**

$$\nabla \times E = \operatorname{curl} E = \frac{\mathrm{d}B}{\mathrm{d}t} = 0.$$

Thus, since E is curl-free, there exist an **electric potential** $\varphi\colon \mathbb{R}^3 \to \mathbb{R}$ (in V) such that

$$E = -\nabla\varphi.$$

Combining this with the Gauss law, we arrive at the **Poisson equation**,

$$\Delta\varphi = \nabla \cdot [\nabla\varphi] = -\frac{\rho}{\varepsilon_0}. \tag{1.1}$$

We can also look at electrostatics in a variational way: We use the norming condition $\varphi(0) = 0$. Then, the electric potential energy $U_E(x;q)$ of a point charge $q > 0$ (in C) at point $x \in \mathbb{R}^3$ in the electric field E is given by the path integral (which does not depend on the path chosen since E is a gradient)

$$U_E(x;q) = -\int_0^x qE \cdot \mathrm{d}s = -\int_0^1 qE(hx)\,\mathrm{d}h = q\varphi(x).$$

Thus, the total **electric energy** of our charge distribution ρ is (the factor $1/2$ is necessary to count mutual reaction forces correctly)

$$U_E := \frac{1}{2}\int_{\mathbb{R}^3} \rho\varphi\,\mathrm{d}x = \frac{\varepsilon_0}{2}\int_{\mathbb{R}^3} (\nabla \cdot E)\varphi\,\mathrm{d}x,$$

which has units of $\mathrm{CV} = \mathrm{J}$. Using $(\nabla \cdot E)\varphi = \nabla \cdot (E\varphi) - E \cdot (\nabla\varphi)$, the divergence theorem, and the natural assumption that φ vanishes at infinity, we get further

$$U_E = \frac{\varepsilon_0}{2}\int_{\mathbb{R}^3} \nabla \cdot (E\varphi) - E \cdot (\nabla\varphi)\,\mathrm{d}x = -\frac{\varepsilon_0}{2}\int_{\mathbb{R}^3} E \cdot (\nabla\varphi)\,\mathrm{d}x = \frac{\varepsilon_0}{2}\int_{\mathbb{R}^3} |\nabla\varphi|^2\,\mathrm{d}x.$$

The integral $\int_\Omega |\nabla\varphi|^2 \, \mathrm{d}x$ is called **Dirichlet integral**.

In Example 2.15 we will see that (1.1) is equivalent to the minimization problem

$$U_E + \int_{\mathbb{R}^3} \rho\varphi \, \mathrm{d}x = \int_{\mathbb{R}^3} \frac{\varepsilon_0}{2} |\nabla\varphi|^2 + \rho\varphi \, \mathrm{d}x \quad \to \quad \min.$$

The second term is the energy stored in the electric field caused by its interaction with the charge density ρ.

1.3 Stationary states in quantum mechanics

The non-relativistic evolution of a quantum mechanical system with N degrees of freedom in an electrical field is described completely through its **wave function** $\Psi \colon \mathbb{R}^N \times \mathbb{R} \to \mathbb{C}$ that satisfies the **Schrödinger equation**

$$\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}t} \Psi(x,t) = \left[\frac{-\hbar}{2\mu} \Delta + V(x,t) \right] \Psi(x,t), \qquad x \in \mathbb{R}^N,\, t \in \mathbb{R},$$

where $\hbar \approx 1.05 \cdot 10^{-34}\,\mathrm{Js}$ is the reduced Planck constant, $\mu > 0$ is the reduced mass, and $V = V(x,t) \in \mathbb{R}$ is the potential energy. The (self-adjoint) operator $H := -(2\mu)^{-1}\hbar\Delta + V$ is called the **Hamiltonian** of the system.

The value of the wave function itself at a given point in spacetime has no physical interpretation, but according to the *Copenhagen Interpretation* of quantum mechanics, $|\Psi(x)|^2$ is the probability density of finding a particle at the point x. In order for $|\Psi|^2$ to be a probability density, we need to impose the side constraint

$$\|\Psi\|_{\mathrm{L}^2} = 1.$$

In particular, $\Psi(x,t)$ has to decay as $|x| \to \infty$.

Of particular interest are the so-called *stationary states*, that is, solutions of the **stationary Schrödinger equation**

$$\left[\frac{-\hbar^2}{2\mu} \Delta + V(x) \right] \Psi(x) = E\Psi(x), \qquad x \in \mathbb{R}^N,$$

where $E > 0$ is an *energy level*. We then solve this *eigenvalue problem* (in a weak sense) for $\Psi \in \mathrm{W}^{1,2}(\mathbb{R}^N;\mathbb{C})$ with $\|\Psi\|_{\mathrm{L}^2} = 1$ (see Appendix A.4 for a collection of facts about Sobolev spaces like $\mathrm{W}^{1,2}$).

If we are just interested in the lowest-energy state, the so-called **ground state**, we can find minimizers of the energy functional

$$\mathscr{E}[\Psi] := \int_{\mathbb{R}^N} \frac{\hbar^2}{2\mu} |\nabla\Psi(x)|^2 + \frac{1}{2} V(x) |\Psi(x)|^2 \, \mathrm{d}x,$$

again under the side constraint

$$\|\Psi\|_{\mathrm{L}^2} = 1.$$

The two parts of the integral above correspond to kinetic and potential energy, respectively. We will continue this investigation in Example 2.39.

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1.4 Hyperelasticity

Elasticity theory is one of the most important theories of **continuum mechanics**, that is, the study of the mechanics of (idealized) continuous media. We will not go into much detail about elasticity modeling here and refer to the book [20] for a thorough introduction.

Consider a body of mass occupying a domain $\Omega \subset \mathbb{R}^3$, called the *reference configuration*. If we deform the body, any *material point* $x \in \Omega$ is mapped into a *spatial point* $y(x) \in \mathbb{R}^3$ and we call $y(\Omega)$ the *deformed configuration*. For a suitable continuum mechanics theory, we also need to require that $y\colon \Omega \to y(\Omega)$ is a differentiable bijection and that it is *orientation-preserving*, i.e. that

$$\det \nabla y(x) > 0 \qquad \text{for all } x \in \Omega.$$

For convenience one also introduces the *displacement*

$$u(x) := y(x) - x.$$

One can show (using the implicit function theorem and the mean value theorem) that the orientation-preserving condition and the invertibility are satisfied if

$$\sup_{x\in\Omega} |\nabla u(x)| < \delta(\Omega) \tag{1.2}$$

for a domain-dependent (small) constant $\delta(\Omega) > 0$.

Next, we need a measure of local "stretching", called a *strain tensor*, which should serve as the parameter to a local energy density. On physical grounds, **rigid body motions**, that is, deformations $u(x) = Rx + u_0$ with a rotation $R \in \mathbb{R}^{3\times 3}$ ($R^T = R^{-1}$ and $\det R = 1$) and $u_0 \in \mathbb{R}^3$, should not cause strain. In this sense, strain measures the deviation of the displacement from a rigid body motion. One popular choice is the **Green–St. Venant strain tensor**[3]

$$E := \frac{1}{2}\big(\nabla u + \nabla u^T + \nabla u^T \nabla u\big). \tag{1.3}$$

We first consider fully nonlinear ("finite strain") elasticity. For our purposes we simply postulate the existence of a stored-energy density $W\colon \mathbb{R}^{3\times 3} \to [0,\infty]$ and an external body force field $b\colon \Omega \to \mathbb{R}^3$ (e.g. gravity) such that

$$\mathscr{F}[y] := \int_\Omega W(\nabla y) - b\cdot y \,\mathrm{d}x$$

represents the total elastic energy stored in the system. If the elastic energy can be written in this way as $\int_\Omega W(\nabla y(x))\,\mathrm{d}x$, we call the material **hyperelastic**. In applications, W is often given as depending on the Green–St. Venant strain tensor E instead of ∇y, but for our mathematical theory, the above form is more convenient. We require several properties of W for arguments $F \in \mathbb{R}^{3\times 3}$:

[3] There is an array of choices for the strain tensor, but they are mostly equivalent within *nonlinear* elasticity.

(i) **The undeformed state costs no energy:** $W(I) = 0$.

(ii) **Invariance under change of frame[4]:** $W(RF) = W(F)$ for all $R \in \mathrm{SO}(3)$.

(iii) **Infinite compression costs infinite energy:** $W(F) \to +\infty$ as $\det F \downarrow 0$.

(iv) **Infinite stretching costs infinite energy:** $W(F) \to +\infty$ as $|F| \to \infty$.

The main problem of nonlinear hyperelasticity is to minimize $\mathscr{F}$ as above over all $y\colon \Omega \to \mathbb{R}^3$ with given boundary values. Of course, it is not a-priori clear in which space we should look for a solution. Indeed, this depends on the growth properties of W. For example, for the prototypical choice

$$W(F) := \mathrm{dist}(F, \mathrm{SO}(3))^2,$$

which however does not satisfy (3) from our list of requirements, we would look for square integrable functions. More realistic in applications are W of the form

$$W(F) := \sum_{i=1}^{M} a_i \,\mathrm{tr}\big[(F^T F)^{\gamma_i/2}\big] + \sum_{j=1}^{N} b_j \,\mathrm{tr}\,\mathrm{cof}\big[(F^T F)^{\delta_j/2}\big] + \Gamma(\det F), \qquad F \in \mathbb{R}^{3\times 3},$$

where $a_i > 0$, $\gamma_i \geq 1$, $b_j > 0$, $\delta_j \geq 1$, and $\Gamma\colon \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ is a convex function with $\Gamma(d) \to +\infty$ as $d \downarrow 0$, $\Gamma(d) = +\infty$ for $s \leq 0$. These materials are called **Ogden materials** and occur in a wide range of applications.

In the setting of *linearized elasticity*, we make the "small strain" assumption that the quadratic term in (1.3) can be neglected and that (1.2) holds. In this case, we work with the **linearized strain tensor**

$$\mathscr{E}u := \frac{1}{2}\big(\nabla u + \nabla u^T\big).$$

In this infinitesimal deformation setting, the displacements that do not create strain are precisely the skew-affine maps[5] $u(x) = Qx + u_0$ with $Q^T = -Q$ and $u_0 \in \mathbb{R}$.

For linearized elasticity we consider an energy of the special "quadratic" form

$$\mathscr{F}[u] := \int_\Omega \frac{1}{2}\mathscr{E}u : \mathbf{C}\mathscr{E}u - b \cdot u \,\mathrm{d}x,$$

where $\mathbf{C}(x) = \mathbf{C}_{ijkl}(x)$ is a symmetric, strictly positive definite ($A : \mathbf{C}(x)A > c|A|^2$ for some $c > 0$) fourth-order tensor, called the **elasticity/stiffness tensor**, and $b\colon \Omega \to \mathbb{R}^3$ is our external body force. Thus we will always look for solutions in $\mathrm{W}^{1,2}(\Omega;\mathbb{R}^3)$.

For homogeneous, isotropic media, $\mathbf{C}$ does not depend on x or the direction of strain, which translates into the additional condition

$$(AR) : \mathbf{C}(x)(AR) = A : \mathbf{C}(x)A \qquad \text{for all } x \in \Omega,\ A \in \mathbb{R}^{3\times 3} \text{ and } R \in \mathrm{SO}(3).$$

[4] Rotating or translating the (inertial) frame of reference must give the same result.

[5] This becomes more meaningful when considering a bit more algebra: The Lie group SO(3) of rotations has as its Lie algebra $\mathrm{Lie}(\mathrm{SO}(3)) = \mathfrak{so}(3)$ the space of all skew-symmetric matrices, which then can be seen as "infinitesimal rotations".

In this case, it can be shown that $\mathscr{F}$ simplifies to

$$\mathscr{F}[u] = \frac{1}{2}\int_\Omega 2\mu|\mathscr{E}u|^2 + \Big(\kappa - \frac{2}{3}\mu\Big)|\operatorname{tr}\mathscr{E}u|^2 - b\cdot u \,\mathrm{d}x$$

for $\mu > 0$ the *shear modulus* and $\kappa > 0$ the *bulk modulus*, which are material constants. For example, for cold-rolled steel $\mu \approx 75$ GPa and $\kappa \approx 160$ GPa.

1.5 Optimal saving and consumption

Consider a capitalist worker earning a (constant) wage w per year, which he can either spend on consumption or save. Denote by $S(t)$ the savings at time t, where $t \in [0,T]$ is in years, with $t = 0$ denoting the beginning of his work life and $t = T$ his retirement. Further, let $C(t)$ be the consumption rate (consumption per time) at time t. On the saved capital, the worker earns interest, say with gross-continuous rate $\rho > 0$, meaning that a capital amount $m > 0$ grows as $\exp(\rho t)m$. If we were given an APR $\rho_1 > 0$ instead of ρ, we could compute $\rho = \ln(1+\rho_1)$. We further assume that salary is paid continuously, not in intervals, for simplicity. So, w is really the rate of pay, given in money per time. Then, the worker's savings evolve according to the differential equation

$$\dot{S}(t) = w + \rho S(t) - C(t). \tag{1.4}$$

Now, assume that our worker is mathematically inclined and wants to optimize the happiness due to consumption in his life by finding the optimal amount of consumption at any given time. Being a pure capitalist, the worker's happiness only depends on his consumption rate C. So, if we denote by $U(C)$ the **utility function**, that is, the "happiness" due to the consumption rate C, our worker wants to find $C\colon [0,T] \to \mathbb{R}$ such that

$$\mathscr{H}[C] := \int_0^T U(C(t))\,\mathrm{d}t$$

is maximized. The choice of U depends on our worker's personality, but it is probably realistic to assume that there is a *law of diminishing returns*, i.e. for twice as much consumption, our worker is happier, but not twice as happy. So, let us assume $U' > 0$ and $U'(C) \to 0$ as $C \to \infty$. Also, we should have $U(0) = -\infty$ (starvation). Moreover, it is realistic for U to be *concave*, which implies that there are no local maxima. One function that satisfies all of these requirements is the logarithm, and so we use

$$U(C) = \ln(C), \qquad C > 0.$$

Let us also assume that the worker starts with no savings, $S(0) = 0$, and at the end of his work life wants to retire with savings $S(T) = S_T \geq 0$. Rearranging (1.4) for $C(t)$ and plugging this into the formula for $\mathscr{H}$, we therefore want to solve the **optimal saving problem**

$$\begin{cases} \mathscr{F}[S] := \displaystyle\int_0^T -\ln(w + \rho S(t) - \dot{S}(t))\,\mathrm{d}t \quad \to \quad \min, \\ S(0) = 0,\ S(T) = S_T \geq 0. \end{cases}$$

This will be solved in Example 2.18.

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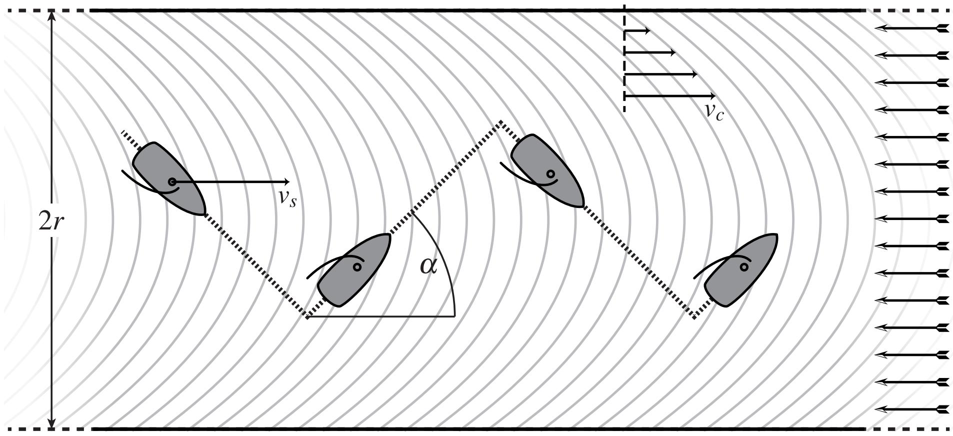

Figure 1.2: Sailing against the wind in a channel.

1.6 Sailing against the wind

Every sailor knows how to sail against the wind by “beating”: One has to sail at an angle of approximately 45° to the wind[6], then tack (turn the bow through the wind) and finally, after the sail has caught the wind on the other side, continue again at approximately 45° to the wind. Repeating this procedure makes the boat follow a zig-zag motion, which gives a net movement directly against the wind, see Figure 1.2. A mathematically inclined sailor might ask the question of “how often to tack”. In an idealized model we can assume that tacking costs no time and that the forward sailing speed v_s of the boat depends on the angle α to the wind as follows (at least qualitatively):

$$v_s(\alpha) = -\cos(4\alpha),$$

which has maxima at $\alpha = \pm 45°$ (in real boats, the maximum might be at a lower angle, i.e. “closer to the wind”). Assume furthermore that our sailor is sailing along a straight river with the current. Now, the current is fastest in the middle of the river and goes down to zero at the banks. In fact, a good approximation would be the formula of *Poiseuille (channel) flow*, which can be derived from the flow equations of fluids: At distance r from the center of the river the current’s flow speed is approximately

$$v_c(r) := v_{\max}\left(1 - \frac{r^2}{R^2}\right),$$

where $R > 0$ is half the width of the river.

[6]The optimal angle depends on the boat. Ask a sailor about this and you will get an evening-filling lecture.

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If we denote by $r(t)$ the distance of our boat from the middle of the channel at time $t \in [0,T]$, then the total speed (called the "velocity made good" in sailing parlance) is

$$\begin{aligned} v(t) &:= v_s(\arctan r'(t)) + v_c(r(t)) \\ &= -\cos(4 \arctan r'(t)) + v_{\max}\left(1 - \frac{r(t)^2}{R^2}\right). \end{aligned}$$

The key to understand this problem is now the fact that $a \mapsto -\cos(4 \arctan a)$ has two maxima at $a = \pm 1$. We say that this function is a **double-well potential**.

The total forward distance traveled over the time interval $[0,T]$ is

$$\int_0^T v(t)\,\mathrm{d}t = \int_0^T -\cos(4 \arctan r'(t)) + v_{\max}\left(1 - \frac{r(t)^2}{R^2}\right)\mathrm{d}t.$$

If we also require the initial and terminal conditions $r(0) = r(T) = 0$, we arrive at the **optimal beating problem**:

$$\begin{cases} \mathscr{F}[r] := \displaystyle\int_0^T \cos(4 \arctan r'(t)) - v_{\max}\left(1 - \frac{r(t)^2}{R^2}\right)\mathrm{d}t \quad \to \quad \min, \\ r(0) = r(T) = 0, \qquad |r(t)| \le R. \end{cases}$$

Our intuition tells us that in this idealized model, where tacking costs no time, we should be tacking as "infinitely fast" in order to stay in the middle of the river. Later, once we have advanced tools at our disposal, we will make this idea precise, see Example 5.6.

Chapter 2

Convexity

In the introduction we got a glimpse of the many applications of minimization problems in physics, technology, and economics. In this chapter we start to develop the abstract theory. Consider a minimization problem of the form

$$\begin{cases} \mathscr{F}[u] := \displaystyle\int_\Omega f(x, u(x), \nabla u(x)) \,\mathrm{d}x \quad \to \quad \min, \\ \text{over all } u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m) \text{ with } u|_{\partial\Omega} = g. \end{cases}$$

Here, and throughout the course (if not otherwise stated) we will make the standard assumption that $\Omega \subset \mathbb{R}^d$ is a **bounded Lipschitz domain**, that is, Ω is open, bounded, and has a Lipschitz boundary. Furthermore, we assume $1 < p < \infty$, so that the **integrand**

$$f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R},$$

is measurable in the first and continuous in the second and third arguments (hence f is a *Carathéodory* integrand). For the prescribed boundary values g we assume

$$g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m).$$

In this context recall that $\mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$ is the space of all traces of functions $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, see Appendix A.4 for some background on Sobolev spaces. These conditions will be assumed from now on unless stated otherwise. Furthermore, to make the above integral finite, we assume the growth bound

$$|f(x,v,A)| \le M(1 + |v|^p + |A|^p) \qquad \text{for some } M > 0.$$

Below, we will investigate the solvability and regularity properties of this minimization problem; in particular, we will take a close look at the way in which **convexity** properties of f in its gradient (third) argument determine whether $\mathscr{F}$ is *lower semicontinuous*, which is the decisive attribute in the fundamental *Direct Method of the Calculus of Variations*.

We will also consider the partial differential equation associated with the minimization problem, the so-called *Euler–Lagrange equation* that is a necessary (but not sufficient) condition for minimizers. This leads naturally to the question of regularity of solutions. Finally, we also treat invariances, leading to the famous *Noether theorem* before having a glimpse at side constraints.

Most results here are already formulated for vector-valued u, as this has many applications. Some aspects, however, in particular regularity theory, are specific to scalar problems, where u is assumed to be merely real-valued.

2.1 The Direct Method

Fundamental to all of the existence theorems in this text is the conceptionally simple *Direct Method*[1] of the Calculus of Variations. Let X be a topological space[2] (e.g. a Banach space with the strong or weak topology) and let $\mathscr{F}\colon X \to \mathbb{R} \cup \{+\infty\}$ be our **objective functional** satisfying the following two assumptions:

(H1) **Coercivity:** For all $\Lambda \in \mathbb{R}$, the sublevel set $\{\, x \in X \,:\, \mathscr{F}[x] \le \Lambda \,\}$ is sequentially relatively compact, that is, if $\mathscr{F}[x_j] \le \Lambda$ for a sequence $(x_j) \subset X$ and some $\Lambda \in \mathbb{R}$, then (x_j) has a convergent subsequence.

(H2) **Lower semicontinuity:** For all sequences $(x_j) \subset X$ with $x_j \to x$ it holds that

$$\mathscr{F}[x] \le \liminf_{j\to\infty} \mathscr{F}[x_j].$$

Consider the abstract minimization problem

$$\mathscr{F}[x] \to \min \quad \text{over all } x \in X. \tag{2.1}$$

Then, the Direct Method is the following simple strategy:

Theorem 2.1 (Direct Method). *Assume that $\mathscr{F}$ is both coercive and lower semicontinuous. Then, the abstract problem* (2.1) *has at least one solution, that is, there exists $x_* \in X$ with $\mathscr{F}[x_*] = \min\{\, \mathscr{F}[x] \,:\, x \in X \,\}$.*

Proof. Let us assume that there exists at least one $x \in X$ such that $\mathscr{F}[x] < +\infty$; otherwise, any $x \in X$ is a "solution" to the (degenerate) minimization problem.

To construct a minimizer we take a **minimizing sequence** $(x_j) \subset X$ such that

$$\lim_{j\to\infty} \mathscr{F}[x_j] \to \alpha := \inf\bigl\{\, \mathscr{F}[x] \,:\, x \in X \,\bigr\} < +\infty.$$

Then, since convergent sequences are bounded, there exists $\Lambda \in \mathbb{R}$ such that $\mathscr{F}[x_j] \le \Lambda$ for all $j \in \mathbb{N}$. Hence, by the coercivity (H1), the sequence (x_j) is contained in a compact subset

[1] "Direct" refers to the fact that we construct solutions without employing a differential equation.

[2] In fact, we only need a space together with a notion of convergence as all our arguments will involve sequences as opposed to more general topological tools such as nets. In all of this text we use "topology" and "convergence" interchangeably.

of X. Now select a subsequence, which here as in the following we do not renumber, such that

$$x_j \to x_* \in X.$$

By the lower semicontinuity (H2), we immediately conclude

$$\alpha \le \mathscr{F}[x_*] \le \liminf_{j\to\infty} \mathscr{F}[x_j] = \alpha.$$

Thus, $\mathscr{F}[x_*] = \alpha$ and x_* is the sought minimizer. □

Example 2.2. Using the Direct Method, one can easily see that

$$h(t) := \begin{cases} 1-t & \text{if } t < 0, \\ t & \text{if } t \ge 0, \end{cases} \qquad t \in \mathbb{R}\ (= X),$$

has minimizer $t = 0$, despite not even being continuous there.

Despite its nearly trivial proof, the Direct Method is very useful and flexible in applications. Of course, neither coercivity nor lower semicontinuity are necessary for the existence of a minimizer. For instance, we really only needed coercivity and lower semicontinuity along a *minimizing* sequence, but this weaker condition is hard to check in most practical situations; however, such a refined approach can be useful in special cases.

The abstract principle of the Direct Method pushes the difficulty in proving existence of a minimizer toward establishing coercivity and lower semicontinuity. This, however, is a big advantage, since we have many tools at our disposal to establish these two hypotheses separately. In particular, for integral functionals lower semicontinuity is tightly linked to *convexity* properties of the integrand, as we will see throughout this course.

At this point it is crucial to observe how coercivity and lower semicontinuity interact with the *topology* (or, more accurately, *convergence*) in X: If we choose a stronger topology, i.e. one for which there are fewer convergent sequences, then it becomes easier for $\mathscr{F}$ to be lower semicontinuous, but harder for $\mathscr{F}$ to be coercive. The opposite holds if choosing a weaker topology. In the mathematical treatment of a problem we are most likely in a situation where $\mathscr{F}$ and X are given by the concrete problem. We then need to find a suitable topology in which we can establish *both* coercivity and lower semicontinuity.

In this text, X will always be an infinite-dimensional Banach space and we have a real choice between using the strong or weak convergence. Usually, it turns out that coercivity with respect to the strong convergence is *false* since strongly compact sets in infinite-dimensional spaces are very restricted, whereas coercivity with respect to the weak convergence is true under reasonable assumptions. However, whereas strong lower semicontinuity poses few challenges, lower semicontinuity with respect to weakly converging sequences is a delicate matter and we will spend considerable time on this topic.

We will always use the Direct Method in the following version:

Theorem 2.3 (Direct Method for weak convergence). *Let X be a Banach space and let $\mathscr{F} : X \to \mathbb{R} \cup \{+\infty\}$. Assume the following:*

(WH1) ***Weak Coercivity:*** *For all $\Lambda \in \mathbb{R}$, the sublevel set*

$$\{x \in X \,:\, \mathscr{F}[x] \leq \Lambda\} \qquad \textit{is sequentially weakly relatively compact,}$$

that is, if $\mathscr{F}[x_j] \leq \Lambda$ for a sequence $(x_j) \subset X$ and some $\Lambda \in \mathbb{R}$, then (x_j) has a weakly convergent subsequence.

(WH2) ***Weak lower semicontinuity:*** *For all sequences $(x_j) \subset X$ with $x_j \rightharpoonup x$ (weak convergence X) it holds that*

$$\mathscr{F}[x] \leq \liminf_{j \to \infty} \mathscr{F}[x_j].$$

Then, the minimization problem

$$\mathscr{F}[x] \to \min \quad \textit{over all } x \in X,$$

has at least one solution.

Before we move on to the more concrete theory, let us record one further remark: While it might appear as if "nature does not give us the topology" and it is up to mathematicians to "invent" a suitable one, it is remarkable that the topology that turns out to be *mathematically* convenient is also often *physically* relevant. This is for instance expressed in the following observation: The only phenomena which are present in weakly but not strongly converging sequences are *oscillations* and *concentrations*, as can be seen in the classical Vitali Convergence Theorem A.6. On the other hand, these phenomena very much represent physical effects, for example in crystal microstructure. Thus, the fact that we work with the weak convergence means that we have to consider *microstructure* (which might or might not be exhibited), a fact that we will come back to many times throughout this course.

2.2 Functionals with convex integrands

Returning to the minimization problem at the beginning of this chapter, we first consider the minimization problem for the simpler functional

$$\mathscr{F}[u] := \int_\Omega f(x, \nabla u(x)) \,\mathrm{d}x$$

over all $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ for some $1 < p < \infty$ to be chosen later (depending on lower growth properties of f). The following lemma together with (2.3) allows us to conclude that $\mathscr{F}[u]$ is indeed well-defined.

Lemma 2.4. *Let $f \colon \Omega \times \mathbb{R}^N \to \mathbb{R}$ be a **Carathéodory integrand**, that is,*

(i) $x \mapsto f(x,A)$ is Lebesgue-measurable for all fixed $A \in \mathbb{R}^N$.

(ii) $A \mapsto f(x,A)$ is continuous for all fixed $x \in \Omega$.

Then, for any Lebesgue-measurable function $V\colon \Omega \to \mathbb{R}^N$ *the composition* $x \mapsto f(x, V(x))$ *is Lebesgue-measurable.*

We will prove this lemma via the following "Lusin-type" theorem for Carathéodory functions:

Theorem 2.5 (Scorza Dragoni). *Let* $f\colon \Omega \times \mathbb{R}^N \to \mathbb{R}$ *be Carathéodory. Then, there exists an increasing sequence of compact sets* $S_k \subset \Omega$ $(k \in \mathbb{N})$ *with* $|\Omega \setminus S_k| \downarrow 0$ *such that* $f|_{S_k \times \mathbb{R}^N}$ *is continuous.*

See Theorem 6.35 in [37] for a proof of this theorem, which is rather measure-theoretic and similar to the proof of the Lusin theorem, cf. Theorem 2.24 in [85].

Proof of Lemma 2.4. Let $S_k \subset \Omega$ $(k \in \mathbb{N})$ be as in the Scorza Dragoni Theorem. Set $f_k := f|_{S_k \times \mathbb{R}^N}$ and

$$g_k(x) := f_k(x, V(x)), \qquad x \in S_k.$$

Then, for any open set $E \subset \mathbb{R}$, the pre-image of E under g_k is the set of all $x \in S_k$ such that $(x, V(x)) \in f_k^{-1}(E)$. As f_k is continuous, $f_k^{-1}(E)$ is open and from the Lebesgue measurability of the product function $x \mapsto (x, V(x))$ we infer that $g_k^{-1}(E)$ is a Lebesgue-measurable subset of S_k. We can extend the definition of g_k to all of Ω by setting $g_k(x) := 0$ whenever $x \in \Omega \setminus S_k$. This function is still Lebesgue-measurable as S_k is compact, hence measurable. The conclusion follows from the fact that $g_k(x) \to f(x, V(x))$ and pointwise limits of Lebesgue-measurable functions are themselves Lebesgue-measurable. □

We next investigate coercivity: Let $1 < p < \infty$ The most basic assumption, and the only one we want to consider here, to guarantee coercivity is the following **lower growth (coercivity) bound**:

$$\mu|A|^p - \mu^{-1} \le f(x, A) \qquad \text{for all } x \in \Omega,\, A \in \mathbb{R}^{m \times d} \text{ and some } \mu > 0. \tag{2.2}$$

This coercivity also suggests the exponent p for the definition of the function spaces where we look for solutions. We further assume the **upper growth bound**

$$|f(x, A)| \le M(1 + |A|^p) \qquad \text{for all } x \in \Omega,\, A \in \mathbb{R}^{m \times d} \text{ and some } M > 0, \tag{2.3}$$

which gives finiteness of $\mathscr{F}[u]$ for all $u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m)$.

Proposition 2.6. *If* f *satisfies the lower growth bound* (2.2)*, then* $\mathscr{F}$ *is weakly coercive on the space* $\mathrm{W}^{1,p}_g(\Omega; \mathbb{R}^m) = \{ u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m) \;:\; u|_{\partial\Omega} = g \}$.

Proof. We need to show that any sequence $(u_j) \subset \mathrm{W}^{1,p}_g(\Omega; \mathbb{R}^m)$ with

$$\sup_j \mathscr{F}[u_j] < \infty$$

is weakly relatively compact. From (2.2) we get

$$\mu \cdot \sup_j \int_\Omega |\nabla u_j|^p \,\mathrm{d}x - \frac{|\Omega|}{\mu} \leq \sup_j \mathscr{F}[u_j] < \infty,$$

whereby $\sup_j \|\nabla u_j\|_{\mathrm{L}^p} < \infty$. By the Poincaré–Friedrichs inequality, Theorem A.16 (I), in conjunction with the fact that we fix the boundary values, we therefore get $\sup_j \|u_j\|_{\mathrm{W}^{1,p}} < \infty$. We conclude by the fact that bounded sets in reflexive Banach spaces, like $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ for $1 < p < \infty$, are sequentially weakly relatively compact, see Theorem A.11. □

Having settled the question of (weak) coercivity, we can now turn to investigate the weak lower semicontinuity.

Theorem 2.7. *If $A \mapsto f(x,A)$ is convex for all $x \in \Omega$ and $f(x,A) \geq \kappa$ for some $\kappa \in \mathbb{R}$ (which follows in particular from* (2.2)*), then $\mathscr{F}$ is weakly lower semicontinuous on* $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$.

Proof. Step 1. We first establish that $\mathscr{F}$ is strongly lower semicontinuous, so let $u_j \to u$ in $\mathrm{W}^{1,p}$ and $\nabla u_j \to \nabla u$ almost everywhere (which holds after selecting a non-renumbered subsequence, see Appendix A.2). By assumption we have

$$f(x, \nabla u_j(x)) - \kappa \geq 0.$$

Thus, applying Fatou's Lemma,

$$\liminf_{j\to\infty} \big(\mathscr{F}[u_j] - \kappa|\Omega|\big) \geq \mathscr{F}[u] - \kappa|\Omega|.$$

Thus, $\mathscr{F}[u] \leq \liminf_{j\to\infty} \mathscr{F}[u_j]$. Since this holds for all subsequences, it also follows for our original sequence.

Step 2. To show weak lower semicontinuity take $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $u_j \rightharpoonup u$. We need to show that

$$\mathscr{F}[u] \leq \liminf_{j\to\infty} \mathscr{F}[u_j] =: \alpha. \tag{2.4}$$

Taking a subsequence (not relabeled), we can in fact assume that $\mathscr{F}[u_j]$ *converges* to α.

By the Mazur Lemma A.14, we may find convex combinations

$$v_j = \sum_{n=j}^{N(j)} \theta_{j,n} u_n, \qquad \text{where} \qquad \theta_{j,n} \in [0,1] \quad \text{and} \quad \sum_{n=j}^{N(j)} \theta_{j,n} = 1,$$

such that $v_j \to u$ strongly. Since $f(x,\,\boldsymbol{\cdot}\,)$ is convex,

$$\mathscr{F}[v_j] = \int_\Omega f\left(x, \sum_{n=j}^{N(j)} \theta_{j,n} \nabla u_n(x)\right) \mathrm{d}x \leq \sum_{n=j}^{N(j)} \theta_{j,n} \mathscr{F}[u_n].$$

Now, $\mathscr{F}[u_n] \to \alpha$ as $n \to \infty$ and $n \geq j$ inside the sum. Thus,

$$\liminf_{j\to\infty} \mathscr{F}[v_j] \leq \alpha = \liminf_{j\to\infty} \mathscr{F}[u_j].$$

On the other hand, from the first step and $v_j \to u$ strongly, we have $\mathscr{F}[u] \leq \liminf_{j\to\infty} \mathscr{F}[v_j]$. Thus, (2.4) follows and the proof is finished. □

Comment...

We can summarize our findings in the following theorem:

Theorem 2.8. *Let* $f \colon \Omega \times \mathbb{R}^{m\times d}$ *be a Carathéodory integrand such that* f *satisfies the lower growth bound* (2.2)*, the upper growth bound* (2.3) *and is convex in its second argument. Then, the associated functional* $\mathscr{F}$ *has a minimizer over the space* $\mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$*.*

Proof. This follows immediately from the Direct Method in Banach spaces, Theorem 2.3 together with Proposition 2.6 and Theorem 2.7. $\square$

Example 2.9. The **Dirichlet integral (functional)** is

$$\mathscr{F}[u] := \int_\Omega |\nabla u(x)|^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega).$$

We already encountered this integral functional when considering electrostatics in Section 1.2. It is easy to see that this functional satisfies all requirements of Theorem 2.8 and so there exists a minimizer for any prescribed boundary values $g \in \mathrm{W}^{1/2,2}(\partial\Omega)$.

Example 2.10. In the prototypical problem of linearized elasticity from Section 1.4, we are tasked with the minimization problem

$$\begin{cases} \mathscr{F}[u] := \dfrac{1}{2}\displaystyle\int_\Omega 2\mu|\mathscr{E}u|^2 + \Big(\kappa - \dfrac{2}{3}\mu\Big)|\operatorname{tr}\mathscr{E}u|^2 - fu\,\mathrm{d}x \quad \to \quad \min, \\ \text{over all } u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^3) \text{ with } u|_{\partial\Omega} = g, \end{cases}$$

where $\mu, \kappa > 0$, $f \in \mathrm{L}^2(\Omega;\mathbb{R}^3)$, and $g \in \mathrm{W}^{1/2,2}(\partial\Omega;\mathbb{R}^m)$. It is clear that $\mathscr{F}$ has quadratic growth. Let us first consider the coercivity: For $u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^3)$ with $u|_{\partial\Omega} = g$ we have the following version of the L^2**-Korn inequality**

$$\|u\|^2_{\mathrm{W}^{1,2}} \le C_K\big(\|\mathscr{E}u\|^2_{\mathrm{L}^2} + \|g\|^2_{\mathrm{W}^{1/2,2}}\big).$$

Here, $C_K = C_K(\Omega) > 0$ is a constant. Let us for a simplified estimate assume that $\kappa - \frac{2}{3}\mu \ge 0$ (this is not necessary, but shortens the argument). Then, also using the Young inequality, we get for any $\delta > 0$,

$$\begin{aligned} \mathscr{F}[u] &\ge \mu\|\mathscr{E}u\|^2_{\mathrm{L}^2} - \|f\|_{\mathrm{L}^2}\|u\|_{\mathrm{L}^2} \\ &\ge \mu\big(\|\mathscr{E}u\|^2_{\mathrm{L}^2} + \|g\|^2_{\mathrm{W}^{1/2,2}}\big) - \frac{1}{2\delta}\|f\|^2_{\mathrm{L}^2} - \frac{\delta}{2}\|u\|^2_{\mathrm{L}^2} - \mu\|g\|^2_{\mathrm{W}^{1/2,2}} \\ &\ge \left(\frac{\mu}{C_K} - \frac{\delta}{2}\right)\|u\|^2_{\mathrm{W}^{1,2}} - \frac{1}{2\delta}\|f\|^2_{\mathrm{L}^2} - \mu\|g\|^2_{\mathrm{W}^{1/2,2}}. \end{aligned}$$

Choosing $\delta = \mu/C_K$, we get the coercivity estimate

$$\mathscr{F}[u] \ge \frac{\mu}{2C_K}\|u\|^2_{\mathrm{W}^{1,2}} - \frac{C_K}{2\mu}\|f\|^2_{\mathrm{L}^2} - \mu\|g\|^2_{\mathrm{W}^{1/2,2}}.$$

Hence, $\mathscr{F}[u]$ controls $\|u\|_{\mathrm{W}^{1,2}}$ and our functional is weakly coercive. Moreover, it is clear that our integrand is convex in in the $\mathscr{E}u$-argument (note that the trace function is linear). Hence, Theorem 2.8 yields the existence of a solution $u_* \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^3)$ to our minimization problem of linearized elasticity.

We finish this section with the following converse to Theorem 2.7:

Proposition 2.11. *Let $\mathscr{F}$ be an integral functional with a Carathéodory integrand $f\colon \Omega \times \mathbb{R}^{m\times d} \to \mathbb{R}$ satisfying the upper growth bound* (2.3). *Assume furthermore that $\mathscr{F}$ is weakly lower semicontinuous on $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$. If either $m=1$ or $d=1$ (the* ***scalar case****), then $A \mapsto f(x,A)$ is convex for almost every $x \in \Omega$.*

Proof. We will not prove the full result here, since Proposition 3.27 together with Corollary 3.4 in the next chapter will give a stronger version (including the corresponding result for the vectorial case as well). □

In the **vectorial case**, i.e. $m \neq 1$ or $d \neq 1$, the necessity of convexity is far from being true and there is indeed another "canonical" condition ensuring weak lower semicontinuity, which is weaker than (ordinary) convexity; we will explore this in the next chapter.

2.3 Integrands with u-dependence

If we try to extend the results in the previous section to more general functionals

$$\mathscr{F}[u] := \int_\Omega f(x,u(x),\nabla u(x)) \,\mathrm{d}x \quad \to \quad \min,$$

we discover that our proof strategy of using the Mazur Lemma runs into difficulties: We cannot "pull out" the convex combination inside

$$\int_\Omega f\left(x, \sum_{n=j}^{N(j)} \theta_{j,n} u_n(x), \sum_{n=j}^{N(j)} \theta_{j,n} \nabla u_n(x)\right) \mathrm{d}x$$

anymore. Nevertheless, a lower semicontinuity result analogous to the one for the scalar case turns out to be true:

Theorem 2.12. *Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}$ satisfy the growth bound*

$$|f(x,v,A)| \leq M(1+|v|^p+|A|^p) \qquad \text{for some } M>0,\ 1<p<\infty,$$

and the convexity property

$$A \mapsto f(x,v,A) \text{ is convex for all } (x,v) \in \Omega \times \mathbb{R}^m.$$

Then, the functional

$$\mathscr{F}[u] := \int_\Omega f(x,u(x),\nabla u(x)) \,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m),$$

is weakly lower semicontinuous.

While it would be possible to give an elementary proof of this theorem here, we postpone the verification to the next chapter. There, using more advanced and elegant techniques (blow-ups and Young measures), we will in fact establish a much more general result and see that, when viewed from the right perspective, u-dependence does not pose any additional complications.

Comment...

2.4 The Euler–Lagrange equation

In analogy to the elementary fact that the derivative of a function at its minimizers is zero, we will now derive a necessary condition for a function to be a minimizer. This condition furnishes the connection between the Calculus of Variations and PDE theory and is of great importance for *computing* particular solutions.

Theorem 2.13 (Euler–Lagrange equation (system)). *Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}$ be a Carathéodory integrand that is continuously differentiable in v and A and satisfies the growth bounds*

$$|D_v f(x,v,A)|, |D_A f(x,v,A)| \le M(1+|v|^p+|A|^p), \qquad (x,v,A) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d},$$

for some $M > 0$, $1 \le p < \infty$. If $u_* \in W^{1,p}_g(\Omega;\mathbb{R}^m)$, where $g \in W^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$, minimizes the functional

$$\mathscr{F}[u] := \int_\Omega f(x,u(x),\nabla u(x))\,\mathrm{d}x, \qquad u \in W^{1,p}_g(\Omega;\mathbb{R}^m),$$

then u_* is a weak solution of the following system of PDEs, called the ***Euler–Lagrange equation***[3],

$$\begin{cases} -\operatorname{div}\big[D_A f(x,u,\nabla u)\big] + D_v f(x,u,\nabla u) = 0 & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{2.5}$$

Here we used the common convention to omit the x-arguments whenever this does not cause any confusion in order to curtail the proliferation of x's, for example in $f(x,u,\nabla u) = f(x,u(x),\nabla u(x))$.

Recall that u_* is a **weak solution** of (2.5) if

$$\int_\Omega D_A f(x,u_*,\nabla u_*) : \nabla\psi + D_v f(x,u_*,\nabla u_*)\cdot\psi\,\mathrm{d}x = 0$$

for all $\psi \in C^\infty_c(\Omega;\mathbb{R}^m)$, where

$$D_A f(x,v,A) : B := \lim_{h\downarrow 0} \frac{f(x,v,A+hB) - f(x,v,A)}{h}, \qquad A,B \in \mathbb{R}^{m\times d},$$

is the **directional derivative** of $f(x,v,\bullet)$ at A in direction B, likewise for $D_v f(x,v,A)\cdot w$. The derivatives $D_A f(x,v,A)$ and $D_v f(x,v,A)$ can also be represented as a matrix in $\mathbb{R}^{m\times d}$ and a vector in $\mathbb{R}^m$, respectively, namely

$$D_A f(x,v,A) := \big(\partial_{A^j_k} f(x,v,A)\big)^j_k, \qquad D_v f(x,v,A) := \big(\partial_{v^j} f(x,v,A)\big)^j.$$

Hence we use the notation with the Frobenius matrix vector product ":" (see the appendix) and the scalar product "·". The boundary condition $u = g$ on $\partial\Omega$ is to be understood in the sense of trace.

[3] This is a *system* of PDEs, but we simply speak of it as an "equation".

Comment...

Proof. For all $\psi \in \mathrm{C}_c^\infty(\Omega;\mathbb{R}^m)$ and all $h > 0$ we have

$$\mathscr{F}[u_*] \le \mathscr{F}[u_* + h\psi]$$

since $u_* + h\psi \in \mathrm{W}_g^{1,p}(\Omega;\mathbb{R}^m)$ is admissible in the minimization. Thus,

$$\begin{aligned}
0 &\le \int_\Omega \frac{f(x,u_*+h\psi,\nabla u_*+h\nabla\psi) - f(x,u_*,\nabla u_*)}{h}\,\mathrm{d}x \\
&= \int_\Omega \int_0^1 \frac{1}{h}\frac{\mathrm{d}}{\mathrm{d}t}\big[f(x,u_*+th\psi,\nabla u_*+th\nabla\psi)\big]\,\mathrm{d}t\,\mathrm{d}x \\
&= \int_\Omega \int_0^1 \mathrm{D}_A f(x,u_*+th\psi,\nabla u_*+th\nabla\psi) : \nabla\psi \\
&\qquad\qquad + \mathrm{D}_v f(x,u_*+th\psi,\nabla u_*+th\nabla\psi)\cdot\psi\,\mathrm{d}t\,\mathrm{d}x.
\end{aligned}$$

By the growth bounds on the derivative, the integrand can be seen to have an h-uniform L^1-majorant, namely $C(1+|u_*|^p+|\psi|^p+|\nabla u_*|^p+|\nabla\psi|^p)$ and so we may apply the Lebesgue dominated convergence theorem to let $h \downarrow 0$ under the (double) integral. This yields

$$0 \le \int_\Omega \mathrm{D}_A f(x,u_*,\nabla u_*) : \nabla\psi + \mathrm{D}_v f(x,u_*,\nabla u_*)\cdot\psi\,\mathrm{d}x$$

and we conclude by taking ψ and $-\psi$ in this inequality. □

Remark 2.14. If we want to allow $\psi \in \mathrm{W}_0^{1,p}(\Omega;\mathbb{R}^m)$ in the weak formulation of (2.5), then we need to assume the stronger growth conditions

$$|\mathrm{D}_v f(x,v,A)|, |\mathrm{D}_A f(x,v,A)| \le M(1+|v|^{p-1}+|A|^{p-1})$$

for some $M > 0$, $1 \le p < \infty$.

Example 2.15. Coming back to the Dirichlet integral from Example 2.9, we see that the associated Euler–Lagrange equation is the **Laplace equation**

$$-\Delta u = 0,$$

where $\Delta := \partial_{x_1}^2 + \cdots + \partial_{x_d}^2$ is the **Laplace operator**. Solutions u are called **harmonic functions**. Since the Dirichlet integral is convex, all solutions of the Laplace equation are in fact minimizers. In particular, it can be seen that solutions of the Laplace equation are unique for given boundary values.

Example 2.16. In the linearized elasticity problem from Example 2.10, we may compute the Euler–Lagrange equation as

$$\begin{cases} -\operatorname{div}\Big[2\mu\,\mathscr{E}u + \Big(\kappa - \dfrac{2}{3}\mu\Big)(\operatorname{tr}\mathscr{E}u)I\Big] = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases}$$

Comment...

Sometimes, one also defines the **first variation** $\delta\mathscr{F}[u]$ of $\mathscr{F}$ at $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ as the linear map

$$\delta\mathscr{F}[u] \colon \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m) \to \mathbb{R}$$

through (whenever this exists)

$$\delta\mathscr{F}[u][\psi] := \lim_{h\downarrow 0} \frac{\mathscr{F}[u+h\psi]-\mathscr{F}[u]}{h}, \qquad \psi \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m).$$

Then, the assertion of the previous theorem can equivalently be formulated as

$$\delta\mathscr{F}[u_*] = 0 \qquad \text{if } u_* \text{ minimizes } \mathscr{F} \text{ over } \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m).$$

Of course, this condition is only *necessary* for u to be a minimizer. Indeed, any solution of the Euler–Lagrange equation is called a **critical point** of $\mathscr{F}$, which could be a minimizer, maximizer, or saddle point.

One crucial consequence of the results in this section is that we can use all available PDE methods to study minimizers. Immediately, one can ask about the type of PDE we are dealing with. In this respect we have the following result:

Proposition 2.17. *Let f be twice continuously differentiable in v and A. Also let $A \mapsto f(x,v,A)$ be convex for all $(x,v) \in \Omega\times\mathbb{R}^m$. Then, the Euler–Lagrange equation is an* ***elliptic*** *PDE, that is,*

$$0 \le \mathrm{D}^2_A f(x,v,A)[B,B] := \frac{\mathrm{d}^2}{\mathrm{d}t^2} f(x,v,A+tB)\Big|_{t=0}$$

for all $x \in \Omega$, $v \in \mathbb{R}^m$, and $A,B \in \mathbb{R}^{m\times d}$.

This proposition is not difficult to establish and just needs a bit of notation; we omit the proof.

One main use of the Euler–Lagrange equation is to find concrete solutions of variational problems:

Example 2.18. Recall the optimal saving problem from Section 1.5,

$$\begin{cases} \mathscr{F}[S] := \displaystyle\int_0^T -\ln(w+\rho S(t)-\dot{S}(t))\,\mathrm{d}t \quad \to \quad \min, \\ S(0)=0,\ S(T)=S_T \ge 0. \end{cases}$$

The Euler–Lagrange equation is

$$-\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{1}{w+\rho S(t)-\dot{S}(t)}\right] = -\frac{\rho}{w+\rho S(t)-\dot{S}(t)},$$

which resolves to

$$\frac{\rho\dot{S}(t)-\ddot{S}(t)}{w+\rho S(t)-\dot{S}(t)} = \rho.$$

Comment...

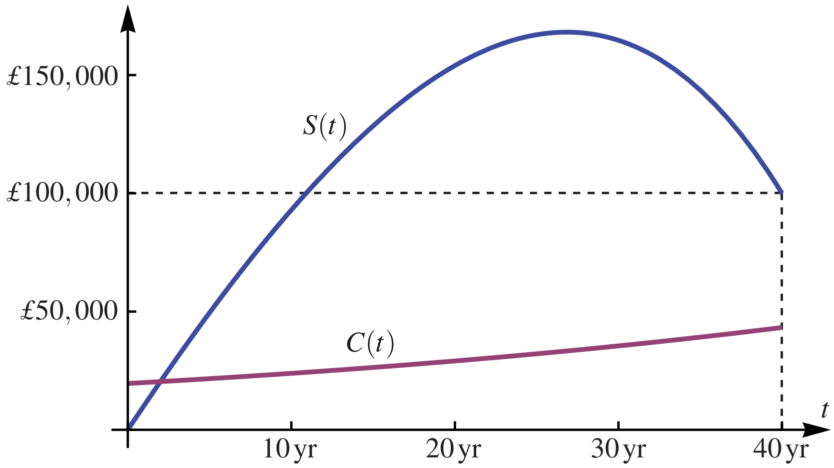

Figure 2.1: The solution to the optimal saving problem.

With the consumption rate $C(t) := w + \rho S(t) - \dot{S}(t)$, this is equivalent to

$$\frac{\dot{C}(t)}{C(t)} = \rho,$$

and so, if $C(0) = C_0$ (to be determined later),

$$w + \rho S(t) - \dot{S}(t) = C(t) = \mathrm{e}^{\rho t} C_0.$$

This differential equation for $S(t)$ can be solved, for example via the *Duhamel formula*, which yields

$$\begin{aligned} S(t) &= \mathrm{e}^{\rho t} \cdot 0 + \int_0^t \mathrm{e}^{\rho(t-s)} (w - \mathrm{e}^{\rho s} C_0)\, \mathrm{d}s \\ &= \frac{\mathrm{e}^{\rho t} - 1}{\rho} w - t \mathrm{e}^{\rho t} C_0. \end{aligned}$$

and C_0 can now be chosen to satisfy the terminal condition $S(T) = S_T$, in fact, $C_0 = (1 - \mathrm{e}^{-\rho T}) w / (\rho T) - \mathrm{e}^{-\rho T} S_T / T$.

Since $C \mapsto -\ln C$ is convex, we know that $S(t)$ as above is a minimizer of our problem. In Figure 2.1 we see the optimal savings strategy for a worker earning a (constant) salary of $w =$ £$30,000$ per year and having a savings goal of $S_T =$ £$100,000$. The worker has to save for approximately 27 years, reaching savings of just over £$168,000$, and then starts withdrawing his savings ($\dot{S}(t) < 0$) for the last 13 years. The worker's consumption $C(t) = \mathrm{e}^{\rho t} C_0$ goes up continuously during the whole working life, making him/her (materially) happy.

Comment...

Example 2.19. For functions $u = u(t,x)\colon \mathbb{R}\times\mathbb{R}^d \to \mathbb{R}$, consider the *action functional*

$$\mathscr{A}[u] := \frac{1}{2}\int_{\mathbb{R}\times\mathbb{R}^d} -|\partial_t u|^2 + |\nabla_x u|^2 \,\mathrm{d}(t,x),$$

where $\nabla_x u$ is the gradient of u with respect to $x \in \mathbb{R}^d$. This functional should be interpreted as the usual Dirichlet integral, see Example 2.9, where, however, we use the Lorentz metric. Then, the Euler–Lagrange equation is the **wave equation**

$$\partial_t^2 u - \Delta u = 0.$$

Notice that $\mathscr{A}$ is not convex and consequently the wave equation is *hyperbolic* instead of elliptic.

It is an important question whether a weak solution of the Euler–Lagrange equation (2.5) is also a **strong solution**, that is, whether $u \in \mathrm{W}^{2,2}(\Omega;\mathbb{R}^m)$ and

$$\begin{cases} -\operatorname{div}\big[\mathrm{D}_A f(x,u(x),\nabla u(x))\big] + \mathrm{D}_v f(x,u(x),\nabla u(x)) = 0 & \text{for a.e. } x\in\Omega, \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{2.6}$$

If $u \in \mathrm{C}^2(\Omega;\mathbb{R}^m)\cap\mathrm{C}(\overline{\Omega};\mathbb{R}^m)$ satisfies this PDE for *every* $x \in \Omega$, then we call u a **classical solution**.

Multiplying (2.6) by a test function $\psi \in \mathrm{C}_c^\infty(\Omega;\mathbb{R}^m)$, integrating over Ω, and using the Gauss–Green Theorem, it follows that any solution of (2.6) also solves (2.5). The converse is also true whenever u is sufficiently regular:

Proposition 2.20. *Let the integrand f be twice continuously differentiable and let $u \in \mathrm{W}^{2,2}(\Omega;\mathbb{R}^m)$ be a weak solution of the Euler–Lagrange equation* (2.5)*. Then, u solves the Euler–Lagrange equation* (2.6) *in the strong sense.*

Proof. If $u \in \mathrm{W}^{2,2}(\Omega;\mathbb{R}^m)$ is a weak solution, then

$$\int_\Omega \mathrm{D}_A f(x,u,\nabla u) : \nabla\psi + \mathrm{D}_v f(x,u,\nabla u)\cdot\psi \,\mathrm{d}x = 0$$

for all $\psi \in \mathrm{C}_c^\infty(\Omega;\mathbb{R}^m)$. Integration by parts (more precisely, the Gauss–Green Theorem) gives

$$\int_\Omega \big\{-\operatorname{div}\big[\mathrm{D}_A f(x,u,\nabla u)\big] + \mathrm{D}_v f(x,u,\nabla u)\big\}\cdot\psi \,\mathrm{d}x = 0,$$

again for all ψ as before. We conclude using the following important lemma. □

Lemma 2.21 (Fundamental lemma of the Calculus of Variations). *Let $\Omega \subset \mathbb{R}^d$ be open. If $g \in \mathrm{L}^1(\Omega)$ satisfies*

$$\int_\Omega g\psi \,\mathrm{d}x = 0 \qquad \textit{for all } \psi \in \mathrm{C}_c^\infty(\Omega),$$

then $g \equiv 0$ almost everywhere.

Comment...

Proof. We can assume that Ω is bounded by considering subdomains if necessary. Also, let g be extended by zero to all of $\mathbb{R}^d$. Fix $\varepsilon > 0$ and let $(\rho_\delta)_\delta \subset \mathrm{C}_c^\infty(B(0,\delta))$ be a family of mollifying kernels, see Appendix A.2 for details. Then, since $\rho_\delta \star g \to g$ in L^1, we may find $h \in (\mathrm{L}^1 \cap \mathrm{C}^\infty)(\mathbb{R}^d)$ with

$$\|g-h\|_{\mathrm{L}^1} \le \frac{\varepsilon}{4} \qquad \text{and} \qquad \|h\|_\infty < \infty.$$

Set $\varphi(x) := h(x)/|h(x)|$ for $h(x) \neq 0$ and $\varphi(x) := 0$ for $h(x) = 0$, so that $h\varphi = |h|$. Then take $\psi = \rho_{\bar{\delta}} \star \varphi \in (\mathrm{L}^1 \cap \mathrm{C}^\infty)(\mathbb{R}^d)$ with

$$\|\varphi - \psi\|_{\mathrm{L}^1} \le \frac{\varepsilon}{2\|h\|_\infty}.$$

Since $|\psi| \le 1$ (this follows from the convolution formula),

$$\begin{aligned}
\|g\|_{\mathrm{L}^1} &\le \|g-h\|_{\mathrm{L}^1} + \int_\Omega h\varphi \,\mathrm{d}x \\
&\le \|g-h\|_{\mathrm{L}^1} + \int_\Omega h(\varphi-\psi) + (h-g)\psi + g\psi \,\mathrm{d}x \\
&\le 2\|g-h\|_{\mathrm{L}^1} + \|h\|_\infty \cdot \|\varphi-\psi\|_{\mathrm{L}^1} + 0 \\
&\le \varepsilon.
\end{aligned}$$

We conclude by letting $\varepsilon \downarrow 0$. □

2.5 Regularity of minimizers

As we saw at the end of the last section, whether a weak solution of the Euler–Lagrange equation is also a strong or even classical solution, depends on its *regularity*, that is, on the amount of differentiability. More generally, one would like to know *how much* regularity we can expect from solutions of a variational problem. Such a question also was the content of David Hilbert's 19th problem [47]:

> *Does every Lagrangian partial differential equation of a regular variational problem have the property of exclusively admitting analytic integrals?*[4]

In modern language, Hilbert asked whether "regular" variational problems (defined below) admit only analytic solutions, i.e. ones that have a local power series representation.

In this section, we will prove some basic regularity assertions, but we will only sketch the solution of Hilbert's 19th problem, as the techniques needed are quite involved. We remark from the outset that regularity results are very sensitive to the dimensions involved, in particular the behavior of the scalar case ($m = 1$) and the vector case ($m > 1$) is fundamentally different. We also only consider the quadratic ($p = 2$) case for reasons of simplicity.

[4] The German original asks "ob jede Lagrangesche partielle Differentialgleichung eines regulären Variationsproblems die Eigenschaft hat, daß sie nur analytische Integrale zuläßt."

In the spirit of Hilbert's 19th problem, call

$$\mathscr{F}[u] := \int_\Omega f(\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m),$$

a **regular variational integral** if $f \in \mathrm{C}^2(\mathbb{R}^{m\times d})$ and there are constants $0 < \mu \le M$ with

$$\mu|B|^2 \le \mathrm{D}_A^2 f(A)[B,B] \le M|B|^2 \qquad \text{for all } A, B \in \mathbb{R}^{m\times d}.$$

In this context,

$$\mathrm{D}_A^2 f(A)[B_1,B_2] = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\mathrm{d}}{\mathrm{d}s} f(A+sB_1+tB_2)\bigg|_{s,t=0} \qquad \text{for all } A, B_1, B_2 \in \mathbb{R}^{m\times d},$$

which is a bilinear form. Clearly, regular variational problems are convex. In fact, integrands f that satisfy the above lower bound $\mu|B|^2 \le \mathrm{D}_A^2 f(A)[B,B]$ are called **strongly convex**. For example, the Dirichlet integral from Example 2.9 is regular. Also, the following Cauchy–Schwarz-type estimate can be shown:

$$\big|\mathrm{D}_A^2 f(A)[B_1,B_2]\big| \le M|B_1||B_2|. \tag{2.7}$$

The basic regularity theorem is the following:

Theorem 2.22 **($\mathrm{W}^{2,2}_{\mathrm{loc}}$-regularity).** *Let $\mathscr{F}$ be a regular variational integral. Then, for any minimizer $u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ of $\mathscr{F}$, it holds that $u \in \mathrm{W}^{2,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^m)$. Moreover, for any $B(x_0,3r) \subset \Omega$ ($x_0 \in \Omega$, $r > 0$) the* ***Caccioppoli inequality***

$$\int_{B(x_0,r)} |\nabla^2 u(x)|^2\,\mathrm{d}x \le \left(\frac{2M}{\mu}\right)^2 \int_{B(x_0,3r)} \frac{|\nabla u(x) - [\nabla u]_{B(x_0,3r)}|^2}{r^2}\,\mathrm{d}x \tag{2.8}$$

holds, where $[\nabla u]_{B(x_0,3r)} := \fint_{B(x_0,3r)} \nabla u\,\mathrm{d}x$. Consequently, the Euler–Lagrange equation holds strongly,

$$-\operatorname{div} \mathrm{D}f(\nabla u) = 0, \qquad \textit{a.e. in } \Omega.$$

For the proof we employ the *difference quotient method*, which is fundamental in regularity theory. Let $u\colon \Omega \to \mathbb{R}^m$, $x \in \Omega$, $k \in \{1,\dots,d\}$, and $h \in \mathbb{R}$. Then, define the **difference quotients**

$$\mathrm{D}_k^h u(x) := \frac{u(x+h\mathrm{e}_k) - u(x)}{h}, \qquad \mathrm{D}^h u := (\mathrm{D}_1^h u, \dots, \mathrm{D}_d^h u),$$

where $\{\mathrm{e}_1,\dots,\mathrm{e}_d\}$ is the standard basis of $\mathbb{R}^d$. The key is the following characterization of Sobolev spaces in terms of difference quotients:

Lemma 2.23. *Let $1 < p < \infty$, $D \subset\subset \Omega \subset \mathbb{R}^d$ be open, and $u \in \mathrm{L}^p(\Omega;\mathbb{R}^m)$.*

(I) If $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, then

$$\|\mathrm{D}_k^h u\|_{\mathrm{L}^p(D)} \le \|\partial_k u\|_{\mathrm{L}^p(\Omega)} \qquad \textit{for all } k \in \{1,\dots,d\},\ |h| < \operatorname{dist}(D,\partial\Omega).$$

Comment...

(II) If for some $0 < \delta < \operatorname{dist}(D, \partial\Omega)$ *it holds that*

$$\|\mathrm{D}_k^h u\|_{\mathrm{L}^p(D)} \le C \qquad \textit{for all } k \in \{1,\dots,d\} \textit{ and all } |h| < \delta,$$

then $u \in \mathrm{W}^{1,p}_{\mathrm{loc}}(\Omega;\mathbb{R}^m)$ *and* $\|\partial_k u\|_{\mathrm{L}^p(D)} \le C$ *for all* $k \in \{1,\dots,d\}$.

Proof. For (I), assume first that $u \in (\mathrm{L}^p \cap \mathrm{C}^1)(\Omega;\mathbb{R}^m)$. In this case, by the Fundamental Theorem of Calculus, at $x \in \Omega$ it holds that

$$\mathrm{D}_k^h u(x) = \frac{1}{h}\int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} u(x + th\mathrm{e}_k)\,\mathrm{d}t = \int_0^1 \partial_k u(x + th\mathrm{e}_k)\,\mathrm{d}t.$$

Thus,

$$\int_D |\mathrm{D}_k^h u|^p\,\mathrm{d}x \le \int_\Omega |\partial_k u|^p\,\mathrm{d}x,$$

from which the assertion follows. The general case follows from the density of $(\mathrm{L}^p \cap \mathrm{C}^1)(\Omega;\mathbb{R}^m)$ in $\mathrm{L}^p(\Omega;\mathbb{R}^m)$.

For (II), we observe that for fixed $k \in \{1,\dots,d\}$ by assumption $(\mathrm{D}_k^h u)_{0<h<\delta}$ is uniformly L^p-bounded. Thus, for an arbitrary fixed sequence of h's tending to zero, there exists a subsequence $h_j \downarrow 0$ with

$$\mathrm{D}_k^{h_j} u \rightharpoonup v_k \quad \text{in } \mathrm{L}^p$$

for some $v_k \in \mathrm{L}^p(\Omega;\mathbb{R}^m)$. Let $\psi \in \mathrm{C}_c^\infty(\Omega;\mathbb{R}^m)$. Using an "integration-by-parts" rule for difference quotients, which is elementary to check, we get

$$\int_D v_k \cdot \psi\,\mathrm{d}x = \lim_{j\to\infty}\int_D \mathrm{D}_k^{h_j} u \cdot \psi\,\mathrm{d}x = -\lim_{j\to\infty}\int_D u \cdot \mathrm{D}_k^{-h_j}\psi\,\mathrm{d}x = -\int_D u \cdot \partial_k \psi\,\mathrm{d}x.$$

Thus, $u \in \mathrm{W}^{1,p}(D;\mathbb{R}^m)$ and $v_k = \partial_k u$. The norm estimate follows from the lower semicontinuity of the norm under weak convergence. □

Proof of Theorem 2.22. The idea is to emulate the a-priori non-existent second derivatives using difference quotients and to derive estimates which allow one to conclude that these difference quotients are in L^2. Then we can conclude by the preceding lemma.

Let $u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ be a minimizer of $\mathscr{F}$. By Theorem 2.13,

$$0 = \int_\Omega \mathrm{D}f(\nabla u) : \nabla\psi\,\mathrm{d}x \tag{2.9}$$

holds for all $\psi \in \mathrm{C}_c^\infty(\Omega;\mathbb{R}^m)$, and then by density also for all $\psi \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$. Here we have used the notation $\mathrm{D}f(A) : B$ for the directional derivative of f at A in direction B. Fix a ball $B(x_0, 3r) \subset \Omega$ and take a Lipschitz cut-off function $\rho \in \mathrm{W}^{1,\infty}(\Omega)$ such that

$$\mathbb{1}_{B(x_0,r)} \le \rho \le \mathbb{1}_{B(x_0,2r)} \qquad \text{and} \qquad |\nabla\rho| \le \frac{1}{r}.$$

Then, for any $k = 1, \dots, d$ and $|h| < r$, we let

$$\psi := \mathrm{D}_k^{-h}\big[\rho^2 \mathrm{D}_k^h (u-a)\big] \quad \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m),$$

where a is an affine function to be chosen later. We may plug ψ into (2.9) to get

$$0 = \int_\Omega \mathrm{D}_k^h(\mathrm{D}f(\nabla u)) : \big[\rho^2 \mathrm{D}_k^h \nabla u + \mathrm{D}_k^h(u-a) \otimes \nabla(\rho^2)\big] \;\mathrm{d}x. \tag{2.10}$$

Here, we used the "integration-by-parts" formula for difference quotients, which is easy to check (also recall $a \otimes b := ab^T$).

Next, we estimate, using the assumptions on f,

$$\begin{aligned}
\mu |\mathrm{D}_k^h \nabla u|^2 &\le \int_0^1 \mathrm{D}^2 f(\nabla u + th\mathrm{D}_k^h \nabla u)[\mathrm{D}_k^h \nabla u, \mathrm{D}_k^h \nabla u] \;\mathrm{d}t \\
&= \frac{1}{h} \mathrm{D}f(\nabla u + th\mathrm{D}_k^h \nabla u) : \mathrm{D}_k^h \nabla u \bigg|_{t=0}^{1} \\
&= \mathrm{D}_k^h(\mathrm{D}f(\nabla u)) : \mathrm{D}_k^h \nabla u.
\end{aligned}$$

From the preceding lemma (both parts), (2.7), and the fact $|a \otimes b| = |a| \cdot |b|$ for our norms, we get

$$\begin{aligned}
\big|\mathrm{D}_k^h(\mathrm{D}f(\nabla u)) : \big[\mathrm{D}_k^h(u-a) \otimes \nabla(\rho^2)\big]\big| &\le \big|\mathrm{D}^2 f(\nabla u)\big[\mathrm{D}_k^h(u-a) \otimes \nabla(\rho^2), \partial_k \nabla u\big]\big| \\
&\le 2M \cdot \rho \cdot |\mathrm{D}_k^h(u-a)| \cdot |\nabla \rho| \cdot |\mathrm{D}_k^h \nabla u|.
\end{aligned}$$

Using the last two estimates, (2.10), and the integration-by-parts formula for difference quotients again,

$$\begin{aligned}
\mu \int_\Omega |\mathrm{D}_k^h \nabla u|^2 \rho^2 \;\mathrm{d}x &\le \int_\Omega \mathrm{D}_k^h(\mathrm{D}f(\nabla u)) : [\rho^2 \mathrm{D}_k^h \nabla u] \;\mathrm{d}x \\
&= -\int_\Omega \mathrm{D}_k^h(\mathrm{D}f(\nabla u)) : \big[\mathrm{D}_k^h(u-a) \otimes \nabla(\rho^2)\big] \;\mathrm{d}x \\
&\le 2M \int_\Omega \rho \cdot |\mathrm{D}_k^h \nabla u| \cdot |\mathrm{D}_k^h(u-a)| \cdot |\nabla \rho| \;\mathrm{d}x.
\end{aligned}$$

Now use the Young inequality to get

$$\mu \int_\Omega |\mathrm{D}_k^h \nabla u|^2 \rho^2 \;\mathrm{d}x \le \frac{\mu}{2} \int_\Omega |\mathrm{D}_k^h \nabla u|^2 \rho^2 \;\mathrm{d}x + \frac{(2M)^2}{2\mu} \int_\Omega |\mathrm{D}_k^h(u-a)|^2 \cdot |\nabla \rho|^2 \;\mathrm{d}x.$$

Absorbing the first term in the left hand side and using the properties of ρ,

$$\int_{B(x_0,r)} |\mathrm{D}_k^h \nabla u|^2 \;\mathrm{d}x \le \left(\frac{2M}{\mu}\right)^2 \int_{B(x_0,2r)} \frac{|\mathrm{D}_k^h(u-a)|^2}{r^2} \;\mathrm{d}x.$$

Now invoke the difference-quotient lemma, part (I), to arrive at

$$\int_{B(x_0,r)} |\mathrm{D}_k^h \nabla u|^2 \;\mathrm{d}x \le \left(\frac{2M}{\mu}\right)^2 \int_{B(x_0,3r)} \frac{|\partial_k(u-a)|^2}{r^2} \;\mathrm{d}x.$$

Applying the lemma again, this time part (II), we get $u \in \mathrm{W}^{2,2}(B(x_0,r);\mathbb{R}^m)$. The Caccioppoli inequality (2.8) follows once we take $\nabla a := [\nabla u]_{B(x_0,3r)}$. $\square$

Comment...

With the $\mathrm{W}^{2,2}_{\mathrm{loc}}$-regularity result at hand, we may use $\psi = \partial_k \tilde{\psi}$, $k = 1, \ldots, d$, as test function in the weak formulation of the Euler–Lagrange equation $-\operatorname{div}[\mathrm{D}f(\nabla u)] = 0$ to conclude that

$$-\operatorname{div}\left[\mathrm{D}^2 f(\nabla u) : \nabla(\partial_k u)\right] = 0 \tag{2.11}$$

holds in the weak sense. Indeed,

$$0 = -\int_\Omega \operatorname{div} \mathrm{D}f(\nabla u) \cdot \partial_k \psi \,\mathrm{d}x = \int_\Omega \operatorname{div}\left[\mathrm{D}^2 f(\nabla u) : \nabla(\partial_k u)\right] \cdot \psi \,\mathrm{d}x$$

for all $\psi \in \mathrm{C}^\infty_c(\Omega;\mathbb{R}^m)$. In the special case that f is quadratic, $\mathrm{D}^2 f$ is constant and the above PDE has the same structure (with $\partial_k u$ as unknown function) as the original Euler–Lagrange equation. Thus it is itself an Euler–Lagrange equation and we can **bootstrap** the regularity theorem to get $u \in \mathrm{W}^{k,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^m)$ for all $k \in \mathbb{N}$, hence $k \in \mathrm{C}^\infty(\Omega;\mathbb{R}^m)$.

Example 2.24. For a minimizer $u \in \mathrm{W}^{1,2}(\Omega)$ of the Dirichlet integral as in Example 2.9, the theory presented here immediately gives $u \in \mathrm{C}^\infty(\Omega)$.

Example 2.25. Similarly, for minimizers $u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^3)$ to the problem of linearized elasticity from Section 1.4 and Example 2.10, we also get $u \in \mathrm{C}^\infty(\Omega;\mathbb{R}^3)$. The reasoning has to be slightly adjusted, however, to take care of the fact that we are dealing with $\mathcal{E}u$ instead of ∇u.

In the general case, however, our Euler–Lagrange equation is *nonlinear* and the above bootstrapping procedure fails. In particular, this includes the problem posed in Hilbert's 19th problem, which was only solved, in the scalar case $m = 1$, by Ennio De Giorgi in 1957 [28] and, using different methods, by John Nash in 1958 [78]. After the proof was improved by Jürgen Moser in 1960/1961 [71, 72], the results are now subsumed in the *De Giorgi–Nash–Moser theory*.

The current standard solution (based on De Giorgi's approach) is based on the **De Giorgi regularity theorem** and the classical **Schauder estimates**, which establish Hölder regularity of weak solutions of a PDE.

Theorem 2.26 (De Giorgi regularity theorem). *Let the function $A\colon \Omega \to \mathbb{R}^{d\times d}$ be* measurable*, symmetric, i.e. $A(x) = A(x)^T$ for $x \in \Omega$, and satisfy the ellipticity and boundedness estimates*

$$\mu|v|^2 \le v^T A(x) v \le M|v|^2, \qquad x \in \Omega, v \in \mathbb{R}^d, \tag{2.12}$$

for constants $0 < \mu \le M$. If $u \in \mathrm{W}^{1,2}(\Omega)$ is a weak solution of

$$-\operatorname{div}[A\nabla u] = 0, \tag{2.13}$$

then $u \in \mathrm{C}^{0,\alpha_0}_{\mathrm{loc}}(\Omega)$, that is, u is α_0-Hölder continuous, for some $\alpha_0 = \alpha_0(d, M/\mu) \in (0,1)$.

Comment...

Theorem 2.27 (Schauder estimate). *Let $A\colon \Omega \to \mathbb{R}^{d\times d}$ be as above but in addition assumed to be of class $\mathrm{C}^{s-1,\alpha}$ for some $s \in \mathbb{N}$ and $\alpha \in (0,1)$. If $u \in \mathrm{W}^{1,2}(\Omega)$ is a weak solution of*

$$-\operatorname{div}[A\nabla u] = 0,$$

then $u \in \mathrm{C}^{s,\alpha}_{\mathrm{loc}}(\Omega)$.

We remark that the Schauder estimates also hold for *systems* of PDEs ($u\colon \Omega \to \mathbb{R}^m$, $m > 1$), but the De Giorgi regularity theorem does not. The proofs of these results are quite involved and reserved for an advanced course on regularity theory, but we establish their arguably most important consequence, namely the solution of Hilbert's 19th problem in the scalar case:

Theorem 2.28 (De Giorgi–Nash–Moser). *Let $\mathscr{F}$ be a regular variational integral and $f \in \mathrm{C}^s(\mathbb{R}^d)$, $s \in \{2,3,\ldots\}$. If $u \in \mathrm{W}^{1,2}(\Omega)$ minimizes $\mathscr{F}$, then it holds that $u \in \mathrm{C}^{s-1,\alpha}_{\mathrm{loc}}(\Omega)$ for some $\alpha \in (0,1)$. In particular, if $f \in \mathrm{C}^\infty(\mathbb{R}^d)$, then $u \in \mathrm{C}^\infty(\Omega)$.*

Proof. We saw in (2.11) that the partial derivative $\partial_k u$, $k = 1,\ldots,d$, of a minimizer $u \in \mathrm{W}^{1,2}(\Omega)$ of $\mathscr{F}$ satisfy (2.13) for

$$A(x) := \mathrm{D}^2 f(\nabla u(x)), \qquad x \in \Omega.$$

From general properties of the Hessian we conclude that $A(x)$ is symmetric and the upper and lower estimates (2.12) on A follow from the respective properties of $\mathrm{D}^2 f$. However, we cannot conclude (yet) any regularity of A beyond measurability. Nevertheless, we may apply the De Giorgi Regularity Theorem 2.26, whereby $\partial_k u \in \mathrm{C}^{0,\alpha_0}_{\mathrm{loc}}(\Omega)$ for some $\alpha_0 \in (0,1)$ and all $k = 1,\ldots,d$. Hence $u \in \mathrm{C}^{1,\alpha_0}_{\mathrm{loc}}(\Omega)$. This is the assertion for $s = 2$.

If $s = 3$, our arguments so far in conjunction with the regularity assumptions on f imply $\mathrm{D}^2 f(\nabla u(x)) \in \mathrm{C}^{0,\alpha_0}_{\mathrm{loc}}(\Omega)$. Hence, the Schauder estimates from Theorem 2.27 apply and yield $\partial_k u \in \mathrm{C}^{1,\alpha_0}_{\mathrm{loc}}(\Omega)$, whereby $u \in \mathrm{C}^{2,\alpha_0}_{\mathrm{loc}}(\Omega) = \mathrm{C}^{s-1,\alpha_0}_{\mathrm{loc}}(\Omega)$. For higher s, this procedure can be iterated until we run out of f-derivatives and we have achieved $u \in \mathrm{C}^{s-1,\alpha_0}_{\mathrm{loc}}(\Omega)$. □

The above argument also yields analyticity of u if f is analytic. Another refinement shows that in the situation of the De Giorgi–Nash–Moser Theorem, we even have $u \in \mathrm{C}^{s-1,\alpha}_{\mathrm{loc}}(\Omega)$ for *all* $\alpha \in (0,1)$. Here one needs the additional Schauder–type result that weak solutions u to $-\operatorname{div}\big[A\nabla(\partial_k u)\big] = 0$ for A continuous are of class $\mathrm{C}^{0,\alpha}_{\mathrm{loc}}$ for *all* $\alpha \in (0,1)$. In the above proof we can apply this result in the case $s = 2$ since $A(x) = \mathrm{D}^2 f(\nabla u(x))$ is continuous. Thus, $u \in \mathrm{C}^{1,\alpha}_{\mathrm{loc}}(\Omega)$ for *any* $\alpha \in (0,1)$. The other parts of the proof are adapted accordingly. See [86] for details and other refinements.

We close our look at regularity theory by considering the vectorial case $m > 1$. It was shown again by De Giorgi in 1968 that if $d = m > 2$, then his regularity theorem does not hold:

Example 2.29 (De Giorgi). Let $d = m > 2$ and define

$$u(x) := |x|^{-\gamma} x, \qquad \gamma := \frac{d}{2}\left(1 - \frac{1}{\sqrt{(2d-2)^2+1}}\right), \qquad x \in B(0,1).$$

Note that $1 < \gamma < \frac{d}{2}$ and so $u \in \mathrm{W}^{1,2}(B(0,1);\mathbb{R}^d)$ but $u \notin \mathrm{L}^\infty_{\mathrm{loc}}(B(0,1);\mathbb{R}^d)$. It can be checked, though, that u solves (2.13) for

$$v^T A(x) v := |v|^2 + \left[\left((d-1)I + d \cdot \frac{x \otimes x}{|x|^2}\right) v\right]^2,$$

which satisfies all assumptions of the De Giorgi Theorem. Moreover, u is a minimizer of the quadratic variational integral

$$\int_{B(0,1)} \nabla u(x)^T A(x) \nabla u(x) \, \mathrm{d}x.$$

However, this is not a regular variational integral because of the x-dependence of the integrand is not smooth.

In principle, this still leaves the possibility that the vectorial analogue of Hilbert's 19th problem has a positive solution, just that its proof would have to proceed along a different avenue than via the De Giorgi Theorem. However, after first results of Nečas in 1975 [79], the current state of the art was established by Šverák and Yan in 2000–2002 [90, 91]. They prove that there exist regular (and C^∞) variational integrals such that minimizers must be:

- non-Lipschitz if $d \geq 3$, $m \geq 5$,
- unbounded if $d \geq 5$, $m \geq 14$.

However, there are also positive results, in particular we have that for $d = 2$ minimizers to regular variational problems are always as regular as the data allows (as in Theorem 2.28), this is the Morrey regularity theorem from 1938, see [68]. If we confine ourselves to Sobolev-regularity, then using the difference quotient technique, we can prove $\mathrm{W}^{2,2+\delta}_{\mathrm{loc}}$-regularity for minimizers of regular variational problems for some *dimension-dependent* $\delta > 0$, this result is originally due to Sergio Campanato. By an embedding theorem this yields $\mathrm{C}^{0,\alpha}$-regularity for some $\alpha \in (0,1)$ when $d \leq 4$. Only in 2008 were Jan Kristensen and Christof Melcher able to establish $\mathrm{W}^{2,2+\delta}_{\mathrm{loc}}$-regularity for a *dimension-independent* $\delta > 0$ (in fact, $\delta = \mu/(50M)$), see [55]. We will learn about further regularity results in Section 3.8. Regularity theory is still an area of active research and new results are constantly discovered.

Finally, we mention that regularity of solutions might in general be extended up to and including the boundary if the boundary is smooth enough, see Section 6.3.2 in [35] and also [44] for results in this direction.

2.6 Lavrentiev gap phenomenon

So far we have chosen the function spaces in which we look for solutions of a minimization problem according to coercivity and growth assumptions. However, at first sight, classically differentiable functions may appear to be more appealing. So the question arises whether the minimum (infimum) value is actually the same when considering different function spaces. Formally, given two spaces $X \subset Y$ such that X is dense in Y and a functional $\mathscr{F}\colon Y \to \mathbb{R} \cup \{+\infty\}$, is

$$\inf_Y \mathscr{F} = \inf_X \mathscr{F} \quad ?$$

This is certainly true if we assume good growth conditions:

Theorem 2.30. *Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}$ be a Caratheéodory function satisfying the growth bound*

$$|f(x,v,A)| \le M(1+|v|^p+|A|^p), \qquad (x,v,A) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d},$$

for some $M > 0$, $1 < p < \infty$. Then, the functional

$$\mathscr{F}[u] := \int_\Omega f(x,u(x),\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$$

is strongly continuous. Consequently,

$$\inf_{\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)} \mathscr{F} = \inf_{\mathrm{C}^\infty(\Omega;\mathbb{R}^m)} \mathscr{F}.$$

Proof. Let $u_j \to u$ in $\mathrm{W}^{1,p}$, whereby $u_j \to u$ and $\nabla u_j \to \nabla u$ almost everywhere (which holds after selecting a non-renumbered subsequence). Then, from the growth assumption we get

$$\mathscr{F}[u_j] = \int_\Omega f(x,u_j,\nabla u_j)\,\mathrm{d}x \le \int_\Omega M(1+|u_j|^p+|\nabla u_j|^p)\,\mathrm{d}x$$

and by the Pratt Theorem A.5,

$$\mathscr{F}[u_j] \to \mathscr{F}[u].$$

This shows the continuity of $\mathscr{F}$ with respect to strong convergence in $\mathrm{W}^{1,p}$.

The assertion about the equality of the infima now follows readily since $\mathrm{C}^\infty(\Omega;\mathbb{R}^m)$ is (strongly) dense in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$. □

If we dispense with the (upper) growth, however, the infimum over different spaces may indeed be different – this is the **Lavrentiev gap phenomenon**, discovered in 1926 by Mikhail Lavrentiev. We here give an example between the spaces $\mathrm{W}^{1,1}$ and $\mathrm{W}^{1,\infty}$ (with boundary conditions) due to Basilio Manià from 1934.

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Example 2.31 (Manià). Consider the minimization problem

$$\begin{cases} \mathscr{F}[u] := \displaystyle\int_0^1 (u(t)^3 - t)^2 \dot{u}(t)^6 \,\mathrm{d}t \quad \to \quad \min, \\ u(0) = 0, \; u(1) = 1 \end{cases}$$

for u from either $\mathrm{W}^{1,1}(0,1)$ or $\mathrm{W}^{1,\infty}(0,1)$ (Lipschitz functions). We claim that

$$\inf_{\mathrm{W}^{1,1}(0,1)} \mathscr{F} < \inf_{\mathrm{W}^{1,\infty}(0,1)} \mathscr{F},$$

where here and in the following these infima are to be taken only over functions u with boundary values $u(0) = 0$, $u(1) = 1$.

Clearly, $\mathscr{F} \geq 0$, and for $u_*(t) := t^{1/3} \in (\mathrm{W}^{1,1} \setminus \mathrm{W}^{1,\infty})(0,1)$ we have $\mathscr{F}[u_*] = 0$. Thus,

$$\inf_{\mathrm{W}^{1,1}(0,1)} \mathscr{F} = 0.$$

On the other hand, for any $u \in \mathrm{W}^{1,\infty}(0,1)$ with $u(0) = 0$, $u(1) = 1$, the Lipschitz-continuity implies that

$$u(t) \leq h(t) := \frac{t^{1/3}}{2} \qquad \text{for all } t \in [0,\tau] \text{ and some } \tau > 0,$$

and $u(\tau) = h(\tau)$. Then, $u(t)^3 - t \leq h(t)^3 - t$ for $t \in [0,\tau]$ and, since both of these terms are negative,

$$(u(t)^3 - t)^2 \geq (h(t)^3 - t)^2 = \frac{7^2}{8^2} t^2 \qquad \text{for all } t \in [0,\tau].$$

We then estimate

$$\mathscr{F}[u] \geq \int_0^\tau (u(t)^3 - t)^2 \,\dot{u}(t)^6 \,\mathrm{d}t \geq \frac{7^2}{8^2} \int_0^\tau t^2 \,\dot{u}(t)^6 \,\mathrm{d}t.$$

Further, by Hölder's inequality,

$$\begin{aligned} \int_0^\tau \dot{u}(t) \,\mathrm{d}t &= \int_0^\tau t^{-1/3} \cdot t^{1/3} \,\dot{u}(t) \,\mathrm{d}t \\ &\leq \left(\int_0^\tau t^{-2/5} \,\mathrm{d}t \right)^{5/6} \cdot \left(\int_0^\tau t^2 \,\dot{u}(t)^6 \,\mathrm{d}t \right)^{1/6} \\ &= \frac{5^{5/6}}{3^{5/6}} \tau^{1/2} \left(\int_0^\tau t^2 \,\dot{u}(t)^6 \,\mathrm{d}t \right)^{1/6}. \end{aligned}$$

Since also

$$\int_0^\tau \dot{u}(t) \,\mathrm{d}t = u(\tau) - u(0) = h(\tau) = \frac{\tau^{1/3}}{2},$$

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we arrive at

$$\mathscr{F}[u] \geq \frac{7^2 3^5}{8^2 5^5 2^6 \tau} \geq \frac{7^2 3^5}{8^2 5^5 2^6} > 0.$$

Thus,

$$\inf_{\mathrm{W}^{1,\infty}(0,1)} \mathscr{F} > 0 = \inf_{\mathrm{W}^{1,1}(0,1)} \mathscr{F}.$$

In a more recent example, Ball & Mizel [13] showed that the problem

$$\begin{cases} \mathscr{F}[u] := \displaystyle\int_{-1}^{1} (t^4 - u(t)^6)^2 |\dot{u}(t)|^{2m} + \varepsilon \dot{u}(t)^2 \,\mathrm{d}t \quad \to \quad \min, \\ u(-1) = \alpha,\ u(1) = \beta, \end{cases}$$

also exhibits a Lavrentiev gap phenomenon between the spaces $\mathrm{W}^{1,2}$ and $\mathrm{W}^{1,\infty}$ if $m \in \mathbb{N}$ satisfies $m > 13$, $\varepsilon > 0$ is sufficiently small, and $-1 \leq \alpha < 0 < \beta \leq 1$. This example is significant because the Ball–Mizel functional is *coercive* on $\mathrm{W}^{1,2}(-1,1)$.

Finally, we note that despite its theoretical significance, the Lavrentiev gap phenomenon is a major obstacle for the numerical approximation of minimization problems. For instance, standard piecewise-affine finite elements are in $\mathrm{W}^{1,\infty}$ and hence in the presence of the Lavrentiev gap phenomenon we *cannot approximate the true solution with such elements*! Thus, one is forced to work with non-conforming elements and other advanced schemes. This issue does not only affect "academic" examples such as the ones above, but is also of great concern in applied problems, such as nonlinear elasticity theory.

2.7 Invariances and the Noether theorem

In physics and other applications of the Calculus of Variations, we are often highly interested in *symmetries* of minimizers or, more generally, critical points. These symmetries manifest themselves in other differential or pointwise relations that are *automatically* satisfied for any critical point. They can be used to identify concrete solutions or are interesting in their own right, for example as "gauge symmetries" in modern physics. In this section, for simplicity we only consider "sufficiently smooth" ($\mathrm{W}^{2,2}_{\mathrm{loc}}$) critical points, which is the natural level of regularity, see Theorem 2.22.

As a first concrete example of a symmetry, consider the Dirichlet integral

$$\mathscr{F}[u] := \int_{\Omega} |\nabla u(x)|^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega),$$

which was introduced in Example 2.9. First we notice that $\mathscr{F}$ is *invariant* under translations in space: Let $u \in (\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}})(\Omega)$, $\tau \in \mathbb{R}$, $k \in \{1, \ldots, d\}$, and set

$$x_\tau := x + \tau \mathrm{e}_k, \qquad u_\tau(x) := u(x + \tau \mathrm{e}_k).$$

Then, for any open $D \subset \mathbb{R}^d$ with $D_\tau := D + \tau \mathrm{e}_k \subset \Omega$, we have the *invariance*

$$\int_D |\nabla u_\tau(x)|^2 \,\mathrm{d}x = \int_{D_\tau} |\nabla u(x_\tau)|^2 \,\mathrm{d}x_\tau. \tag{2.14}$$

The Dirichlet integral also exhibits invariance with respect to *scaling*: For $\lambda > 0$ set

$$x_\lambda := \lambda x, \qquad u_\lambda(x) := \lambda^{(d-2)/2} u(\lambda x), \qquad D_\lambda := \lambda D. \tag{2.15}$$

Then it is not hard to see that (2.14) again holds if we set $\lambda = \mathrm{e}^\tau$ (to allow $\tau \in \mathbb{R}$ as before).

The main result of this section, the *Noether theorem*, roughly says that "differentiable invariances of a functional give rise to conservation laws". More concretely, the two invariances of the Dirichlet integral presented above will yield two additional PDEs that any minimizer of the Dirichlet integral must satisfy.

To make these statement precise in the general case, we need a bit of notation: Let $u \in (\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}})(\Omega;\mathbb{R}^m)$ be a critical point of the functional

$$\mathscr{F}[u] := \int_\Omega f(x, u(x), \nabla u(x)) \,\mathrm{d}x,$$

where $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}$ is twice continuously differentiable. Then, u satisfies the strong Euler–Lagrange equation

$$-\operatorname{div}\big[\mathrm{D}_A f(x,u,\nabla u)\big] + \mathrm{D}_v f(x,u,\nabla u) = 0 \quad \text{a.e. in } \Omega.$$

see Proposition 2.20. We consider u to be extended to all of $\mathbb{R}^d$ (it will not matter below how we extend u).

Our invariance is given through maps $g\colon \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^d$ and $H\colon \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^m$, which will depend on u above, such that

$$g(x,0) = x \qquad \text{and} \qquad H(x,0) = u(x), \qquad x \in \mathbb{R}^d.$$

We also require that g, H are *continuously differentiable* in their second argument τ for almost every $x \in \Omega$. Then set for $x \in \mathbb{R}^d$, $\tau \in \mathbb{R}$, $D \subset\subset \mathbb{R}^d$

$$x_\tau := g(x,\tau), \qquad u_\tau(x) := H(x,\tau), \qquad D_\tau := g(D,\tau).$$

Therefore, g, H can be thought of as a form of *homotopy*.

We call $\mathscr{F}$ **invariant** under the transformation defined by (g,H) if

$$\int_D f(x, u_\tau(x), \nabla u_\tau(x)) \,\mathrm{d}x = \int_{D_\tau} f(x', u(x'), \nabla u(x')) \,\mathrm{d}x'. \tag{2.16}$$

for all $\tau \in \mathbb{R}$ sufficiently small and all open $D \subset \mathbb{R}^d$ such that $D_\tau \subset\subset \Omega$.

The main result of this section goes back to Emmy Noether[5] and is considered one of the most important mathematical theorems ever proved. Its pivotal idea of systematically generating conservation laws from invariances has proved to be immensely influential in modern physics.

[5] The German mathematician Emmy Noether (23 March 1882 – 14 April 1935) has been called the most important woman in the history of mathematics by Einstein and others.

Theorem 2.32 (Noether). *Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}$ be twice continuously differentiable and satisfy the growth bounds*

$$|\mathrm{D}_v f(x,v,A)|, |\mathrm{D}_A f(x,v,A)| \le M(1+|v|^p+|A|^p), \qquad (x,v,A) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d},$$

for some $M > 0$, $1 \le p < \infty$, and let the associated functional $\mathscr{F}$ be invariant under the transformation defined by (g,H) as above, for which we assume that there exist a p-majorant $g \in \mathrm{L}^p(\Omega)$ such that

$$|\partial_\tau H(x,\tau)|, |\partial_\tau g(x,\tau)| \le g(x) \qquad \text{for a.e. } x \in \Omega \text{ and all } \tau \in \mathbb{R}. \tag{2.17}$$

Then, for any critical point $u \in (\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}})(\Omega;\mathbb{R}^m)$ of $\mathscr{F}$ the conservation law

$$\operatorname{div}\big[\mu(x)\cdot \mathrm{D}_A f(x,u,\nabla u) - \nu(x)\cdot f(x,u,\nabla u)\big] = 0 \qquad \text{for a.e. } x \in \Omega. \tag{2.18}$$

holds, where

$$\mu(x) := \partial_\tau H(x,0) \in \mathbb{R}^m \qquad \text{and} \qquad \nu(x) := \partial_\tau g(x,0) \in \mathbb{R}^d \qquad (x \in \Omega)$$

*are the **Noether multipliers**.*

Corollary 2.33. *If $f = f(v,A)\colon \mathbb{R}\times\mathbb{R}^d \to \mathbb{R}$ does not depend on x, then every critical point $u \in (\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}})(\Omega)$ of $\mathscr{F}$ satisfies*

$$\sum_{i=1}^d \partial_i \Big[(\partial_k u)\cdot \partial_{A_k} f(u,\nabla u) - \delta_{ik} f(u,\nabla u)\Big] = 0 \tag{2.19}$$

for all $k = 1,\ldots,d$.

Proof. Differentiate (2.16) with respect to τ and then set $\tau = 0$. To be able to differentiate the left hand side under the integral, we use the growth assumptions on the derivatives $|\mathrm{D}_v f(x,v,A)|, |\mathrm{D}_A f(x,v,A)|$ and (2.17) to get a uniform (in τ) majorant, which allows to move the differentiation under the intgral sign. For the right hand side, we need to employ the formula for the differentiation of an integral with respect to a moving domain (this is a special case of the *Reynolds transport theorem*),

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\int_{D_\tau} f(x,u,\nabla u)\,\mathrm{d}x = \int_{\partial D_\tau} f(x,u,\nabla u)\partial_\tau g(x,\tau)\cdot n\,\mathrm{d}\sigma,$$

where n is the unit outward normal; this formula can be checked in an elementary way. Abbreviating for readability $F := f(x,u(x),\nabla u(x))$, $\mathrm{D}_A F := \mathrm{D}_A f(x,u(x),\nabla u(x))$, and $\mathrm{D}_v F := \mathrm{D}_v f(x,u(x),\nabla u(x))$, we get

$$\int_D \mathrm{D}_A F : \nabla\mu + \mathrm{D}_v F\cdot\mu\,\mathrm{d}x = \int_{\partial D} F\nu\cdot n\,\mathrm{d}\sigma.$$

Applying the Gauss–Green theorem,

$$\begin{aligned}\int_D \big[-\operatorname{div}\mathrm{D}_A F + \mathrm{D}_v F\big]\cdot\mu\,\mathrm{d}x &= \int_{\partial D}\big[-\mu\cdot\mathrm{D}_A F + \nu F\big]\cdot n\,\mathrm{d}\sigma \\ &= \int_D \operatorname{div}\big[-\mu\cdot\mathrm{D}_A F + \nu F\big]\,\mathrm{d}x\end{aligned}$$

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Now, the Euler–Lagrange equation $-\operatorname{div} \mathrm{D}_A F + \mathrm{D}_v F = 0$ immediately implies

$$\int_D \operatorname{div}\big[-\mu \cdot \mathrm{D}_A F + \nu F\big] \, \mathrm{d}x = 0.$$

and varying D we conclude that (2.18) holds.

The corollary follows by considering the invariance $x_\tau := x + \tau \mathrm{e}_k$, $u_\tau(x) := u(x + \tau \mathrm{e}_k)$, $k = 1, \dots, d$. □

Example 2.34 (Brachistochrone problem). In the brachistochrone problem presented in Section 1.1, we were tasked to minimize the functional

$$\mathscr{F}[y] := \int_0^1 \sqrt{\frac{1+(y')^2}{-y}} \, \mathrm{d}x$$

over all $y \colon [0,1] \to \mathbb{R}$ with $y(0) = 0$, $y(1) = \bar{y} < 0$. The integrand $f(v,a) = \sqrt{-(1+a^2)/v}$ is independent of x, hence from (2.19) we get

$$y' \cdot \mathrm{D}_a f(y,y') - f(y,y') = \text{const} = -\frac{1}{\sqrt{2r}}$$

for some $r > 0$ (positive constants lead to inadmissible y). So,

$$\frac{(y')^2}{\sqrt{1+(y')^2} \cdot \sqrt{-y}} - \frac{\sqrt{1+(y')^2}}{\sqrt{-y}} = -\frac{1}{\sqrt{2r}},$$

which we transform into the differential equation

$$(y')^2 = -\frac{2r}{y} - 1.$$

The solution of this differential equation is called an *inverted cycloid* with radius $r > 0$, which is the curve traced out by a fixed point on a sphere of radius r (touching the y-axis at the beginning) that rolls to the right on the bottom of the x-axis, see Figure 2.2. It can be written in *parametric form* as

$$\begin{aligned} x(t) &= r(t - \sin t), \\ y(t) &= -r(1 - \cos t), \end{aligned} \qquad t \in \mathbb{R}.$$

The radius $r > 0$ has to be chosen as to satisfy the boundary conditions on *one* cycloid segment. This is the solution of the brachistochrone problem.

Example 2.35. We return to our canonical example, the Dirichlet integral, see Example 2.9. We know from Example 2.24 that critical points u are smooth. We noticed at the beginning of this section that the Dirichlet integral is invariant under the scaling transformation (2.15) with $\lambda = \mathrm{e}^\tau$. By the Noether theorem, this yields the (non-obvious) conservation law

$$\operatorname{div}\big[(2\nabla u(x) \cdot x + (d-2)u)\nabla u - |\nabla u|^2 x\big] = 0.$$

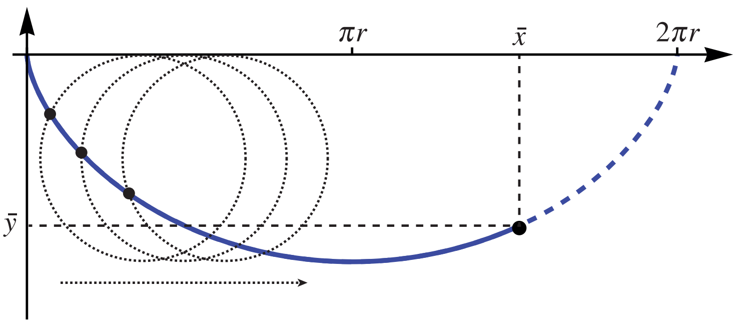

Figure 2.2: The brachistochrone curve (here, $\bar{x} = 1$).

Integrating this over $B(0,r) \subset \Omega$, $r > 0$, and using the Gauss–Green theorem, we get

$$(d-2)\int_{B(0,r)} |\nabla u(x)|^2 \,\mathrm{d}x = r\int_{\partial B(0,r)} |\nabla u(x)|^2 - 2\left(\nabla u(x)\cdot\frac{x}{|x|}\right)^2 \,\mathrm{d}\sigma$$

and then, after some computations,

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r^{d-2}}\int_{B(0,r)} |\nabla u(x)|^2 \,\mathrm{d}x\right) = \frac{2}{r^{d-2}}\int_{\partial B(0,r)} \left(\nabla u(x)\cdot\frac{x}{|x|}\right)^2 \,\mathrm{d}\sigma \geq 0.$$

This *monotonicity formula* implies that

$$\frac{1}{r^{d-2}}\int_{B(0,r)} |\nabla u(x)|^2 \,\mathrm{d}x \quad \text{is increasing in } r > 0.$$

Any harmonic function ($-\Delta u = 0$) that is defined on all of $\mathbb{R}^d$ satisfies this monotonicity formula. For example, this allows us to draw the conclusion that if $d \geq 3$, then u cannot be compactly supported. The formula also shows that the growth around a singularity at zero (or anywhere else by translation) has to behave “more smoothly” than $|x|^{-2}$ (in fact, we already know that solutions are C^∞). While these are not particularly strong remarks, they serve to illustrate how Noether’s theorem restricts the candidates for solutions.

The last example exhibited a conservation law that was not obvious from the Euler–Lagrange equation. While in principle it could have been derived directly, the Noether theorem gave us a *systematic* way to find this conservation law from an invariance.

2.8 Side constraints

In many minimization problems, the class of functions over which we minimize our functional may be restricted to include one or several **side constraints**. To establish existence of a minimizer also in these cases, we first need to extend the Direct Method:

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Theorem 2.36 (Direct Method with side constraint). *Let X be a Banach space and let $\mathscr{F}, \mathscr{G} \colon X \to \mathbb{R} \cup \{+\infty\}$. Assume the following:*

(WH1) ***Weak Coercivity of $\mathscr{F}$:*** *For all $\Lambda \in \mathbb{R}$, the sublevel set*

$$\{ x \in X \ : \ \mathscr{F}[x] \leq \Lambda \} \qquad \text{is sequentially weakly compact,}$$

that is, if $\mathscr{F}[x_j] \leq \Lambda$ for a sequence $(x_j) \subset X$, then (x_j) has a weakly convergent subsequence.

(WH2) ***Weak lower semicontinuity of $\mathscr{F}$:*** *For all sequences $(x_j) \subset X$ with $x_j \rightharpoonup x$ it holds that*

$$\mathscr{F}[x] \leq \liminf_{j \to \infty} \mathscr{F}[x_j].$$

(WH3) ***Weak continuity of $\mathscr{G}$:*** *For all sequences $(x_j) \subset X$ with $x_j \rightharpoonup x$ it holds that*

$$\mathscr{G}[x_j] \to \mathscr{G}[x].$$

Assume also that there exists at least one $x_0 \in X$ with $\mathscr{G}[x_0] = 0$. Then, the minimization problem

$$\mathscr{F}[x] \to \min \quad \text{over all } x \in X \text{ with } \mathscr{G}[x] = \alpha \in \mathbb{R},$$

has a solution.

Proof. The proof is almost exactly the same as the one for the standard Direct Method in Theorem 2.3. However, we need to select the x_j for an infimizing sequence with $\mathscr{G}[x_j] = \alpha$. Then, by (WH3), this property also holds for any weak limit x_* of a subsequence of the x_j's. □

A large class of side constraints can be treated using the following simple result:

Lemma 2.37. *Let $g \colon \Omega \times \mathbb{R}^m \to \mathbb{R}$ be a Carathéodory integrand such that there exists $M > 0$ with*

$$|g(x,v)| \leq M(1 + |v|^p), \qquad (x,v) \in \Omega \times \mathbb{R}^m.$$

Then, $\mathscr{G} \colon \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m) \to \mathbb{R}$ defined through

$$\mathscr{G}[u] := \int_\Omega g(x, u(x)) \, \mathrm{d}x.$$

is weakly continuous.

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Proof. Let $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$, whereby after selecting a subsequence and employing the Rellich–Kondrachov Theorem A.18 and Lemma A.3, $u_j \to u$ in L^p (strongly) and almost everywhere. By assumption we have

$$\pm g(x,v) + M(1+|v|^p) \geq 0.$$

Thus, applying Fatou's Lemma separately to these two integrands, we get

$$\liminf_{j\to\infty} \left(\pm \mathscr{G}[u_j] + \int_\Omega M(1+|u_j|^p)\,\mathrm{d}x \right) \geq \pm \mathscr{G}[u] + \int_\Omega M(1+|u|^p)\,\mathrm{d}x.$$

Since $\|u_j\|_{\mathrm{L}^p} \to \|u\|_{\mathrm{L}^p}$, we can combine these two assertions to get $\mathscr{G}[u_j] \to \mathscr{G}[u]$. Since this holds for all subsequences, it also follows for our original sequence. □

We will later see more complicated side constraints, in particular constraints involving the determinant of the gradient.

The following result is very important for the calculation of minimizers under side constraints. It effectively says that at a minimizer u_* of $\mathscr{F}$ subject to the side constraint $\mathscr{G}[u_*] = 0$, the derivatives of $\mathscr{F}$ and $\mathscr{G}$ have to be *parallel*, in analogy with the finite-dimensional situation.

Theorem 2.38 (Lagrange multipliers). *Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}$, $g\colon \Omega \times \mathbb{R}^m \to \mathbb{R}$ be Carathéodory integrands and satisfy the growth bounds*

$$\begin{aligned} |f(x,v,A)| &\leq M(1+|v|^p+|A|^p), \\ |g(x,v)| &\leq M(1+|v|^p), \qquad\qquad (x,v,A) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d}, \end{aligned}$$

for some $M > 0$, $1 \leq p < \infty$, and let f, g be continuously differentiable in v and A. Suppose that $u_* \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$, where $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$, minimizes the functional

$$\mathscr{F}[u] := \int_\Omega f(x,u(x),\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m),$$

under the side constraint

$$\mathscr{G}[u] := \int_\Omega g(x,u(x))\,\mathrm{d}x = \alpha \in \mathbb{R}.$$

Assume furthermore the consistency condition

$$\delta\mathscr{G}[u_*][v] = \int_\Omega \mathrm{D}_v g(x,u_*(x)) \cdot v(x)\,\mathrm{d}x \neq 0 \tag{2.20}$$

*for at least one $v \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$. Then, there exists a **Lagrange multiplier** $\lambda \in \mathbb{R}$ such that u_* is a weak solution of the system of PDEs*

$$\begin{cases} -\operatorname{div}\big[\mathrm{D}_A f(x,u,\nabla u)\big] + \mathrm{D}_v f(x,u,\nabla u) = \lambda \mathrm{D}_v g(x,u) & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{2.21}$$

Proof. The proof is structurally similar to its finite-dimensional analogue.

Let $u_* \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ be a minimizer of $\mathscr{F}$ under the side constraint $\mathscr{G}[u] = 0$. From the consistency condition (2.20) we infer that there exists $w \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$ such that

$$\delta\mathscr{G}[u_*][w] = \int_\Omega \mathrm{D}_v g(x,u_*) \cdot w \, \mathrm{d}x = 1.$$

Now fix any variation $v \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$ and define for $s,t \in \mathbb{R}$,

$$H(s,t) := \mathscr{G}[u_* + sv + tw].$$

It is not difficult to see that H is continuously differentiable in both s and t (this uses strong continuity of the integral, see Theorem 2.30) and

$$\partial_s H(s,t) = \delta\mathscr{G}[u_* + sv + tw][v],$$
$$\partial_t H(s,t) = \delta\mathscr{G}[u_* + sv + tw][w].$$

Thus, by definition of w,

$$H(0,0) = \alpha \qquad \text{and} \qquad \partial_t H(0,0) = 1.$$

By the implicit function theorem, there exists a function $\tau\colon \mathbb{R} \to \mathbb{R}$ such that $\tau(0) = 0$ and

$$H(s,\tau(s)) = \alpha \qquad \text{for small } |s|.$$

The chain rule yields for such s,

$$0 = \partial_s[H(s,\tau(s))] = \partial_s H(s,\tau(s)) + \partial_t H(s,\tau(s))\tau'(s),$$

whereby

$$\tau'(0) = -\partial_s H(0,0) = -\int_\Omega \mathrm{D}_v g(x,u_*) \cdot v \, \mathrm{d}x. \tag{2.22}$$

Now define for small $|s|$ as above,

$$J(s) := \mathscr{F}[u_* + sv + \tau(s)w].$$

We have

$$\mathscr{G}[u_* + sv + \tau(s)w] = H(s,\tau(s)) = \alpha$$

and the continuously differentiable function J has a minimum at $s = 0$ by the minimization property of u_*. Thus, with the shorthand notations $\mathrm{D}_A f = \mathrm{D}_A f(x,u_*,\nabla u_*)$ and $\mathrm{D}_v f = \mathrm{D}_v f(x,u_*,\nabla u_*)$,

$$0 = J'(0) = \int_\Omega \mathrm{D}_A f : (\nabla v + \tau'(0)\nabla w) + \mathrm{D}_v f \cdot (v + \tau'(0)w) \, \mathrm{d}x.$$

Rearranging and using (2.22), we get

$$\int_\Omega \mathrm{D}_A f : \nabla v + \mathrm{D}_v f \cdot v \,\mathrm{d}x = -\tau'(0) \int_\Omega \mathrm{D}_A f : \nabla w + \mathrm{D}_v f \cdot w \,\mathrm{d}x$$
$$= \lambda \int_\Omega \mathrm{D}_v g(x, u_*) \cdot v \,\mathrm{d}x, \tag{2.23}$$

where we have defined

$$\lambda := \int_\Omega \mathrm{D}_A f : \nabla w + \mathrm{D}_v f \cdot w \,\mathrm{d}x.$$

Since (2.23) shows that u_* is a weak solution of (2.21), the proof is finished. □

Example 2.39 (Stationary Schrödinger equation). When looking for ground states in quantum mechanics as in Section 1.3, we have to minimize

$$\mathscr{E}[\Psi] := \int_{\mathbb{R}^N} \frac{\hbar^2}{2\mu} |\nabla\Psi|^2 + \frac{1}{2} V(x) |\Psi|^2 \,\mathrm{d}x$$

over all $\Psi \in \mathrm{W}^{1,2}(\mathbb{R}^N;\mathbb{C})$ under the side constraint

$$\|\Psi\|_{\mathrm{L}^2} = 1.$$

From Theorem 2.36 in conjunction with Lemma 2.37 (slightly extended to also apply to the whole space $\Omega = \mathbb{R}^N$) we see that this problem always has at least one solution. Theorem 2.38 (likewise extended) yields that this Ψ satisfies

$$\left[\frac{-\hbar^2}{2\mu}\Delta + V(x)\right]\Psi(x) = E\Psi(x), \qquad x \in \Omega,$$

for some Lagrange multiplier $E \in \mathbb{R}$, which is precisely the *stationary Schrödinger equation*. One can also show that $E > 0$ is the smallest eigenvalue of the operator $\Psi \mapsto [\frac{-\hbar^2}{2\mu}\Delta + V(x)]\Psi$.

2.9 Notes and bibliographical remarks

If a given problem has some convexity, this is invariable very useful and the analysis can draw on a vast theory. We have not mentioned the general field of *Convex Analysis*, see for example the classic texts [83] or [34]. In this area, however, the emphasis is more on extending the convex theory to "non-smooth" situations, see for example the monographs [84] and [66, 67].

For the u-dependent variational integrals, the growth in the u-variable can be improved up to q-growth, where $q < p/(p-d)$ (the Sobolev embedding exponent for $\mathrm{W}^{1,p}$). Moreover, we can work with the more general growth bounds $|f(x,v,A)| \le M(g(x) + |v|^q + |A|^p)$, with $g \in \mathrm{L}^1(\Omega;[0,\infty))$ and $q < p/(p-d)$. For reasons of simplicity, we have omitted these generalization here.

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The Fundamental Lemma of Calculus of Variations 2.21 is due to Du Bois-Raymond and is sometimes named after him.

Difference quotients were already considered by Newton, their application to regularity theory is due to Nirenberg in the 1940s and 1950s. Many of the fundamental results of regularity theory (albeit in a non-variational context) can be found in [43]. The book by Giusti [44] contains theory relevant for variational questions. A nice framework for Schauder estimates is [86]. De Giorgi's 1968 counterexample is from [29].

Noether's theorem has many ramifications and can be put into a very general form in Hamiltonian systems and Lie-group theory. The idea is to study groups of symmetries and their actions. For an introduction into this diverse field see [80].

Example 2.35 about the monotonicity formula is from [36].

Side-constraints can also take the form of inequalities, which is often the case in *Optimization Theory* or *Mathematical Programming*. This leads to Karush–Kuhn–Tucker conditions or obstacle problems.

Chapter 3

Quasiconvexity

We saw in the last chapter that convexity of the integrand implies lower semicontinuity for integral functionals. Moreover, we saw in Proposition 2.11 that also the converse is true in the scalar case $d = 1$ or $m = 1$. In the vectorial case $(d, m > 1)$, however, it turns out that one can find weakly lower semicontinuous integral functionals whose integrand is non-convex, the following is the most prominent one: Let $p \geq d$ ($\Omega \subset \mathbb{R}^d$ a bounded Lipschitz domain as usual) and define

$$\mathscr{F}[u] := \int_\Omega \det \nabla u(x) \, \mathrm{d}x, \qquad u \in \mathrm{W}_0^{1,p}(\Omega;\mathbb{R}^d).$$

Then, we can argue using Stokes' Theorem (this can also be computed in a more elementary way, see Lemma 3.8 below):

$$\mathscr{F}[u] = \int_\Omega \det \nabla u \, \mathrm{d}x = \int_\Omega du^1 \wedge \cdots \wedge du^d = \int_{\partial\Omega} u^1 \wedge du^2 \wedge \cdots \wedge du^d = 0$$

because $u \in \mathrm{W}_0^{1,d}(\Omega;\mathbb{R}^d)$ is zero on the boundary $\partial\Omega$. Thus, $\mathscr{F}$ is in fact constant on $\mathrm{W}_0^{1,p}(\Omega;\mathbb{R}^d)$, hence trivially weakly lower semicontinuous. However, the determinant function is far from being convex if $d \geq 2$; for instance we can easily convex-combine a matrix with positive determinant from two singular ones. But we can also find examples not involving singular matrices: For

$$A := \begin{pmatrix} -1 & -2 \\ 2 & 1 \end{pmatrix}, \qquad B := \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}, \qquad \frac{1}{2}A + \frac{1}{2}B = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix},$$

we have $\det A = \det B = 3$, but $\det(A/2 + B/2) = 4$.

Furthermore, convexity of the integrand is not compatible with one of the most fundamental principles of continuum mechanics, the *frame-indifference* or *objectivity*. Assume that our integrand $f = f(A)$ is **frame-indifferent**, that is,

$$f(RA) = f(A) \qquad \text{for all } A \in \mathbb{R}^{d \times d},\ R \in \mathrm{SO}(d),$$

where $SO(d)$ is the set of $(d \times d)$-orthogonal matrices with determinant 1 (i.e. rotations). Furthermore, assume that every purely compressive or purely expansive deformation costs energy, i.e.

$$f(\alpha I) > f(I) \qquad \text{for all } \alpha \neq 1, \tag{3.1}$$

which is very reasonable in applications. Then, f cannot be convex: Let us for simplicity assume $d = 2$. Set, for a fixed $\gamma \in (0, 2\pi)$,

$$R := \begin{pmatrix} \cos\gamma & -\sin\gamma \\ \sin\gamma & \cos\gamma \end{pmatrix} \quad \in \mathrm{SO}(2).$$

Then, if f was convex, for $M := (R + R^T)/2 = (\cos\gamma)I$ we would get

$$f((\cos\gamma)I) = f(M) \leq \frac{1}{2}\big(f(R) + f(R^T)\big) = f(I),$$

contradicting (3.1). Sharper arguments are available, but the essential conclusion is the same: convexity is not suitable for many variational problems originating from continuum mechanics.

3.1 Quasiconvexity

We are now ready to replace convexity with a more general notion, which has become one of the cornerstones of the modern Calculus of Variations. This is the notion of quasiconvexity, introduced by Charles B. Morrey in the 1950s: A locally bounded Borel-measurable function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is called **quasiconvex** if

$$h(A) \leq \fint_{B(0,1)} h(A + \nabla\psi(z))\,\mathrm{d}z \tag{3.2}$$

for all $A \in \mathbb{R}^{m\times d}$ and all $\psi \in \mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^m)$. Here, $\fint_{B(0,1)} := |B(0,1)|^{-1}\int$, where $B(0,1)$ is the unit ball in $\mathbb{R}^d$.

Let us give a physical interpretation to the notion of quasiconvexity: Let $d = m = 3$ and assume that

$$\mathscr{F}[y] := \int_{B(0,1)} h(\nabla y(x))\,\mathrm{d}x, \qquad y \in \mathrm{W}^{1,\infty}(B(0,1);\mathbb{R}^3),$$

models the physical energy of an elastically deformed body, whose deformation from the reference configuration is given through $y\colon \Omega = B(0,1) \to \mathbb{R}^3$ (see Section 1.4 for more details on this modeling). A special class of deformations are the *affine* ones, say $a(x) = y_0 + Ax$ for some $y_0 \in \mathbb{R}^3$, $A \in \mathbb{R}^{3\times 3}$. Then, quasiconvexity of f entails that

$$\mathscr{F}[a] = \int_{B(0,1)} h(A)\,\mathrm{d}x \leq \int_{B(0,1)} h(A + \nabla\psi(z))\,\mathrm{d}z$$

for all $\psi \in \mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^3)$. This means that the affine deformation a is always energetically favorable over the *internally distorted* deformation $a(x) + \psi(x)$; this is very often a

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reasonable assumption for real materials. This argument also holds on any other domain by Lemma 3.2 below.

To justify its name, we also need to convince ourselves that quasiconvexity is indeed a notion of *convexity*: For $A \in \mathbb{R}^{m\times d}$ and $V \in \mathrm{L}^1(B(0,1);\mathbb{R}^{m\times d})$ with $\int_{B(0,1)} V(x)\,\mathrm{d}x = 0$ define the probability measure $\mu \in \mathbf{M}^1(\mathbb{R}^{m\times d})$ via its *action* as an element of $\mathrm{C}_0(\mathbb{R}^{m\times d})^*$ as follows (recall that $\mathbf{M}(\mathbb{R}^{m\times d}) \cong \mathrm{C}_0(\mathbb{R}^{m\times d})^*$ by the Riesz Representation Theorem A.10):

$$\langle h,\mu\rangle := \fint_{B(0,1)} h(A+V(x))\,\mathrm{d}x \qquad \text{for } h \in \mathrm{C}_0(\mathbb{R}^{m\times d}).$$

This μ is easily seen to be an element of the dual space to $\mathrm{C}_0(\mathbb{R}^{m\times d})$, and in fact μ is a probability measure: For the boundedness we observe $|\langle h,\mu\rangle| \le \|h\|_\infty$, whereas the positivity $\langle h,\mu\rangle \ge 0$ for $h \ge 0$ and the normalization $\langle \mathbb{1},\mu\rangle = 1$ follow at once. The barycenter $[\mu]$ of μ is

$$[\mu] = \langle \mathrm{id},\mu\rangle = A + \fint_{B(0,1)} V(x)\,\mathrm{d}x = A.$$

Therefore, if h is convex, we get from then Jensen inequality, Lemma A.9,

$$h(A) = h([\mu]) \le \langle h,\mu\rangle = \fint_{B(0,1)} h(A+V(x))\,\mathrm{d}x.$$

In particular, (3.2) holds if we set $V(x) := \nabla\psi(x)$ for any $\psi \in \mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^m)$. Thus, we have shown:

Proposition 3.1. *All convex functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ are quasiconvex.*

However, quasiconvexity is weaker than classical convexity, at least if $d,m \ge 2$. The determinant function and, more generally, minors are quasiconvex, as will be proved in the next section, but these minors (except for the (1×1)-minor) are not convex. We will see many further quasiconvex functions in the course of this and the next chapter. The relation between (quasi)convexity and Jensen's inequality plays a major role in the theory, as we will see soon.

Other basic properties of quasiconvexity are the following:

Lemma 3.2. *In the definition of quasiconvexity we can replace the domain $B(0,1)$ by any bounded Lipschitz domain $\Omega' \subset \mathbb{R}^d$. Furthermore, if h has p-growth, i.e. $|h(A)| \le M(1+|A|^p)$ for some $1 \le p < \infty$, $M > 0$, then in the definition of quasiconvexity we can replace testing with ψ of class $\mathrm{W}_0^{1,\infty}$ by ψ of class $\mathrm{W}_0^{1,p}$.*

Proof. To see the first statement, we will prove the following: If $\psi \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, satisfies $\psi|_{\partial\Omega} = Ax$ for some $A \in \mathbb{R}^{m\times d}$, then there exists $\tilde\psi \in \mathrm{W}^{1,p}(\Omega';\mathbb{R}^m)$ with $\tilde\psi|_{\partial\Omega'} = Ax$ and, if one of the following integrals exists and is finite,

$$\fint_\Omega h(\nabla\psi)\,\mathrm{d}x = \fint_{\Omega'} h(\nabla\tilde\psi)\,\mathrm{d}y \qquad \text{for all measurable } h\colon \mathbb{R}^{m\times d} \to \mathbb{R}. \tag{3.3}$$

In particular, $\|\nabla\psi\|_{\mathrm{L}^p} = \|\nabla\tilde{\psi}\|_{\mathrm{L}^p}$. Clearly, this implies that the definition of quasiconvexity is independent of the domain.

To see (3.3), take a Vitali cover of Ω' with re-scaled versions of Ω, see Theorem A.8,

$$\Omega' = Z \uplus \biguplus_{k=1}^{N} \Omega(a_k, r_k), \qquad |Z| = 0,$$

with $a_k \in \Omega$, $r_k > 0$, $\Omega(a_k, r_k) := a_k + r_k\Omega$, where $k = 1, \ldots, N \in \mathbb{N}$. Then let

$$\tilde{\psi}(y) := \begin{cases} r_k\psi\left(\dfrac{y-a_k}{r_k}\right) + Aa_k & \text{if } y \in \Omega(a_k, r_k), \\ 0 & \text{otherwise,} \end{cases} \qquad y \in \Omega'.$$

It holds that $\tilde{\psi} \in \mathrm{W}^{1,\infty}(\Omega'; \mathbb{R}^m)$. We compute for any measurable $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$,

$$\begin{aligned} \int_{\Omega'} h(\nabla\tilde{\psi}) \,\mathrm{d}y &= \sum_k \int_{\Omega(a_k,r_k)} h\left(\nabla\psi\left(\frac{y-a_k}{r_k}\right)\right) \mathrm{d}y \\ &= \sum_k r_k^d \int_\Omega h(\nabla\psi) \,\mathrm{d}x \\ &= \frac{|\Omega'|}{|\Omega|} \int_\Omega h(\nabla\psi) \,\mathrm{d}x \end{aligned}$$

since $\sum_k r_k^d = |\Omega'|/|\Omega|$. This shows (3.3).

The second assertion follows since $\mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^m)$ is dense in $\mathrm{W}_0^{1,p}(B(0,1);\mathbb{R}^m)$ and under a p-growth assumption, the integral functional

$$\psi \mapsto \fint_\Omega h(\nabla\psi) \,\mathrm{d}x, \qquad \psi \in \mathrm{W}_0^{1,p}(B(0,1);\mathbb{R}^m),$$

is well-defined and continuous, the latter for example by Pratt's Theorem A.5, see the proof of Theorem 2.30 for a similar argument. □

An even weaker notion of convexity is the following one: A locally bounded Borel-measurable function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is called **rank-one convex** if it is convex along any rank-one line, that is,

$$h\big(\theta A + (1-\theta)B\big) \le \theta h(A) + (1-\theta)h(B)$$

for all $A, B \in \mathbb{R}^{m\times d}$ with $\operatorname{rank}(A-B) \le 1$ and all $\theta \in (0,1)$. In this context, recall that a matrix $M \in \mathbb{R}^{m\times d}$ has rank one if and only if $M = a \otimes b = ab^T$ for some $a \in \mathbb{R}^m$, $b \in \mathbb{R}^d$.

Proposition 3.3. *If $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is quasiconvex, then it is rank-one convex.*

Proof. Let $A, B \in \mathbb{R}^{m\times d}$ with $B - A = a \otimes n$ for $a \in \mathbb{R}^m$ and $n \in \mathbb{R}^d$ with $|n| = 1$. Denote by Q_n a unit cube ($|Q_n| = 1$) centered at the origin and with one side normal to n. By Lemma 3.2 we can require h to satisfy

$$h(M) \le \fint_{Q_n} h(\nabla\psi(z)) \,\mathrm{d}z \tag{3.4}$$

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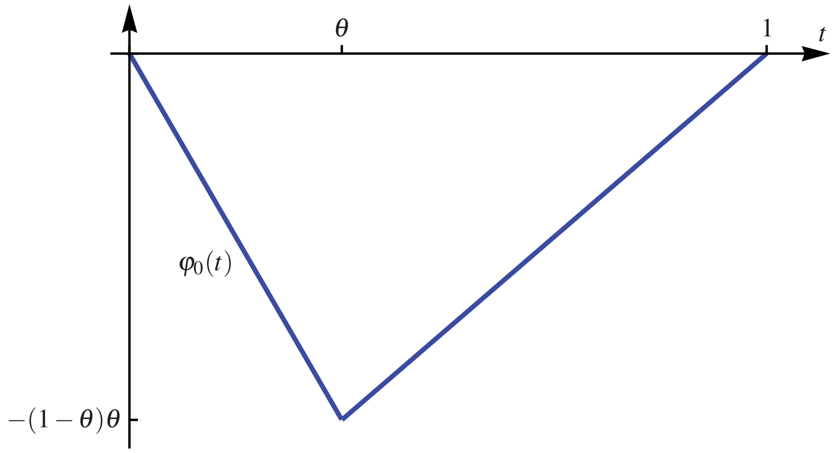

Figure 3.1: The function φ_0.

for all $M \in \mathbb{R}^{m\times d}$ and all $\psi \in \mathrm{W}^{1,\infty}(Q_n;\mathbb{R}^m)$ with $\psi(x) = Mx$ on ∂Q_n (in the sense of trace).

For fixed $\theta \in (0,1)$, we now let $M := \theta A + (1-\theta)B$ and define the sequence of test functions $\varphi_j \in \mathrm{W}^{1,\infty}(Q_n;\mathbb{R}^m)$ as follows:

$$\varphi_j(x) := Mx + \frac{1}{j}\varphi_0\big(jx\cdot n - \lfloor jx\cdot n\rfloor\big)a, \qquad x \in Q_n,$$

where $\lfloor s\rfloor$ denotes the largest integer below or equal to $s \in \mathbb{R}$, and

$$\varphi_0(t) := \begin{cases} -(1-\theta)t & \text{if } t \in [0,\theta], \\ \theta t - \theta & \text{if } t \in (\theta,1], \end{cases} \qquad t \in [0,1).$$

Such φ_j are called **laminations** in direction n. See Figures 3.1 and 3.2 for φ_0 and φ_j, respectively.

We calculate

$$\nabla\varphi_j(x) = \begin{cases} M - (1-\theta)a\otimes n = A & \text{if } jx\cdot n - \lfloor jx\cdot n\rfloor \in (0,\theta), \\ M + \theta a\otimes n = B & \text{if } jx\cdot n - \lfloor jx\cdot n\rfloor \in (\theta,1). \end{cases}$$

Thus,

$$\lim_{j\to\infty} \fint_{Q_n} h(\nabla\varphi_j(x))\,\mathrm{d}x = \theta h(A) + (1-\theta)h(B).$$

Also notice that $\varphi_j \overset{*}{\rightharpoondown} a$ in $\mathrm{W}^{1,\infty}$ for $a(x) := Mx$ since φ_0 is uniformly bounded. We will show below that we may replace the sequence (φ_j) with a sequence $(\psi_j) \subset \mathrm{W}^{1,\infty}(Q_n;\mathbb{R}^m)$ with the additional property $\psi(x) = Mx$ on ∂Q_n, but such that still

$$\lim_{j\to\infty} \fint_{Q_n} h(\nabla\psi_j(x))\,\mathrm{d}x = \theta h(A) + (1-\theta)h(B).$$

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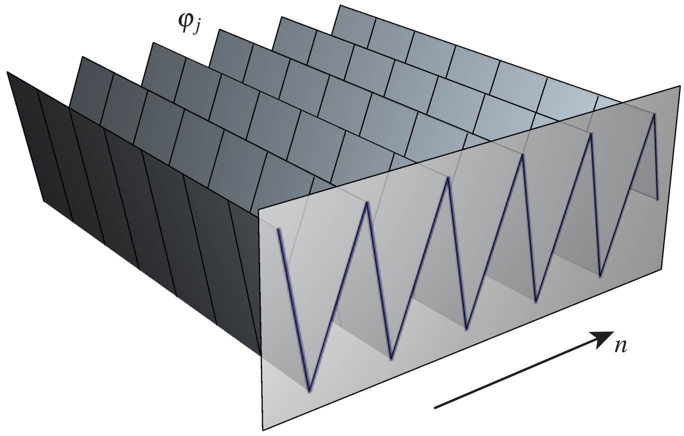

Figure 3.2: The laminate φ_j.

Combining this with (3.4) allows us to conclude

$$h\big(\theta A + (1-\theta)B\big) \le \theta h(A) + (1-\theta)h(B)$$

and h is indeed rank-one convex.

It remains to construct the sequence (ψ_j) from (φ_j), for which we employ a standard **cut-off construction**: Take a sequence $(\rho_j) \subset \mathrm{C}_c^\infty(Q_n;[0,1])$ of cut-off functions such that for $G_j := \{ x \in \Omega : \rho_j(x) = 1 \}$ it holds that $|Q_n \setminus G_j| \to 0$ as $j \to \infty$. With $a(x) = Mx$ set

$$\psi_{j,k} := \rho_j \varphi_k + (1-\rho_j)a,$$

which lies in $\mathrm{W}^{1,\infty}(Q_n;\mathbb{R}^m)$ and satisfies $\psi_{j,k}(x) = Mx$ near ∂Q_n. Also,

$$\nabla \psi_{j,k} = \rho_j \nabla \varphi_k + (1-\rho_j)M + (\varphi_k - a) \otimes \nabla \rho_j.$$

Since $\mathrm{W}^{1,\infty}(Q_n;\mathbb{R}^m)$ embeds compactly into $\mathrm{L}^\infty(Q_n;\mathbb{R}^m)$ by the Rellich-Kondrachov theorem (or the classical Arzéla–Ascoli theorem), we have $\varphi_k \to a$ uniformly. Thus, in

$$\|\nabla \psi_{j,k}\|_{\mathrm{L}^\infty} \le \|\nabla \varphi_k\|_{\mathrm{L}^\infty} + |M| + \|(\varphi_k - a) \otimes \nabla \rho_j\|_{\mathrm{L}^\infty}$$

the last term vanishes as $k \to \infty$. As the $\nabla \varphi_k$ furthermore are uniformly L^∞-bounded, we can for every $j \in \mathbb{N}$ choose $k(j) \in \mathbb{N}$ such that $\|\nabla \psi_{j,k(j)}\|_{\mathrm{L}^\infty}$ is bounded by a constant that is independent of j. Then, as h is assumed to be locally bounded, this implies that there exists $C > 0$ (again independent of j) with

$$\|h(\nabla \varphi_j)\|_{\mathrm{L}^\infty} + \|h(\nabla \psi_{j,k(j)})\|_{\mathrm{L}^\infty} \le C.$$

Hence, for $\psi_j := \psi_{j,k(j)}$, we may estimate

$$\begin{aligned}\lim_{j\to\infty}\int_{Q_n}|h(\nabla\varphi_j)-h(\nabla\psi_j)|\,\mathrm{d}x &= \lim_{j\to\infty}\int_{Q_n\setminus G_j}|h(\nabla\varphi_j)|+|h(\nabla\psi_{j,k(j)})|\,\mathrm{d}x\\ &\le \lim_{j\to\infty} C|Q_n\setminus G_j| = 0.\end{aligned}$$

This shows that above we may replace (φ_j) by (ψ_j). $\square$

Since for $d=1$ or $m=1$, rank-one convexity obviously is equivalent to convexity, we have for the *scalar* case:

Corollary 3.4. *If $h\colon \mathbb{R}^{m\times d}\to\mathbb{R}$ is quasiconvex and $d=1$ or $m=1$, then h is convex.*

The following is a standard example:

Example 3.5 (Alibert–Dacorogna–Marcellini). For $d=m=2$ and $\gamma\in\mathbb{R}$ define

$$h_\gamma(A) := |A|^2\bigl(|A|^2-2\gamma\det A\bigr), \qquad A\in\mathbb{R}^{2\times 2}.$$

For this function it is known that

- h_γ is convex if and only if $|\gamma|\le \dfrac{2\sqrt{2}}{3}\approx 0.94$,
- h_γ is rank-one convex if and only if $|\gamma|\le \dfrac{2}{\sqrt{3}}\approx 1.15$,
- h_γ is quasiconvex if and only if $|\gamma|\le\gamma_{\mathrm{QC}}$ for some $\gamma_{\mathrm{QC}}\in\left(1,\dfrac{2}{\sqrt{3}}\right]$.

It is currently unknown whether $\gamma_{\mathrm{QC}}=\frac{2}{\sqrt{3}}$. We do not prove these statements here, see Section 5.3.8 in [25] for the (involved!) details.

We end this section with the following regularity theorem for rank-one convex (and hence also for quasiconvex) functions:

Proposition 3.6. *If $h\colon\mathbb{R}^{m\times d}\to\mathbb{R}$ is rank-one convex (and locally bounded), then it is locally Lipschitz continuous.*

Proof. We will prove the stronger *quantitative bound*

$$\operatorname{lip}(h;B(A_0,r)) \le \sqrt{\min(d,m)}\cdot\frac{\operatorname{osc}(h;\overline{B(A_0,6r)})}{3r}, \tag{3.5}$$

where

$$\operatorname{lip}(h;B(A_0,r)) := \sup_{A,B\in B(A_0,r)}\frac{|h(A)-h(B)|}{|A-B|}$$

is the *Lipschitz constant* of h on the ball $B(A_0,r) \subset \mathbb{R}^{m\times d}$, and

$$\operatorname{osc}(h;B(A_0,r)) := \sup_{A,B\in B(A_0,r)} |h(A)-h(B)|$$

is the *oscillation* of h on $B(A_0,r)$. By the local boundedness, the oscillation is bounded on every ball, and so, locally, the Lipschitz constant is finite.

To show (3.5), let $A,B \in B(x_0,r)$ and assume first that $\operatorname{rank}(A-B) \le 1$. Define the point $C \in \mathbb{R}^{m\times d}$ as the intersection of $\partial B(A_0,2r)$ with the ray starting at B and going through A. Then, because h is convex along this ray,

$$\frac{|h(A)-h(B)|}{|A-B|} \le \frac{|h(C)-h(B)|}{|C-B|} \le \frac{\operatorname{osc}(h;\overline{B(A_0,2r)})}{r} =: \alpha(2r). \tag{3.6}$$

For general $A,B \in B(A_0,r)$, use the (real) singular value decomposition to write

$$B-A = \sum_{i=1}^{\min(d,m)} \sigma_i P(\mathrm{e}_i \otimes \mathrm{e}_i)Q^T,$$

where $\sigma_i \ge 0$ is the i'th singular value, and $P \in \mathbb{R}^{m\times m}$, $Q \in \mathbb{R}^{d\times d}$ are orthogonal matrices. Set

$$A_k := A + \sum_{i=1}^{k-1} \sigma_i P(\mathrm{e}_i \otimes \mathrm{e}_i)Q^T, \qquad k = 1,\dots,\min(d,m)+1,$$

for which we have $A_1 = A$, $A_{\min(d,m)+1} = B$. We estimate (recall that we are employing the Frobenius norm $|M| := \sqrt{\sum_{j,k}(M^j_k)^2} = \sqrt{\sum_i \sigma_i(M)^2}$)

$$|A_k - A_0| \le |A-A_0| + \sqrt{\sum_{i=1}^{k-1} \sigma_i^2} \le |A-A_0| + |B-A| < 3r$$

and

$$\sum_{k=1}^{\min(d,m)} |A_k - A_{k+1}|^2 = \sum_{k=1}^{\min(d,m)} \sigma_i^2 = |A-B|^2.$$

Applying (3.6) to $A_k, A_{k+1} \in B(A_0,3r)$, $k = 1,\dots,\min(d,m)$, we get

$$\begin{aligned}
|h(A)-h(B)| &\le \sum_{k=1}^{\min(d,m)} |h(A_k)-h(A_{k+1})| \\
&\le \alpha(6r) \sum_{k=1}^{\min(d,m)} |A_k - A_{k+1}| \\
&\le \alpha(6r)\sqrt{\min(d,m)} \cdot \left[\sum_{k=1}^{\min(d,m)} |A_k - A_{k+1}|^2\right]^{1/2} \\
&= \alpha(6r)\sqrt{\min(d,m)} \cdot |A-B|.
\end{aligned}$$

This yields (3.5). $\square$

Comment...

Remark 3.7. An improved argument (see Lemma 2.2 in [12]), where one orders the singular values in a particular way, allows to establish the better estimate

$$\operatorname{lip}(h; B(A_0,r)) \le \sqrt{\min(d,m)} \cdot \frac{\operatorname{osc}(h; \overline{B(A_0,2r)})}{r}.$$

3.2 Null-Lagrangians

The determinant is quasiconvex, but it is only one representative of a larger class of canonical examples of quasiconvex, but not convex, functions: In this section, we will investigate the properties of minors (subdeterminants) as integrands. Let for $r \in \{1,2,\dots,\min(d,m)\}$,

$$I \in P(m,r) := \left\{ (i_1,i_2,\dots,i_r) \in \{1,\dots,m\}^r \;\mathbf{:}\; i_1 < i_2 < \dots < i_r \right\}$$

and $J \in P(d,r)$ be **ordered multi-indices**. Then, a $(r \times r)$**-minor** $M \colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is a function of the form

$$M(A) = M_J^I(A) := \det\bigl(A_J^I\bigr),$$

where A_J^I is the (square) matrix consisting of the I-rows and J-columns of A. The number $r \in \{1,\dots,\min(d,m)\}$ is called the **rank** of the minor M.

The first result shows that all minors are **null-Lagrangians**, which by definition is the class of integrands $h \colon \mathbb{R}^{m\times d}$ such that $\int_\Omega h(\nabla u)\,\mathrm{d}x$ only depends on the boundary values of u.

Lemma 3.8. *Let $M \colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be a $(r\times r)$-minor, $r \in \{1,\dots,\min(d,m)\}$. If $u,v \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, $p \ge r$, with $u - v \in \mathrm{W}_0^{1,p}(\Omega;\mathbb{R}^m)$, then*

$$\int_\Omega M(\nabla u)\,\mathrm{d}x = \int_\Omega M(\nabla v)\,\mathrm{d}x.$$

Proof. In all of the following, we will assume that u,v are smooth and $\operatorname{supp}(u-v) \subset\subset \Omega$, which can be achieved by approximation (incorporating a cut-off procedure), see Theorem. A.19. We also need that taking the minor M of the gradient commutes with strong convergence, i.e. the strong continuity of $u \mapsto M(\nabla u)$ in $\mathrm{W}^{1,p}$ for $p \ge r$; this follows by Hadamard's inequality $|M(A)| \le |A|^r$ and Pratt's Theorem A.5. Finally, we assume (again by approximation) that $\operatorname{supp}(u-v) \subset\subset \Omega$.

All first-order minors are just the entries of $\nabla u, \nabla v$ and the result follows from

$$\int_\Omega \nabla u\,\mathrm{d}x = \int_{\partial\Omega} u \cdot n\,\mathrm{d}\sigma = \int_{\partial\Omega} v \cdot n\,\mathrm{d}\sigma = \int_\Omega \nabla v\,\mathrm{d}x$$

since $u - v \in \mathrm{W}_0^{1,p}(\Omega;\mathbb{R}^m)$.

For higher-order minors, the crucial observation is that minors of gradients can be written as *divergences*, which we will establish below. So, if $M(\nabla u) = \operatorname{div} G(u,\nabla u)$ for some $G(u,\nabla u) \in \mathrm{C}^\infty(\Omega;\mathbb{R}^d)$, then, since $\operatorname{supp}(u-v) \subset\subset \Omega$,

$$\int_\Omega M(\nabla u)\,\mathrm{d}x = \int_{\partial\Omega} G(u,\nabla u)\cdot n\,\mathrm{d}\sigma = \int_{\partial\Omega} G(v,\nabla v)\cdot n\,\mathrm{d}\sigma = \int_\Omega M(\nabla v)\,\mathrm{d}x$$

and the result follows.

We first consider the physically most relevant cases $d = m \in \{2,3\}$.

For $d = m = 2$ and $u = (u^1, u^2)^T$, the only second-order minor is the Jacobian determinant and we easily see from the fact that second derivatives of smooth functions commute that

$$\begin{aligned}\det \nabla u &= \partial_1 u^1 \partial_2 u^2 - \partial_2 u^1 \partial_1 u^2 \\ &= \partial_1 (u^1 \partial_2 u^2) - \partial_2 (u^1 \partial_1 u^2) \\ &= \operatorname{div}\big(u^1 \partial_2 u^2, -u^1 \partial_1 u^2\big).\end{aligned}$$

For $d = m = 3$, consider a second-order minor $M^{\neg k}_{\neg l}(A)$, i.e. the determinant of A after deleting the k'th row and l'th column. Then, analogously to the situation in dimension two, we get, using *cyclic* indices $k, l \in \{1,2,3\}$,

$$\begin{aligned}M^{\neg k}_{\neg l}(\nabla u) &= (-1)^{k+l}\big[\partial_{l+1} u^{k+1} \partial_{l+2} u^{k+2} - \partial_{l+2} u^{k+1} \partial_{l+1} u^{k+2}\big] \\ &= (-1)^{k+l}\big[\partial_{l+1}(u^{k+1} \partial_{l+2} u^{k+2}) - \partial_{l+2}(u^{k+1} \partial_{l+1} u^{k+2})\big].\end{aligned} \tag{3.7}$$

For the three-dimensional Jacobian determinant, we will show

$$\det \nabla u = \sum_{l=1}^{3} \partial_l \big(u^1 (\operatorname{cof} \nabla u)^1_l\big), \tag{3.8}$$

where we recall that $(\operatorname{cof} A)^k_l = (-1)^{k+l} M^{\neg k}_{\neg l}(A)$. To see this, use the Cramer formula $(\det A) I = A (\operatorname{cof} A)^T$, which holds for any matrix $A \in \mathbb{R}^{n \times n}$, to get

$$\det \nabla u = \sum_{l=1}^{3} \partial_l u^1 \cdot (\operatorname{cof} \nabla u)^1_l.$$

Then, our formula (3.8) follows from the **Piola identity**

$$\operatorname{div} \operatorname{cof} \nabla u = 0, \tag{3.9}$$

which can be verified directly from the expression (3.7) for $M^{\neg k}_{\neg l}(\nabla u)$. Hence, (3.8) holds.

For general dimensions d, m we use the notation of differential geometry to tame the multilinear algebra involved in the proof. So, let M be a $(r \times r)$-minor. Reordering $x^1, \ldots, x^d$ and $u^1, \ldots, u^m$, we can assume without loss of generality that M is a *principal* minor, i.e. M is the determinant of the top-left $(r \times r)$-submatrix. Then, in differential geometry it is well-known that

$$\begin{aligned}M(\nabla u)\, dx^1 \wedge \cdots \wedge dx^d &= du^1 \wedge \cdots \wedge du^r \wedge dx^{r+1} \wedge \cdots \wedge dx^d \\ &= d(u^1 \wedge du^2 \wedge \cdots \wedge du^r \wedge dx^{r+1} \wedge \cdots \wedge dx^d).\end{aligned}$$

Thus, Stokes Theorem from Differential Geometry gives

$$\begin{aligned}\int_\Omega M(\nabla u)\, dx^1 \wedge \cdots \wedge dx^d &= \int_\Omega d(u^1 \wedge du^2 \wedge \cdots \wedge du^r \wedge dx^{r+1} \wedge \cdots \wedge dx^d) \\ &= \int_{\partial\Omega} u^1 \wedge du^2 \wedge \cdots \wedge du^r \wedge dx^{r+1} \wedge \cdots \wedge dx^d.\end{aligned}$$

Therefore, $\int_\Omega M(\nabla u)\, dx^1 \wedge \cdots \wedge dx^d$ only depends on the values of u around $\partial\Omega$. □

Comment...

Consequently, we immediately have:

Corollary 3.9. *All $(r \times r)$-minors $M\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ are **quasiaffine**, that is, both M and $-M$ are quasiconvex.*

We will later use this property to define a large and useful class of quasiconvex functions, namely the *polyconvex* ones; this is the topic of the next chapter.

Further, minors enjoy a surprising continuity property:

Lemma 3.10 (Weak continuity of minors). *Let $M\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be an $(r\times r)$-minor, $r \in \{1,\dots,\min(d,m)\}$, and $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ where $r < p \le \infty$. If*

$$u_j \rightharpoonup u \quad \text{in } \mathrm{W}^{1,p} \qquad (\text{or } u_j \overset{*}{\rightharpoonup} u \text{ in } \mathrm{W}^{1,\infty}).$$

then

$$M(\nabla u_j) \rightharpoonup M(\nabla u) \quad \text{in } \mathrm{L}^{p/r} \qquad (\text{or } M(\nabla u_j) \overset{*}{\rightharpoonup} M(\nabla u) \text{ in } \mathrm{W}^{1,\infty}).$$

Proof. For $d = m \in \{2,3\}$, we use the special structure of minors as divergences, as exhibited in the proof of Lemma 3.8. For a (2×2)-minor $M_{\neg l}^{\neg k}$ in three dimensions (that is, deleting the k'th row and l'th column; in two dimensions there is only one (2×2)-minor, the determinant, but we still use the same notation), we rely on (3.7) to observe that with cyclic indices $k,l \in \{1,2,3\}$,

$$\int_\Omega M_{\neg l}^{\neg k}(\nabla u_j)\psi \,\mathrm{d}x = -(-1)^{k+l}\int_\Omega (u_j^{k+1}\partial_{l+2}u_j^{k+2})\partial_{l+1}\psi - (u_j^{k+1}\partial_{l+1}u_j^{k+2})\partial_{l+2}\psi \,\mathrm{d}x$$

for all $\psi \in \mathrm{C}_c^\infty(\Omega)$ and then by density also for all $\psi \in \mathrm{L}^{p/r}(\Omega)^* \cong \mathrm{L}^{p/(p-r)}(\Omega)$. Since $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$, we have $u_j \to u$ strongly in L^p. The above expression consists of products of one L^p-weakly and one L^p-strongly convergent factor as well as a uniformly bounded function under the integral. Hence the integral is weakly continuous and as $j\to\infty$ converges to

$$\int_\Omega M_{\neg l}^{\neg k}(\nabla u)\psi \,\mathrm{d}x.$$

This finishes the proof for $d = m = 2$.

For $d = m = 3$, we additionally need to consider the determinant. However, as a consequence of the two-dimensional reasoning, $\operatorname{cof}\nabla u_j \rightharpoonup \operatorname{cof}\nabla u$ in $\mathrm{L}^{p/2}$. Then, (3.8) implies

$$\int_\Omega \det\nabla u_j\,\psi\,\mathrm{d}x = -\sum_{l=1}^{3}\int_\Omega \big[u_j^1(\operatorname{cof}\nabla u_j)_l^1\big]\partial_l\psi\,\mathrm{d}x$$

and by a similar reasoning as above, this converges to

$$-\sum_{l=1}^{3}\int_\Omega \big[u^1(\operatorname{cof}\nabla u)_l^1\big]\partial_l\psi\,\mathrm{d}x = \int_\Omega \det\nabla u\,\psi\,\mathrm{d}x.$$

In the general case, one proceeds by induction. We omit the details. □

It can also be shown that *any* quasiaffine function can be written as an affine function of all the minors of the argument. This characterization of quasiaffine functions is due to Ball [5], a different proof (also including further characterizing statements) can be found in Theorem 5.20 of [25].

3.3 Young measures

We now introduce a very versatile tool for the study of minimization problems – the *Young measure*, named after its inventor Laurence Chisholm Young. In this chapter, we will only use it as a device to organize the proof of lower semicontinuity for quasiconvex integral functions, see Theorems 3.25, 3.29, but later it will also be important as an object in its own right.

Let us motivate this device by the following central issue: Assume that we have a weakly convergent sequence $v_j \rightharpoonup v$ in $\mathrm{L}^2(\Omega)$, where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, and an integral functional

$$\mathscr{F}[v] := \int_\Omega f(x, v(x)) \,\mathrm{d}x$$

with $f \colon \Omega \times \mathbb{R} \to \mathbb{R}$ continuous and bounded, say. Then, we may want to know the limit of $\mathscr{F}[v_j]$ as $j \to \infty$, for instance if (v_j) is a minimizing sequence. In fact, this will be cornerstone of the proof of weak lower semicontinuity for integral functionals with quasiconvex integrand.

Equivalently, we could ask for the L^∞-weak* limit of the sequence of compound functions $g_j(x) := f(x, v_j(x))$. It is easy to see that the weak* limit in general is *not* equal to $x \mapsto f(x, v(x))$. For example, in $\Omega = (0,1)$, consider the sequence

$$v_j(x) := \begin{cases} a & \text{if } jx - \lfloor jx \rfloor \leq \theta, \\ b & \text{otherwise,} \end{cases} \qquad x \in \Omega,$$

where $a, b \in \mathbb{R}$, $\theta \in (0,1)$, and $\lfloor s \rfloor$ is the largest integer below $s \in \mathbb{R}$. Then, if $f(x,a) = \alpha \in \mathbb{R}$ and $f(x,b) = \beta \in \mathbb{R}$ (and smooth and bounded elsewhere), we see $v_j \overset{*}{\rightharpoonup} \theta a + (1-\theta)b$ and

$$f(x, v_j(x)) \;\overset{*}{\rightharpoonup}\; h(x) = \theta\alpha + (1-\theta)\beta.$$

The right-hand side, however, is not necessarily $f(x, \theta a + (1-\theta)b)$. Instead, we could write

$$h(x) = \int f(x, v) \,\mathrm{d}\nu_x(v)$$

with a x-parametrized family of probability measures $\nu_x \in \mathbf{M}^1(\mathbb{R})$, namely

$$\nu_x = \theta\delta_\alpha + (1-\theta)\delta_\beta, \qquad x \in (0,1).$$

This family $(\nu_x)_{x\in\Omega}$ is the "Young measure generated by the sequence (v_j)".

We start with the following fundamental result:

Theorem 3.11 (Fundamental theorem of Young measure theory). *Let $(V_j) \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$ be a bounded sequence, where $1 \leq p \leq \infty$. Then, there exists a subsequence (not relabeled) and a family $(\nu_x)_{x\in\Omega} \subset \mathbf{M}^1(\mathbb{R}^N)$ of probability measures, called the **p-Young measure generated by the sequence** (V_j), such that the following assertions are true:*

(I) The family (ν_x) is weakly measurable, that is, for all Carathéodory integrands $f\colon \Omega \times \mathbb{R}^N \to \mathbb{R}$, the compound function*

$$x \mapsto \int f(x,A)\, \mathrm{d}\nu_x(A), \qquad x \in \Omega,$$

is Lebesgue-measurable.

(II) If $1 \leq p < \infty$, it holds that

$$\langle\!\langle |\bullet|^p, \nu \rangle\!\rangle := \int_\Omega \int |A|^p \, \mathrm{d}\nu_x(A)\, \mathrm{d}x < \infty$$

or, if $p = \infty$, there exists a compact set $K \subset \mathbb{R}^N$ such that

$$\operatorname{supp} \nu_x \subset K \quad \textit{for a.e. } x \in \Omega.$$

(III) For all Carathéodory integrands $f\colon \Omega \times \mathbb{R}^N \to \mathbb{R}$ such that the family $(f(x,V_j(x)))_j$ is uniformly L^1-bounded and equiintegrable, it holds that

$$f(x,V_j(x)) \rightharpoonup \left(x \mapsto \int f(x,A)\, \mathrm{d}\nu_x(A) \right) \quad \textit{in } \mathrm{L}^1(\Omega). \tag{3.10}$$

For parametrized measures $\nu = (\nu_x)_{x\in\Omega}$ that satisfy (I) and (II) above, we write $\nu = (\nu_x) \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$, the latter being called the **set of p-Young measures**. In symbols we express the generation of a Young measure ν by as sequence (V_j) as

$$V_j \xrightarrow{\mathbf{Y}} \nu.$$

Furthermore, the convergence (3.10) can equivalently be expressed as (absorb a test function for weak convergence into f)

$$\int_\Omega f(x,V_j(x))\, \mathrm{d}x \to \int_\Omega \int f(x,A)\, \mathrm{d}\nu_x(A)\, \mathrm{d}x =: \langle\!\langle f, \nu \rangle\!\rangle.$$

In this context, recall from the Dunford–Pettis Theorem A.13 that $(f(x,V_j(x)))_j$ is equiintegrable if and only it is weakly relatively compact in $\mathrm{L}^1(\Omega)$.

Remark 3.12. Some assumptions in the fundamental theorem can be weakened: For the existence of a Young measure, we only need to require $\lim_{K\uparrow\infty} \sup_j |\{|V_j| \geq K\}| = 0$, which for example follows from $\sup_j \int_\Omega |V_j|^p \, \mathrm{d}x < \infty$ for some $p > 0$. Of course, in this case (II) needs to be suitably adapted.

Proof. We associate with each V_j an **elementary Young measure** $(\delta_{V_j}) \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ as follows:

$$(\delta_{V_j})_x := \delta_{V_j(x)} \qquad \text{for a.e. } x \in \Omega. \tag{3.11}$$

Below we will show the following **Young measure compactness principle**, which is also useful in other contexts. We only formulate the principle for $1 \le p < \infty$ for reasons of convenience, but it holds (suitably adapted) also for $p = \infty$.

> *Let* $(\nu^{(j)})_j \subset \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ *be a sequence of p-Young measures such that*
>
> $$\sup_j \langle\!\langle |\cdot|^p, \nu^{(j)} \rangle\!\rangle = \sup_j \int_\Omega \int |A|^p \,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x < \infty. \tag{3.12}$$
>
> *Then, there exists a subsequence of the* (ν_j) *(not relabeled) and* $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ *such that*
>
> $$\lim_{j\to\infty} \langle\!\langle f, \nu^{(j)} \rangle\!\rangle = \langle\!\langle f, \nu \rangle\!\rangle \tag{3.13}$$
>
> *for all Carathéodory* $f\colon \Omega \times \mathbb{R}^N \to \mathbb{R}$ *for which the sequence of functions* $x \mapsto \langle f(x,\cdot), \nu_x^{(j)} \rangle$ *is uniformly* L^1*-bounded and the equiintegrability condition*
>
> $$\sup_j \langle\!\langle f(x,A)\mathbb{1}_{\{|f(x,A)| \ge m\}}, \nu^{(j)} \rangle\!\rangle \;\to\; 0 \qquad \textit{as} \qquad m \to \infty. \tag{3.14}$$
>
> *holds. Moreover,*
>
> $$\langle\!\langle |\cdot|^p, \nu \rangle\!\rangle \le \liminf_{j\to\infty} \langle\!\langle |\cdot|^p, \nu^{(j)} \rangle\!\rangle.$$

In the theorem, our assumption of L^p-boundedness of (V_j) corresponds to (3.12), so (I)–(III) follow immediately from the compactness principle applied to our sequence $(\delta_{V_j})_j$ of elementary Young measures from (3.11) above once we realize that for the sequence of Young measures $(\delta_{V_j})_j$ the condition (3.14) corresponds precisely to the equiintegrability of $(f(x,V_j(x)))_j$.

It remains to show the compactness principle. To this end define the measures

$$\mu^{(j)} := \mathscr{L}_x^d \llcorner \Omega \otimes \nu_x^{(j)},$$

which is just a shorthand notation for the Radon measures $\mu^{(j)} \in \mathbf{M}(\Omega \times \mathbb{R}^N) \cong \mathrm{C}_0(\Omega \times \mathbb{R}^N)^*$ defined through their action

$$\langle f, \mu^{(j)} \rangle = \int_\Omega \int f(x,A)\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x \qquad \text{for all } f \in \mathrm{C}_0(\Omega \times \mathbb{R}^N).$$

So, for the $\nu^{(j)} = \delta_{V_j}$ from (3.11), we recover the familiar form

$$\langle f, \mu^{(j)} \rangle = \int_\Omega f(x, V_j(x))\,\mathrm{d}x \qquad \text{for all } f \in \mathrm{C}_0(\Omega \times \mathbb{R}^N),$$

but in the compactness principle we might be in a more general situation.

Clearly, every so-defined $\mu^{(j)}$ is a *positive* measure and

$$\big|\langle f, \mu^{(j)} \rangle\big| \le |\Omega| \cdot \|f\|_\infty,$$

whereby the $(\mu^{(j)})$ constitute a uniformly bounded sequence in the dual space $\mathrm{C}_0(\Omega \times \mathbb{R}^N)^*$. Thus, by the Sequential Banach–Alaoglu Theorem A.12, we can select a (non-relabeled) subsequence such that with a $\mu \in \mathrm{C}_0(\Omega \times \mathbb{R}^N)^*$ it holds that

$$\langle f, \mu^{(j)} \rangle \to \langle f, \mu \rangle \qquad \text{for all } f \in \mathrm{C}_0(\Omega \times \mathbb{R}^N). \tag{3.15}$$

Next we will show that μ can be written as $\mu = \mathscr{L}_x^d \llcorner \Omega \otimes \nu_x$ for a weakly* measurable parametrized family $\nu = (\nu_x)_{x \in \Omega} \subset \mathbf{M}^1(\mathbb{R}^N)$ of probability measures.

Theorem 3.13 (Disintegration). *Let $\mu \in \mathbf{M}(\Omega \times \mathbb{R}^N)$ be a finite Radon measure. Then, there exists a weakly* measurable family $(\nu_x)_{x \in \Omega} \subset \mathbf{M}^1(\mathbb{R}^N)$ of probability measures such that with $\kappa(A) := \mu(A \times \mathbb{R}^N)$,*

$$\mu = \kappa(\mathrm{d}x) \otimes \nu_x,$$

that is,

$$\langle f, \mu \rangle = \int_\Omega \int f(x,A) \,\mathrm{d}\nu_x(A) \,\mathrm{d}\kappa(x) \qquad \textit{for all } f \in \mathrm{C}_0(\Omega \times \mathbb{R}^N).$$

A proof of this measure-theoretic result can be found in Theorem 2.28 of [2].

In our situation, we have for $f(x,v) := \varphi(x)$ with arbitrary $\varphi \in \mathrm{C}_0(\Omega)$ that

$$\langle \varphi, \mu \rangle = \lim_{j \to \infty} \langle \varphi, \mu^{(j)} \rangle = \int_\Omega \varphi(x) \,\mathrm{d}x,$$

and so $\kappa = \mathscr{L}^d \llcorner \Omega$. Hence, the Disintegration Theorem immediately yields the weak* measurability of (ν_x) and thus $\lim_{j\to\infty} \langle\langle f, \nu^{(j)} \rangle\rangle = \langle\langle f, \nu \rangle\rangle$ for $f \in \mathrm{C}_0(\Omega \times \mathbb{R}^N)$, which is (3.13).

Next, we show (3.13) for the case of f merely Carathéodory and bounded and such that there exists a compact set $K \subset \mathbb{R}^N$ with $\operatorname{supp} f \subset \Omega \times K$. We invoke the Scorza Dragoni Theorem 2.5 to get an increasing sequence of compact sets $S_k \subset\subset \Omega$ ($k \in \mathbb{N}$) with $|\Omega \setminus S_k| \downarrow 0$ such that $f|_{S_k \times \mathbb{R}^N}$ is continuous. Let $f_k \in \mathrm{C}(\Omega \times \mathbb{R}^N)$ be an extension of $f|_{S_k \times \mathbb{R}^N}$ to all of $\Omega \times \mathbb{R}^N$ with the property that the f_k are uniformly bounded. Indeed, since K is bounded and $|f_k(x,A)| \le M(1+|A|^p)$, f is bounded and hence we may require uniform boundedness of the f_k.

Now, since $f_k \in \mathrm{C}_0(\Omega \times \mathbb{R}^N)$, (3.15) implies

$$\langle f_k(x,\,\cdot\,), \nu_x^{(j)} \rangle \rightharpoonup \langle f_k(x,\,\cdot\,), \nu_x \rangle \quad \text{in } \mathrm{L}^1 \qquad \text{as } j \to \infty.$$

Thus, we also get

$$\langle f(x,\,\cdot\,), \nu_x^{(j)} \rangle \rightharpoonup \langle f(x,\,\cdot\,), \nu_x \rangle \quad \text{in } \mathrm{L}^1(S_k) \qquad \text{as } j \to \infty.$$

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On the other hand,

$$\int_\Omega \bigl|\langle f(x,\bullet),\nu_x^{(j)}\rangle - \mathbb{1}_{S_k}\langle f(x,\bullet),\nu_x^{(j)}\rangle\bigr| \,\mathrm{d}x \le \int_{\Omega\setminus S_k} \bigl|\langle f(x,\bullet),\nu_x^{(j)}\rangle\bigr| \,\mathrm{d}x$$

and this converges to zero as $k\to\infty$, uniformly in j, by the uniform boundedness of f_k; the same estimate holds with ν in place of $\nu^{(j)}$. Therefore, we may conclude

$$\langle f(x,\bullet),\nu_x^{(j)}\rangle \rightharpoonup \langle f(x,\bullet),\nu_x\rangle \quad \text{in } \mathrm{L}^1(\Omega) \qquad \text{as } j\to\infty,$$

which directly implies (3.13) (for f with $\operatorname{supp} f \subset \Omega\times K$).

Finally, to remove the restriction of compact support in A, we remark that it suffices to show the assertion under the additional constraint that $f\ge 0$ by considering the positive and negative part separately. Then, in case f positive, choose for any $m\in\mathbb{N}$ a function $\rho_m\in \mathrm{C}_c^\infty(\mathbb{R}^N;[0,1])$ with $\rho_m\equiv 1$ on $B(0,m)$ and $\operatorname{supp}\rho_m\subset B(0,2m)$. Set

$$f^m(x,A) := \rho_m(|A|^{p/2})\rho_m(f(x,A))f(x,A).$$

Then,

$$\begin{aligned}
E_{j,m} &:= \int_\Omega \langle f(x,\bullet),\nu_x^{(j)}\rangle - \langle f^m(x,\bullet),\nu_x^{(j)}\rangle \,\mathrm{d}x \\
&\le \int_\Omega \int \bigl[1-\rho_m(|A|^{p/2})\rho_m(f(x,A))\bigr] f(x,A) \,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x \\
&\le \iint_{\{|A|\,\ge\, m^{2/p} \text{ or } f(x,A)\,\ge\, m\}} f(x,A)\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x \\
&\le M\int_\Omega\int_{\{|A|\ge m^{2/p}\}} m\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x + \int_{\{f(x,A)\ge m\}} f(x,A)\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x \\
&\le \frac{M}{m}\sup_j \langle\!\langle |A|^p,\nu^{(j)}\rangle\!\rangle + \sup_j \langle\!\langle f(x,A)\mathbb{1}_{\{|f(x,A)|\ge m\}},\nu^{(j)}\rangle\!\rangle.
\end{aligned}$$

Both of these terms converge to zero as $m\to\infty$ by the assumption from the compactness principle, in particular (3.12) and the equiintegrability condition (3.14).

As the Young measure convergence holds for f^m by the previous proof step, we get

$$\begin{aligned}
\limsup_{j\to\infty}\bigl(\langle\!\langle f,\nu^{(j)}\rangle\!\rangle - \langle\!\langle f^m,\nu\rangle\!\rangle\bigr) &= \limsup_{j\to\infty}\bigl(\langle\!\langle f^m,\nu^{(j)}\rangle\!\rangle - \langle\!\langle f^m,\nu\rangle\!\rangle\bigr) + E_{j,m}\bigr) \\
&\le \sup_j E_{j,m}.
\end{aligned}$$

Then, since $f^m\uparrow f$ and $\sup_j E_{j,m}$ vanishes as $m\to\infty$ and $f\ge 0$, we get that indeed

$$\langle\!\langle f,\nu^{(j)}\rangle\!\rangle \to \langle\!\langle f,\nu\rangle\!\rangle \qquad \text{as} \qquad j\to\infty.$$

The last assertion to be shown in the compactness principle is, for $1\le p<\infty$,

$$\langle\!\langle |\bullet|^p,\nu\rangle\!\rangle \le \liminf_{j\to\infty}\langle\!\langle |\bullet|^p,\nu^{(j)}\rangle\!\rangle.$$

For $K \in \mathbb{N}$ define $|A|_K := \min\{|A|, K\} \le |A|$. Then, by the Young measure representation for equiintegrable integrands,

$$\liminf_{j\to\infty} \langle\!\langle |\cdot|^p, \nu^{(j)} \rangle\!\rangle \ge \lim_{j\to\infty} \langle\!\langle |\cdot|_K^p, \nu^{(j)} \rangle\!\rangle = \langle\!\langle |\cdot|_K^p, \nu \rangle\!\rangle \qquad \text{for all } K \in \mathbb{N}.$$

We conclude by letting $K \uparrow \infty$ and using the monotone convergence lemma. □

Remark 3.14. The Fundamental Theorem can also be proved in a more functional analytic way as follows: Let $\mathrm{L}^\infty_{w*}(\Omega; \mathbf{M}(\mathbb{R}^N))$ be the set of essentially bounded weakly* measurable functions defined on Ω with values in the Radon measures $\mathbf{M}(\mathbb{R}^N)$. It turns out (see for example [7] or [30, 31]) that $\mathrm{L}^\infty_{w*}(\Omega; \mathbf{M}(\mathbb{R}^N))$ is the dual space to $\mathrm{L}^1(\Omega; \mathrm{C}_0(\mathbb{R}^N))$. One can show that the maps $\nu^{(j)} = (x \mapsto \nu_x^{(j)})$ form a uniformly bounded set in $\mathrm{L}^\infty_{w*}(\Omega; \mathbf{M}(\mathbb{R}^N))$ and by the Sequential Banach–Alaoglu Theorem A.12 we can again conclude the existence of a weak* limit point ν of the $\nu^{(j)}$'s. The extended representation of limits for Carathéodory f follows as before.

It is known that *every* weakly* measurable parametrized measure $(\nu_x)_{x\in\Omega} \subset \mathbf{M}^1(\mathbb{R}^N)$ with $(x \mapsto \langle |\cdot|^p, \nu_x \rangle) \in \mathrm{L}^1(\Omega)$ $(1 \le p < \infty)$ can be generated by a sequence $(u_j) \subset \mathrm{L}^p(\Omega; \mathbb{R}^N)$ (one starts first by approximating Dirac masses and then uses the approximation of a general measure by linear combinations of Dirac masses, a spatial gluing argument completes the reasoning). Thus, in the definition of $\mathbf{Y}^p(\Omega; \mathbb{R}^N)$ we did not need to include (III) from the Fundamental Theorem.

A Young measure $\nu = (\nu_x) \in \mathbf{Y}^p(\Omega; \mathbb{R}^N)$ is called **homogeneous** if $\nu_x = \nu_y$ for a.e. $x, y \in \Omega$; by a mild abuse of notation we then simply write ν in place of ν_x.

Before we come to further properties of Young measures, let us consider a few examples:

Example 3.15. In $\Omega := (0,1)$ define $u := \mathbb{1}_{(0,1/2)} - \mathbb{1}_{(1/2,1)}$ and extend this function periodically to all of $\mathbb{R}$. Then, the functions $u_j(x) := u(jx)$ for $j \in \mathbb{N}$ (see Figure 3.3) generate the homogeneous Young measure $\nu \in \mathbf{Y}^\infty((0,1); \mathbb{R})$ with

$$\nu_x = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_{+1} \qquad \text{for a.e. } x \in (0,1).$$

Indeed, for $\varphi \otimes h \in \mathrm{C}_0((0,1) \times \mathbb{R})$ we have that φ is uniformly continuous, say $|\varphi(x) - \varphi(y)| \le \omega(|x-y|)$ with a modulus of continuity $\omega\colon [0,\infty) \to [0,\infty)$ (that is, ω is continuous, increasing, and $\omega(0) = 0$). Then,

$$\begin{aligned} \lim_{j\to\infty} \int_0^1 \varphi(x) h(u_j(x)) \,\mathrm{d}x &= \lim_{j\to\infty} \sum_{k=0}^{j-1} \left(\int_{k/j}^{(k+1)/j} \varphi(k/j) h(u_j(x)) \,\mathrm{d}x + \frac{\omega(1/j)\|h\|_\infty}{j} \right) \\ &= \lim_{j\to\infty} \sum_{k=0}^{j-1} \frac{1}{j} \varphi(k/j) \int_0^1 h(u(y)) \,\mathrm{d}y \\ &= \int_0^1 \varphi(x) \,\mathrm{d}x \cdot \left(\frac{1}{2} h(-1) + \frac{1}{2} h(+1) \right) \end{aligned}$$

since the Riemann sums converge to the integral of φ. The last line implies the assertion.

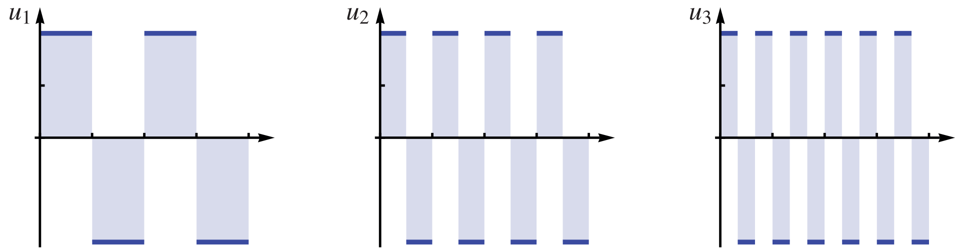

Figure 3.3: An oscillating sequence.

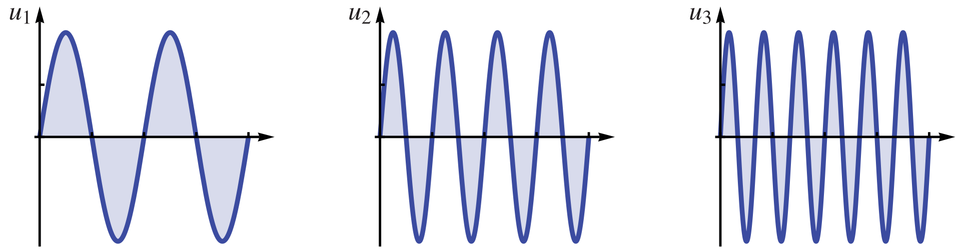

Figure 3.4: Another oscillating sequence.

Example 3.16. Take $\Omega := (0,1)$ again and let $u_j(x) = \sin(2\pi jx)$ for $j \in \mathbb{N}$ (see Figure 3.4). The sequence (u_j) generates the homogeneous Young measure $\nu \in \mathbf{Y}^\infty((0,1);\mathbb{R})$ with

$$\nu_x = \frac{1}{\pi\sqrt{1-y^2}} \mathscr{L}^1_y \llcorner (-1,1) \qquad \text{for a.e. } x \in (0,1),$$

as should be plausible from the oscillating sequence (there is more mass close to the horizontal axis than farther away).

Example 3.17. Take a bounded Lipschitz domain $\Omega \subset \mathbb{R}^2$, let $A, B \in \mathbb{R}^{2\times 2}$ with $B - A = a \otimes n$ (this is equivalent to $\operatorname{rank}(A - B) \leq 1$), where $a, n \in \mathbb{R}^2$, and for $\theta \in (0,1)$ define

$$u(x) := Ax + \left(\int_0^{x\cdot n} \chi(t) \,\mathrm{d}t \right) a, \qquad x \in \mathbb{R}^2,$$

where $\chi := \mathbb{1}_{\bigcup_{z\in\mathbb{Z}}[z,z+1-\theta)}$. If we let $u_j(x) := u(jx)/j$, $x \in \Omega$, then (∇u_j) generates the homogeneous Young measure $\nu \in \mathbf{Y}^\infty(\Omega;\mathbb{R}^2)$ with

$$\nu_x = \theta\delta_A + (1-\theta)\delta_B \qquad \text{for a.e. } x \in \Omega.$$

This example can also be extended to include multiple scales, cf. [73].

The Fundamental Theorem has many important consequences, two of which are the following:

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Lemma 3.18. *Let $(V_j) \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$, $1 < p \le \infty$, be a bounded sequence generating the Young measure $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$. Then,*

$$V_j \rightharpoonup V \quad \text{in } \mathrm{L}^p(\Omega;\mathbb{R}^N) \qquad \text{for} \qquad V(x) := [\nu_x] = \langle \mathrm{id}, \nu_x \rangle.$$

Proof. Bounded sequences in L^p with $p > 1$ are weakly relatively compact and it suffices to identify the limit of any weakly convergent subsequence. From the Dunford–Pettis Theorem A.13 it follows that any such sequence is in fact equiintegrable (in L^1). Now simply apply assertion (III) of the Fundamental Theorem for the integrand $f(x,A) := A$ (or, more pedantically, $f^i_j(A) := A^i_j$ for $i = 1,\dots,d$, $m = 1,\dots,m$). □

The preceding lemma does not hold for $p = 1$. One example is $V_j := j\mathbb{1}_{(0,1/j)}$, which *concentrates*.

Proposition 3.19. *Let $(V_j) \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$, $1 \le p \le \infty$, be a bounded sequence generating the Young measure $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$, and let $f\colon \Omega \times \mathbb{R}^N \to [0,\infty)$ be any Carathéodory integrand (not necessarily satisfying the equiintegrability property in (III) of the Fundamental Theorem). Then, it holds that*

$$\liminf_{j\to\infty} \int_\Omega f(x,V_j(x))\,\mathrm{d}x \ge \langle\!\langle f,\nu \rangle\!\rangle.$$

Proof. For $M \in \mathbb{N}$ define $f_M(x,A) := \min(f(x,A),M)$. Then, (III) from the Fundamental Theorem is applicable with f_M and we get

$$\int_\Omega f_M(x,V_j(x))\,\mathrm{d}x \to \langle\!\langle f_M,\nu \rangle\!\rangle = \int_\Omega \int f_M(x,A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x.$$

Since $f \ge f_M$, we have

$$\liminf_{j\to\infty} \int_\Omega f(x,V_j(x))\,\mathrm{d}x \;\ge\; \langle\!\langle f_M,\nu \rangle\!\rangle.$$

We conclude by letting $M \uparrow \infty$ and using the monotone convergence lemma. □

Another important feature of Young measures is that they allow us to read off whether or not our sequence converges in measure:

Proposition 3.20. *Let $(\nu_x) \subset \mathbf{M}^1(\mathbb{R}^N)$ be the Young measure generated by a bounded sequence $(V_j) \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$, $1 \le p \le \infty$. Let furthermore $K \subset \mathbb{R}^N$ be compact. Then,*

$$\operatorname{dist}(V_j(x),K) \to 0 \text{ in measure} \qquad \text{if and only if} \qquad \operatorname{supp}\nu_x \subset K \text{ for a.e. } x \in \Omega.$$

Moreover,

$$V_j \to V \text{ in measure} \qquad \text{if and only if} \qquad \nu_x = \delta_{V(x)} \text{ for a.e. } x \in \Omega.$$

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Proof. We only show the second assertion, the proof of the first one is similar. Take the bounded, positive Carathéodory integrand

$$f(x,A) := \frac{|A - V(x)|}{1 + |A - V(x)|}, \qquad x \in \Omega, A \in \mathbb{R}^N.$$

Then, for all $\delta \in (0,1)$, the Markov inequality and the Fundamental Theorem on Young measures yield

$$\begin{aligned}\limsup_{j\to\infty} \bigl|\bigl\{ x \in \Omega \,:\, f(x,V_j(x)) \geq \delta \bigr\}\bigr| &\leq \lim_{j\to\infty} \frac{1}{\delta} \int_\Omega f(x,V_j(x)) \,\mathrm{d}x \\ &= \frac{1}{\delta} \int_\Omega \bigl\langle f(x,\cdot), \nu_x \bigr\rangle \,\mathrm{d}x.\end{aligned}$$

On the other hand,

$$\begin{aligned}\int_\Omega \bigl\langle f(x,\cdot), \nu_x \bigr\rangle \,\mathrm{d}x &= \lim_{j\to\infty} \int_\Omega f\bigl(x,V_j(x)\bigr) \,\mathrm{d}x \\ &\leq \delta|\Omega| + \limsup_{j\to\infty} \bigl|\bigl\{ x \in \Omega \,:\, f(x,V_j(x)) \geq \delta \bigr\}\bigr|.\end{aligned}$$

The preceding estimates show that $f(x,V_j(x))$ converges to zero in measure if and only if $\langle f(x,\cdot), \nu_x \rangle = 0$ for $\mathscr{L}^d$-almost every $x \in \Omega$, which is the case if and only if $\nu_x = \delta_{V(x)}$ almost everywhere. We conclude by observing that $f(x,V_j(x))$ converges to zero in measure if and only if V_j converges to V in measure. □

3.4 Gradient Young measures

The most important subclass of Young measures for our purposes is that of Young measures that can be generated by a sequence of *gradients*: Let $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^{m\times d})$ (use $\mathbb{R}^N = \mathbb{R}^{m\times d}$ in the theory of the last section). We say that ν is a *p*-**gradient Young measure**, in symbols $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, if there exists a bounded sequence $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ such that $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$, i.e. the sequence (∇u_j) generates ν. Note that it is *not* necessary that every sequence that generates ν is a sequence of gradients (which, in fact, is *never* the case). We have already considered examples of gradient Young measures in the previous section, see in particular Example 3.17.

Is it the case that every Young measure is a gradient Young measure? The answer is no, as can be easily seen: Let $V \in (\mathrm{L}^p \cap \mathrm{C}^\infty)(\Omega;\mathbb{R}^{m\times d})$ with $\operatorname{curl} V(x) \neq 0$ for some $x \in \Omega$. Then, $\nu_x := \delta_{V(x)}$ cannot be a gradient Young measure since for such measures it must hold that $[\nu]$ is a gradient itself: if there existed a sequence $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$, then $\nabla u_j \rightharpoonup V$ by Lemma 3.18, and it would follow that $\operatorname{curl} V = 0$. Consequently, for every $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ there must exist $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $[\nu] = \nabla u$. In this case we call any such u an **underlying deformation** of ν.

We first show the following technical, but immensely useful, result about gradient Young measures. Recall that we assume Ω to be open, bounded, and to have a Lipschitz boundary (to make the trace well-defined).

Lemma 3.21. *Let $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, $1 < p < \infty$, and let $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ be an underlying deformation of ν, i.e. $[\nu] = \nabla u$. Then, there exists a sequence $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ such that*

$$u_j|_{\partial\Omega} = u|_{\partial\Omega} \qquad \text{and} \qquad \nabla u_j \xrightarrow{\mathbf{Y}} \nu.$$

Furthermore, in addition we can ensure that (∇u_j) is p-equiintegrable.

Proof. Step 1. Since Ω is assumed to have a Lipschitz boundary, we can extend a generating sequence (∇v_j) for ν to all of $\mathbb{R}^d$, so we assume $(v_j) \subset \mathrm{W}^{1,p}(\mathbb{R}^d;\mathbb{R}^m)$ with $\sup_j \|v_j\|_{\mathrm{W}^{1,p}} \leq C < \infty$.

In the following, we need the **maximal function** $\mathscr{M}f \colon \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ of $f \colon \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$, which is defined as

$$\mathscr{M}f(x_0) := \sup_{r>0} \fint_{B(x_0,r)} |f(x)| \,\mathrm{d}x, \qquad x_0 \in \mathbb{R}^d.$$

We quote the following two results about the maximal function whose proofs can be found in [87]:

(i) $\|\mathscr{M}f\|_{\mathrm{L}^p} \leq C\|\mathscr{M}f\|_{\mathrm{L}^p}$, where $C = C(d,p)$ is a constant.

(ii) For every $M > 0$ and $f \in \mathrm{W}^{1,p}(\mathbb{R}^d;\mathbb{R}^m)$, the maximal function $\mathscr{M}f$ is Lipschitz-continuous on the set $\{\mathscr{M}(|f| + |\nabla f|) < M\}$ and its Lipschitz-constant is bounded by CM, where $C = C(d,m,p)$ is a dimensional constant.

With this tool at hand, consider the sequence

$$V_j := \mathscr{M}(|v_j| + |\nabla v_j|), \qquad j \in \mathbb{N}.$$

By (i) above, (V_j) is uniformly bounded in $\mathrm{L}^p(\Omega)$ and we may select a (non-renumbered) subsequence such that V_j generates a Young measure $\mu \in \mathbf{Y}^p(\Omega)$.

Next, define for $h \in \mathbb{N}$ the (nonlinear) *truncation operator T_h*,

$$T_h v := \begin{cases} v & \text{if } |v| \leq h, \\ h\frac{v}{|v|} & \text{if } |v| > h, \end{cases} \qquad v \in \mathbb{R},$$

and let $(\varphi_l)_l \subset \mathrm{C}^\infty(\Omega)$ be a countable family that is dense in $\mathrm{C}(\Omega)$.

Then, for fixed $h \in \mathbb{N}$, the sequence $(T_h V_j)_j$ is uniformly bounded in $\mathrm{L}^\infty(\Omega)$ and so, by Young measure representation,

$$\lim_{h\to\infty}\lim_{j\to\infty} \int_\Omega \varphi_l(x)|T_h V_j(x)|^p \,\mathrm{d}x = \lim_{h\to\infty} \int_\Omega \varphi_l(x) \int |T_h v|^p \,\mathrm{d}\mu_x(v) \,\mathrm{d}x = \int_\Omega \varphi_l(x) \big\langle |\cdot|^p, \mu_x \big\rangle \,\mathrm{d}x,$$

where in the last step we used the monotone convergence lemma. Thus, for some diagonal sequence $W_k := T_k V_{j(k)}$, we have that

$$\lim_{k\to\infty} \int_\Omega \varphi_l(x)\, |W_k(x)|^p \,\mathrm{d}x = \int_\Omega \varphi_l(x) \,\big\langle |\cdot|^p, \mu_x \big\rangle \,\mathrm{d}x \qquad \text{for all } l \in \mathbb{N}.$$

Thus, $|W_k|^p \rightharpoonup \langle|\bullet|^p, \mu_x\rangle$ in L^1 and $\{|W_k|^p\}_k$ is an equiintegrable family by the Dunford–Pettis Theorem. We can see also the equiintegrability in a more elementary way as follows: For a Borel subset $A \subset \Omega$ take $\varphi \in C^\infty(\Omega)$ with $\mathbb{1}_A \le \varphi$ and estimate

$$\limsup_{k\to\infty} \int_A |W_k(x)|^p \,dx \le \lim_{k\to\infty} \int_\Omega \varphi(x)\,|W_k(x)|^p \,dx = \int_\Omega \varphi(x)\big\langle|\bullet|^p, \mu_x\big\rangle \,dx$$

by the L^1-convergence of $|W_k|^p$ to $x \mapsto \langle|\bullet|^p, \mu_x\rangle$. The last integral tends to $\int_A \langle|\bullet|^p, \mu_x\rangle \,dx$ as $\varphi \downarrow \mathbb{1}_A$, which vanishes if $|A| \to 0$. Thus the family $\{W_k\}_k$ is p-equiintegrable.

By (ii) above, $v_k = v_{j(k)}$ is Ck-Lipschitz on the set $E_k := \{W_k \le k\}$ and by a well-known result for Lipschitz functions due to Kirszbraun, we may extend v_k to a function $w_k \colon \mathbb{R}^d \to \mathbb{R}^m$ that is *globally* Lipschitz continuous with the same Lipschitz constant Ck and $w_k = v_k$ in E_k. Thus,

$$|\nabla w_k| \le CW_k \quad \text{in } \Omega.$$

Thus, $\{|\nabla w_k|\}_k$ inherits the p-equiintegrable from $\{W_k\}_k$.

Moreover, by the Markov inequality,

$$|\Omega \setminus E_k| \le \frac{\|W_k\|_{L^p}^p}{k^p} \to 0 \qquad \text{as } k \to \infty.$$

Therefore, for all $\varphi \in C_0(\Omega)$ and $h \in C_0(\mathbb{R}^m)$,

$$\int_\Omega |\varphi h(\nabla w_k) - \varphi h(\nabla v_k)| \,dx \le \|\varphi h\|_\infty |\Omega \setminus E_k| \ \to\ 0.$$

Thus, since all such φ, h determine Young measure convergence (see the proof of the Fundamental Theorem 3.11), we have shown that (∇w_k) generates the same Young measure ν as (∇v_j) and additionally the family $\{\nabla w_k\}_k$ is p-equiintegrable.

Step 2. Since $W^{1,p}(\Omega;\mathbb{R}^m)$ embeds compactly into the space $L^p(\Omega;\mathbb{R}^m)$ by the Rellich-Kondrachov Theorem A.18, we have $w_k \to u$ in L^p. Let moreover $(\rho_j) \subset C_c^\infty(\Omega;[0,1])$ be a sequence of cut-off functions such that for $G_j := \big\{x \in \Omega \,:\, \rho_j(x) = 1\big\}$ it holds that $|\Omega \setminus G_j| \to 0$ as $j \to \infty$. For

$$u_{j,k} := \rho_j w_k + (1-\rho_j)u \quad \in W_u^{1,p}(\Omega;\mathbb{R}^m)$$

we observe $\nabla u_{j,k} = \rho_j \nabla w_k + (1-\rho_j)\nabla u + (w_k - u) \otimes \nabla \rho_j$.

We only consider the case $p < \infty$ in the following; the proof for the case $p = \infty$ is in fact easier. For every Carathéodory function $f \colon \Omega \times \mathbb{R}^{m\times d} \to \mathbb{R}$ with $|f(x,A)| \le M(1+|A|^p)$ we have

$$\begin{aligned}
&\limsup_{k\to\infty} \int_\Omega \big|f(x,\nabla w_k(x)) - f(x,\nabla u_{j,k}(x))\big| \,dx \\
&\qquad \le \limsup_{k\to\infty} CM \int_{\Omega\setminus G_j} 2 + 2|\nabla w_k|^p + |\nabla u|^p + |w_k - u|^p \cdot |\nabla \rho_j|^p \,dx \\
&\qquad \le \limsup_{k\to\infty} CM \int_{\Omega\setminus G_j} 2 + 2|\nabla w_k|^p + |\nabla u|^p \,dx,
\end{aligned}$$

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where $C > 0$ is a constant. However, by assumption, the last integrand is equiintegrable and hence

$$\lim_{j\to\infty} \limsup_{k\to\infty} \int_\Omega \bigl| f(x, \nabla w_k(x)) - f(x, \nabla u_{j,k}(x)) \bigr| \,\mathrm{d}x = 0.$$

We can now select a diagonal sequence $u_j = u_{j,k(j)}$ such that (∇u_j) generates ν and satisfies all requirements from the statement of the lemma. Note that the equiintegrability is not affected by the cut-off procedure. □

The following result is now easy to prove:

Lemma 3.22 (Jensen-type inequality). *Let $\nu \in \mathbf{GY}^p(\Omega; \mathbb{R}^{m\times d})$ be a homogeneous Young measure, i.e. $\nu_x = \nu \in \mathbf{M}^1(\mathbb{R}^{m\times d})$ for almost every $x \in \Omega$, and let $B := [\nu] \in \mathbb{R}^{m\times d}$ be its barycenter (which is a constant). Then, for all quasiconvex functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with $|h(A)| \le M(1 + |A|^p)$ ($M > 0$, $1 < p < \infty$) it holds that*

$$h(B) \le \int h \,\mathrm{d}\nu. \tag{3.16}$$

Notice that the conclusion of this lemma is trivial (and also holds for general Young measures, not just the gradient ones) if h is *convex*. Thus, also (3.16) expresses the convexity in the notion of quasiconvexity.

Proof. Let $(u_j) \subset \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m)$ with $\nabla u_j \overset{\mathbf{Y}}{\to} \nu$, $u_j|_{\partial\Omega}(x) = Bx$ for $x \in \partial\Omega$ (in the sense of trace) and (∇u_j) p-equiintegrable, the latter two conditions being realizable by Lemma 3.21. Then, from the definition of quasiconvexity, we get

$$h(B) \le \fint_\Omega h(\nabla u_j) \,\mathrm{d}x.$$

Passing to the Young measure limit as $j \to \infty$ on the right-hand side, for which we note that $h(\nabla u_j)$ is equiintegrable by the growth assumption on h, we arrive at

$$h(B) \le \fint_\Omega \int h \,\mathrm{d}\nu \,\mathrm{d}x = \int h \,\mathrm{d}\nu,$$

which already is the sought inequality. □

The last result will be of crucial importance in proving weak lower semicontinuity in the next section. It is remarkable that the converse also holds, i.e. the validity of (3.16) for *all* quasiconvex h with p-growth precisely characterizes the class of (homogeneous) gradient Young measures in the class of all (homogeneous) Young measures; this is the content of the Kinderlehrer–Pedregal Theorem 5.9, which we will meet in Chapter 5.

3.5 Gradient Young measures and rigidity

To round off the introduction to Young measures let us return to the question if every Young measure is a gradient Young measure, but now with the additional requirement that the Young measure in question is *homogeneous*. Then, the barycenter is constant and hence always the gradient of an affine function, so the simple reasoning from the beginning of the section no longer applies. Still, there are homogeneous Young measures that are not gradients, but proving that no generating sequence of gradients can be found, is not so trivial. One possible way would be to show that there is a quasiconvex function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ such that the Jensen-type inequality of Lemma 3.22 fails; this is, however, not easy.

Let us demonstrate this assertion instead through the so-called **(approximate) two-gradient problem**: Let $A, B \in \mathbb{R}^{m\times d}$, $\theta \in (0,1)$ and consider the homogeneous Young measure

$$\nu := \theta\delta_A + (1-\theta)\delta_B \quad \in \mathbf{Y}^\infty(B(0,1);\mathbb{R}^{m\times d}). \tag{3.17}$$

We know from Example 3.17 that for $\operatorname{rank}(A-B) \le 1$, ν is a gradient Young measure. In any case, $\operatorname{supp}\nu = \{A,B\}$ and by Proposition 3.20 it follows that $\operatorname{dist}(V_j,\{A,B\}) \to 0$ in measure for any generating sequence $V_j \xrightarrow{\mathbf{Y}} \nu$.

The following is the first instance of so-called *rigidity* results, which play a prominent role in the field:

Theorem 3.23 (Ball–James). *Let $\Omega \subset \mathbb{R}^d$ be open, bounded, and* convex. *Also, let $A, B \in \mathbb{R}^{m\times d}$.*

(I) Let $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ satisfy the ***exact two-gradient inclusion***

$$\nabla u \in \{A,B\} \quad \text{a.e. in } \Omega.$$

(a) If $\operatorname{rank}(A-B) \ge 2$, then $\nabla u = A$ a.e. or $\nabla u = B$ a.e.

(b) If $B - A = a \otimes n$ ($a \in \mathbb{R}^m$, $n \in \mathbb{R}^d \setminus \{0\}$), then there exists a Lipschitz function $h\colon \mathbb{R} \to \mathbb{R}$ with $h' \in \{0,1\}$ almost everywhere, and a constant $v_0 \in \mathbb{R}^m$ such that

$$u(x) = v_0 + Ax + h(x\cdot n)a.$$

(II) Let $\operatorname{rank}(A-B) \ge 2$ and assume that $u_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ is uniformly bounded and satisfies the ***approximate two-gradient inclusion***

$$\operatorname{dist}\big(\nabla u_j,\{A,B\}\big) \to 0 \quad \text{in measure.}$$

Then,

$$\nabla u_j \to A \quad \text{in measure} \qquad \text{or} \qquad \nabla u_j \to B \quad \text{in measure.}$$

Comment...

Proof. *Ad (I) (a).* Assume after a translation that $B = 0$ and thus $\operatorname{rank} A \geq 2$. Then, $\nabla u = Ag$ for a scalar function $g \colon \Omega \to \mathbb{R}$. Mollifying u (i.e. working with $u_\varepsilon := \rho_\varepsilon \star u$ instead of u and passing to the limit $\varepsilon \downarrow 0$), we may assume that in fact $g \in \mathrm{C}^1(\Omega)$.

The idea of the proof is that the curl of ∇u vanishes, expressed as follows: for all $i, j = 1, \ldots, d$ and $k = 1, \ldots, m$, it holds that

$$\partial_i (\nabla u)^k_j = \partial_{ij} u^k = \partial_{ji} u^k = \partial_j (\nabla u)^k_i$$

where M^k_j denotes the element in the k'th row and j'th column of the matrix M. For our special $\nabla u = Ag$, this reads as

$$A^k_j \partial_i g = A^k_i \partial_j g. \tag{3.18}$$

Under the assumptions of (I) (a), we claim that $\nabla g \equiv 0$. If otherwise $\xi(x) := \nabla g(x) \neq 0$ for some $x \in \Omega$, then with $a_k(x) := A^k_j / \xi_j(x)$ ($k = 1, \ldots, m$) for any j such that $\xi_j(x) \neq 0$, where $a_k(x)$ is well-defined by the relation (3.18), we have

$$A^k_j = a_k(x) \xi_j(x), \qquad \text{i.e.} \qquad A = a(x) \otimes \xi(x).$$

This, however, is impossible if $\operatorname{rank} A \geq 2$. Hence, $\nabla g \equiv 0$ and u is an affine function. This property is also stable under mollification.

Ad (I) (b). As in (I) (a), we assume $\nabla u = Ag$ ($B = 0$) and $A = a \otimes n$. Pick any $b \perp n$. Then,

$$\frac{\mathrm{d}}{\mathrm{d}t} u(x + tb) \bigg|_{t=0} = \nabla u(x) b = [a n^T b] g(x) = 0.$$

This implies that u is constant in direction b. As $b \perp n$ was arbitrary and Ω is assumed convex, $u(x)$ can only depend on $x \cdot n$. This implies the claim.

Ad (II). Assume once more that $B = 0$ and that there exists a (2×2)-minor M with $M(A) \neq 0$. By assumption there are sets $D_j \subset \Omega$ with

$$\nabla u_j - A \mathbb{1}_{D_j} \to 0 \quad \text{in measure.}$$

Let us also assume that we have selected a subsequence such that

$$u_j \overset{*}{\rightharpoonup} u \quad \text{in } \mathrm{W}^{1,\infty} \qquad \text{and} \qquad \mathbb{1}_{D_j} \overset{*}{\rightharpoonup} \chi \quad \text{in } \mathrm{L}^\infty.$$

In the following we use that under an assumption of uniform L^∞-boundedness convergence in measure implies weak* convergence in L^∞. Indeed, for any $\psi \in \mathrm{L}^1(\Omega)$ and any $\varepsilon > 0$ we have

$$\begin{aligned} &\int_\Omega (\nabla u_j - A \mathbb{1}_{D_j}) \psi \,\mathrm{d}x \\ &\qquad \leq \big| \big\{ x \in \Omega \,:\, |\nabla u_j(x) - A \mathbb{1}_{D_j}(x)| > \varepsilon \big\} \big| \cdot \|\nabla u_j - A \mathbb{1}_{D_j}\|_{\mathrm{L}^\infty} \cdot \|\psi\|_{\mathrm{L}^1} + \varepsilon \|\psi\|_{\mathrm{L}^1} \\ &\qquad \to 0 + \varepsilon \|\psi\|_{\mathrm{L}^1} \end{aligned}$$

As $\varepsilon > 0$ is arbitrary, we have indeed $\nabla u_j - A\mathbb{1}_{D_j} \overset{*}{\rightharpoonup} 0$ in L^∞.

Now,

$$\operatorname*{w*-lim}_{j\to\infty} M(\nabla u_j) = \operatorname*{w*-lim}_{j\to\infty} M(A\mathbb{1}_{D_j}) = M(A)\cdot \operatorname*{w*-lim}_{j\to\infty} \mathbb{1}_{D_j} = M(A)\chi.$$

By a similar argument, $\nabla u_j \overset{*}{\rightharpoonup} \nabla u = A\chi$, and so, by the weak* continuity of minors proved in Lemma 3.10,

$$\operatorname*{w*-lim}_{j\to\infty} M(\nabla u_j) = M(\nabla u) = M(A\chi) = M(A)\chi^2.$$

Thus, $\chi = \chi^2$ and we can conclude that there exists $D \subset \Omega$ such that $\chi = \mathbb{1}_D$ and $\nabla u = A\mathbb{1}_D$. Furthermore, combining the above convergence assertions,

$$\nabla u_j \to \nabla u = A\mathbb{1}_D \quad \text{in measure.}$$

Finally, (I) (a) implies that $\nabla u = A$ or $\nabla u = 0 = B$ a.e. in Ω, finishing the proof. □

With this result at hand it is easy to see that our example (3.17) cannot be a gradient Young measure if $\operatorname{rank}(A-B) \geq 2$: If there was a sequence of gradients $\nabla u_j \overset{\mathbf{Y}}{\to} \nu$, then by the reasoning above we would have $\operatorname{dist}(\nabla u_j, \{A,B\}) \to 0$, but from (II) of the Ball–James rigidity theorem, we would get $\nabla u_j \to A$ or $\nabla u_j \to B$ in measure, either one of which yields a contradiction by Proposition 3.20.

3.6 Lower semicontinuity

After our detour through Young measure theory, we now return to the central subject of this chapter, namely to minimization problems of the form

$$\begin{cases} \mathscr{F}[u] := \displaystyle\int_\Omega f(x, \nabla u(x))\,\mathrm{d}x \quad \to \quad \min, \\ \text{over all } u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m) \text{ with } u|_{\partial\Omega} = g, \end{cases}$$

where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $1 < p < \infty$, the integrand $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}$ has p-growth,

$$|f(x,A)| \leq M(1+|A|^p) \qquad \text{for some } M > 0,$$

and $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$ specifies the boundary values. In Chapter 2 we solved this problem via the Direct Method of the Calculus of Variations, a coercivity result, and, crucially, the Lower Semicontinuity Theorem 2.7. In this section, we recycle the Direct Method and the coercivity result, but extend lower semicontinuity to *quasi*convex integrands; some motivation for this was given at the beginning of the chapter.

Let us first consider how we could approach the proof of lower semicontinuity (it should be clear that the proof via Mazur's Lemma that we used for the convex lower semicontinuity theorem, does not extend). Assume we have a sequence $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ such that $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$ and we want to show lower semicontinuity of our functional $\mathscr{F}$. If we assume that our (norm-bounded) sequence ∇u_j also generates the gradient Young measure $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, which is true up to a subsequence, and that the sequence of integrands $(f(x,\nabla u_j(x)))_j$ is equiintegrable, then we already have a limit:

$$\mathscr{F}[u_j] \to \int_\Omega \int f(x,A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x.$$

This is useful, because it now suffices to show the inequality

$$\int f(x,\cdot)\,\mathrm{d}\nu_x \geq f(x,\nabla u(x))\,\mathrm{d}x$$

for almost every $x \in \Omega$, to conclude lower semicontinuity. Now, this inequality is close to the Jensen-type inequality (3.16) for gradient Young measures from Lemma 3.22, and we already explained how this is related to (quasi)convexity. The only difference in comparison to (3.16) is that here we need to "localize" in $x \in \Omega$ and make ν_x a gradient Young measure in its own right. This is accomplished via the fundamental **blow-up technique**:

Lemma 3.24 (Blow-up technique). *Let $\nu = (\nu_x) \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ be a gradient Young measure. Then, for almost every $x_0 \in \Omega$, $\nu_{x_0} \in \mathbf{GY}^p(B(0,1);\mathbb{R}^{m\times d})$ is a homogeneous gradient Young measure in its own right.*

Proof. We start with a general property of Young measures: Let $\{\varphi_k\}_k$ and $\{h_l\}_l$ be countable dense subsets of $\mathrm{C}_0(\Omega)$ and $\mathrm{C}_0(\mathbb{R}^N)$, respectively. Assume furthermore that for a uniformly norm-bounded sequence $(V_j) \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$ and $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ we have

$$\lim_{j\to\infty} \int_\Omega \varphi_k(x) h_l(V_j(x))\,\mathrm{d}x = \int_\Omega \varphi_k(x) \int h_l\,\mathrm{d}\nu_x\,\mathrm{d}x \qquad \text{for all } k,l \in \mathbb{N}.$$

Then we claim that $V_j \xrightarrow{\mathbf{Y}} \nu$. This is in fact clear once we realize that the set of linear combinations of the functions $f_{k,l} := \varphi_k \otimes h_l$ (that is, $f_{k,l}(x,A) := \varphi_k(x)h_l(A)$) is dense in $\mathrm{C}_0(\Omega\times\mathbb{R}^N)$ and testing with such functions determines Young measure convergence, see the proof of the Fundamental Theorem 3.11.

Now let $x_0 \in \Omega$ be a Lebesgue point of all the functions $x \mapsto \langle h_l, \nu_x\rangle$ ($l \in \mathbb{N}$), that is,

$$\lim_{r\downarrow 0} \int_{B(0,1)} \bigl|\langle h_l, \nu_{x_0+ry}\rangle - \langle h_l, \nu_{x_0}\rangle\bigr|\,\mathrm{d}y = 0.$$

By Theorem A.7, almost every $x \in \Omega$ has this property. Then, at such a Lebesgue point x_0, set

$$v_j^{(r)}(y) := \frac{u_j(x_0+ry) - [u_j]_{B(x_0,r)}}{r}, \qquad y \in B(0,1),$$

where $[u]_{B(x_0,r)} := \fint_{B(x_0,r)} u\,\mathrm{d}x$.

For φ_k, h_l from the collections exhibited at the beginning of the proof, we get

$$\begin{aligned}\int_{B(0,1)} \varphi_k(y) h_l(\nabla v_j^{(r)}(y))\,\mathrm{d}y &= \int_{B(0,1)} \varphi_k(y) h_l(\nabla u_j(x_0+ry))\,\mathrm{d}y \\ &= \frac{1}{r^d}\int_{B(x_0,r)} \varphi_k\Big(\frac{x-x_0}{r}\Big) h_l(\nabla u_j(x))\,\mathrm{d}x\end{aligned}$$

after a change of variables. Letting $j \to \infty$ and then $r \downarrow 0$, we get

$$\begin{aligned}\lim_{r\downarrow 0}\lim_{j\to\infty}\int_{B(0,1)} \varphi_k(y) h_l(\nabla v_j^{(r)}(y))\,\mathrm{d}y &= \lim_{r\downarrow 0}\frac{1}{r^d}\int_{B(x_0,r)} \varphi_k\Big(\frac{x-x_0}{r}\Big)\big\langle h_l, \nu_x\big\rangle\,\mathrm{d}x \\ &= \lim_{r\downarrow 0}\int_{B(0,1)} \varphi_k(y)\big\langle h_l, \nu_{x_0+ry}\big\rangle\,\mathrm{d}x \\ &= \int_{B(0,1)} \varphi_k(y)\big\langle h_l, \nu_{x_0}\big\rangle\,\mathrm{d}y,\end{aligned}$$

the last convergence following from the Lebesgue point property of x_0. Moreover,

$$\int_{B(0,1)} |\nabla v_j^{(r)}|^p\,\mathrm{d}y = \int_{B(0,1)} |\nabla u_j(x_0+ry)|^p\,\mathrm{d}y = \frac{1}{r^d}\int_{B(x_0,r)} |\nabla u_j(x)|^p\,\mathrm{d}x$$

and the last integral is uniformly bounded in j (for fixed r). Denote by $\lambda \in \mathbf{M}(\overline{\Omega};\mathbb{R}^m)$ the weak* limit of the measures $|\nabla u_j|^p \mathscr{L}^d \llcorner \Omega$, which exists after taking a subsequence. If we require of x_0 additionally that

$$\limsup_{r\downarrow 0} \frac{\lambda(B(x_0,r))}{r^d} < \infty,$$

which hold at $\mathscr{L}^d$-almost every $x_0 \in \Omega$, then

$$\limsup_{r\downarrow 0}\lim_{j\to\infty}\int_{B(0,1)} |\nabla v_j^{(r)}|^p\,\mathrm{d}y < \infty.$$

Since also $[v_j^{(r)}]_{B(0,1)} = 0$, the Poincaré inequality from Theorem A.16 yields that there exists a diagonal sequence $w_j := v_{J(j)}^{R(j)}$ that is uniformly bounded in $\mathrm{W}^{1,p}(B(0,1);\mathbb{R}^m)$ and such that for all $k,l \in \mathbb{N}$,

$$\lim_{j\to\infty}\int_{B(0,1)} \varphi_k(y) h_l(\nabla w_j(y))\,\mathrm{d}y = \int_{B(0,1)} \varphi_k(y)\big\langle h_l, \nu_{x_0}\big\rangle\,\mathrm{d}y.$$

Therefore, $\nabla w_j \xrightarrow{\mathbf{Y}} \nu_{x_0}$, where we now understand ν_{x_0} as a homogeneous (gradient) Young measure on $B(0,1)$. □

The proof of weak lower semicontinuity is then straightforward:

Theorem 3.25. *Let $f\colon \Omega \times \mathbb{R}^{m\times d} \to \mathbb{R}$ be a Carathéodory integrand with $|f(x,A)| \le M(1+|A|^p)$ for some $M > 0$, $1 < p < \infty$. Assume furthermore that $f(x,\cdot)$ is quasiconvex for every fixed $x \in \Omega$. Then the corresponding functional $\mathscr{F}$ is weakly lower semicontinuous on $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$.*

Proof. Let $(\nabla u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$. Assume that (∇u_j) generates the gradient Young measure $\nu = (\nu_x) \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, for which it holds that $[\nu] = \nabla u$. This is only true up to a (non-relabeled) subsequence, but if we can establish the lower semicontinuity for every such subsequence it follows that the result also holds for the original sequence.

From Proposition 3.19 we get

$$\liminf_{j\to\infty} \int_\Omega f(x, \nabla u_j(x))\,\mathrm{d}x \geq \langle\!\langle f, \nu \rangle\!\rangle = \int_\Omega \int f(x,A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x.$$

Now, for almost every $x \in \Omega$, we can consider ν_x as a homogeneous Young measure in $\mathbf{GY}^p(B(0,1);\mathbb{R}^{m\times d})$ by the Blow-up Lemma 3.24. Thus, the Jensen-type inequality from Lemma 3.22 reads

$$\int f(x,A)\,\mathrm{d}\nu_x(A) \geq f(x, \nabla u(x)) \qquad \text{for a.e. } x \in \Omega.$$

Combining, we arrive at

$$\liminf_{j\to\infty} \mathscr{F}[u_j] \geq \mathscr{F}[u],$$

which is what we wanted to show. □

At this point is is worthwhile to reflect on the role of Young measures in the proof of the preceding result, namely that they allowed us to split the argument into two parts: First, we passed to the limit in the functional via the Young measure. Second, we established a Jensen-type inequality, which then yielded the lower semicontinuity inequality. It is remarkable that the Young measure preserves exactly the “right” amount of information to serve as an intermediate object.

We can now sum up and prove existence to our minimization problem:

Theorem 3.26. *Let $f\colon \Omega \times \mathbb{R}^{m\times d} \to \mathbb{R}$ be a Carathéodory integrand such that f satisfies the lower growth bound*

$$\mu|A|^p - \mu^{-1} \leq f(x,A) \qquad \text{for some } \mu > 0,$$

the upper growth bound

$$|f(x,A)| \leq M(1+|A|^p) \qquad \text{for some } M > 0,$$

and is quasiconvex in its second argument. Then, the associated functional $\mathscr{F}$ has a minimizer over the space $\mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$, where $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$.

Proof. This follows directly by combining the Direct Method from Theorem 2.3 with the coercivity result in Proposition 2.6 and the Lower Semicontinuity Theorem 3.25. □

The following result shows that quasiconvexity is also necessary for weak lower semicontinuity; we only state and show this for x-independent integrands, but we note that it also holds for x-dependent ones by a localization technique.

Proposition 3.27. *Let $f \colon \mathbb{R}^{m\times d} \to \mathbb{R}$ satisfy the p-growth condition.*

$$|f(x,A)| \le M(1+|A|^p) \qquad \text{for some } M>0.$$

If the associated functional

$$\mathscr{F}[u] := \int_\Omega f(\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m),$$

is weakly lower semicontinuous (with or without fixed boundary values), then f is quasiconvex.

Proof. We may assume that $B(0,1) \subset\subset \Omega$, otherwise we can translate and rescale the domain. Let $A \in \mathbb{R}^{m\times d}$ and $\psi \in \mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^m)$. We need to show

$$h(A) \le \fint_{B(0,1)} h(A+\nabla\psi(z))\,\mathrm{d}z.$$

Take for every $j \in \mathbb{N}$ a Vitali cover of $B(0,1)$, see Theorem A.8,

$$B(0,1) = Z \uplus \biguplus_{k=1}^{N(j)} B(a_k^{(j)}, r_k^{(j)}), \qquad |Z| = 0,$$

with $a_k^{(j)} \in B(0,1)$, $r_k^{(j)} \le 1/j$ $(k=1,\ldots,N(j))$, and fix a smooth function $h \colon \Omega\setminus B(0,1) \to \mathbb{R}^m$ with $h(x) = Ax$ for $x \in \partial B(0,1)$ (and $h|_{\partial\Omega}$ equal to the prescribed boundary values if there are any). Define

$$u_j(x) := Ax + r_k^{(j)}\psi\left(\frac{x-a_k^{(j)}}{r_k^{(j)}}\right) \quad \text{if } x \in B(a_k^{(j)}, r_k^{(j)}),$$

and $u_j := h$ on $\Omega\setminus B(0,1)$. Then, it is not hard to see that $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$ for

$$u(x) = \begin{cases} Ax & \text{if } x \in B(0,1), \\ h(x) & \text{if } x \in \Omega\setminus B(0,1), \end{cases} \qquad x \in \Omega.$$

Thus, the lower semicontinuity yields, after eliminating the constant part of the functional on $\Omega\setminus B(0,1)$,

$$\begin{aligned}
\int_{B(0,1)} f(A)\,\mathrm{d}x &\le \liminf_{j\to\infty} \int_{B(0,1)} f(\nabla u_j(x))\,\mathrm{d}x \\
&= \liminf_{j\to\infty} \sum_k \int_{B(a_k^{(j)}, r_k^{(j)})} f\left(A + \nabla\psi\left(\frac{x-a_k^{(j)}}{r_k^{(j)}}\right)\right)\mathrm{d}x \\
&= \liminf_{j\to\infty} \sum_k r_k^{(j)} \int_{B(0,1)} f(A+\nabla\psi(y))\,\mathrm{d}y \\
&= \int_{B(0,1)} f(A+\nabla\psi(y))\,\mathrm{d}y
\end{aligned}$$

since $\sum_k r_k^{(j)} = 1$. This is nothing else than quasiconvexity. □

We note that also for quasiconvex integrands, our results about the Euler–Lagrange equations hold just the same (if we assume the same growth conditions).

3.7 Integrands with u-dependence

One very useful feature of our Young measure approach is that it allows to derive a lower semicontinuity result for u-dependent integrands with minimal additional effort. So consider

$$\begin{cases} \mathscr{F}[u] := \int_\Omega f(x, u(x), \nabla u(x)) \,\mathrm{d}x \quad \to \quad \min, \\ \text{over all } u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m) \text{ with } u|_{\partial\Omega} = g, \end{cases}$$

where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $1 < p < \infty$, and the Carathéodory integrand $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}$ satisfies the p-growth bound

$$|f(x,v,A)| \le M(1 + |v|^p + |A|^p) \qquad \text{for some } M > 0, \tag{3.19}$$

and $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$.

The main idea of our approach is to consider Young measures generated by the pairs $(u_j, \nabla u_j) \in \mathbb{R}^{m+md}$. We have the following result:

Lemma 3.28. *Let $(u_j) \subset \mathrm{L}^p(\Omega;\mathbb{R}^M)$ and $(V_j) \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$ be norm-bounded sequences such that*

$$u_j \to u \text{ pointwise a.e.} \qquad \text{and} \qquad V_j \overset{\mathbf{Y}}{\to} \nu = (\nu_x) \in \mathbf{Y}^p(\Omega;\mathbb{R}^N).$$

Then, $(u_j, V_j) \overset{\mathbf{Y}}{\to} \mu = (\mu_x) \in \mathbf{Y}^p(\Omega;\mathbb{R}^{M+N})$ with

$$\mu_x = \delta_{u(x)} \otimes \nu_x \qquad \text{for a.e. } x \in \Omega,$$

that is,

$$\int_\Omega f(x, u_j(x), V_j(x)) \,\mathrm{d}x \to \int_\Omega \int f(x, u(x), A) \,\mathrm{d}\nu_x(A) \,\mathrm{d}x \tag{3.20}$$

for all Carathéodory $f\colon \Omega \times \mathbb{R}^M \times \mathbb{R}^N \to \mathbb{R}$ satisfying the p-growth bound (3.19).

Proof. By a similar density argument as in the proof of Lemma 3.24, it suffices to show the convergence (3.20) for $f(x,v,A) = \varphi(x)\psi(v)h(A)$, where $\varphi \in \mathrm{C}_0(\Omega)$, $\psi \in \mathrm{C}_0(\mathbb{R}^M)$, $h \in \mathrm{C}_0(\mathbb{R}^N)$. We already know by assumption

$$h(V_j) \overset{*}{\rightharpoonup} \big(x \mapsto \langle h, \nu_x \rangle\big) \quad \text{in } \mathrm{L}^\infty.$$

Furthermore, $\psi(u_j) \to \psi(u)$ almost everywhere and thus strongly in L^1 since ψ is bounded. Thus, since the product of a L^∞-weakly* converging sequence and a L^1-strongly converging sequence converges itself weakly* in the sense of measures,

$$\int_\Omega \varphi(x)\psi(u_j(x))h(V_j(x)) \,\mathrm{d}x \to \int_\Omega \varphi(x)\psi(u(x))\langle h, \nu_x \rangle \,\mathrm{d}x,$$

which is (3.20) for our special f. This already finishes the proof. □

Comment...

The trick of the preceding lemma is that in our situation, where $V_j = \nabla u_j \xrightarrow{\mathbf{Y}} \nu \in \mathbf{GY}(\Omega;\mathbb{R}^{m\times d})$, it allows us to "freeze" $u(x)$ in the integrand. Then, we can apply the Jensen-type inequality from Lemma 3.22 just as we did in Theorem 3.25 to get:

Theorem 3.29. *Let $f\colon \Omega\times\mathbb{R}^m\times\mathbb{R}^{m\times d}\to\mathbb{R}$ be a Carathéodory integrand with p-growth, i.e. $|f(x,v,A)| \le M(1+|v|^p+|A|^p)$ for some $M>0$, $1<p<\infty$. Assume furthermore that $f(x,v,\cdot)$ is quasiconvex for every fixed $(x,v)\in\Omega\times\mathbb{R}^m$. Then, the corresponding functional $\mathscr{F}$ is weakly lower semicontinuous on $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$.*

Remark 3.30. It is not too hard to extend the previous theorem to integrands f satisfying the more general growth condition

$$|f(x,v,A)| \le M(1+|v|^q+|A|^p)$$

for a $1\le q<p/(d-p)$. By the Sobolev Embedding Theorem A.17, $u_j\to u$ in L^q (strongly) for all such q and thus Lemma 3.28 and then Theorem 3.29 can be suitably generalized.

3.8 Regularity of minimizers

At the end of Section 2.5 we discussed the failure of regularity for vector-valued problems. In particular, we mentioned various counterexamples to regularity. Upon closer inspection, however, it turns out that in all counterexamples, the points where regularity fails form a closed set. This is not a coincidence.

Another issue was that all regularity theorems discussed so far require (strong) *convexity* of the integrand and our discussion so far has shown that convexity is not a good notion for vector-valued problems (however, much of the early regularity theory for vector-valued problems focused on convex problems, we will not elaborate on this aspect here). To remedy this, we need a new notion, which will take over from strong convexity in regularity theory: A locally bounded Borel-measurable function $h\colon\mathbb{R}^{m\times d}\to\mathbb{R}$ is called **strongly (2-)quasiconvex** if there exists $\mu>0$ such that

$$A\mapsto h(A)-\mu|A|^2 \quad \text{is quasiconvex.}$$

Equivalently, we may require that

$$\mu\int_{B(0,1)}|\nabla\psi(z)|^2\,\mathrm{d}z \le \int_{B(0,1)} h(A+\nabla\psi(z))-h(A)\,\mathrm{d}z$$

for all $A\in\mathbb{R}^{m\times d}$ and all $\psi\in\mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^m)$.

The most well-known regularity result in this situation is by Lawrence C. Evans from 1986 (there is significant overlap with work by Acerbi & Fusco in the same year):

Theorem 3.31 (Evans partial regularity theorem). *Let $f\colon\mathbb{R}^{m\times d}\to\mathbb{R}$ be of class C^2, strongly quasiconvex, and assume that there exists $M>0$ such that*

$$\mathrm{D}_A^2 f(A)[B,B] \le M|B|^2 \qquad \text{for all } A,B\in\mathbb{R}^{m\times d}.$$

Let $u \in \mathrm{W}^{1,2}_g(\Omega;\mathbb{R}^m)$ *be a minimizer of* $\mathscr{F}$ *over* $\mathrm{W}^{1,2}_g(\Omega;\mathbb{R}^m)$ *where* $g \in \mathrm{W}^{1/2,2}(\partial\Omega;\mathbb{R}^m)$. *Then, there exists a relatively closed* ***singular set*** $\Sigma_u \subset \Omega$ *with* $|\Sigma_u| = 0$ *and it holds that* $u \in \mathrm{C}^{1,\alpha}_{\mathrm{loc}}(\Omega \setminus \Sigma_u)$ *for all* $\alpha \in (0,1)$.

The proof is long and builds on earlier work of Almgren and De Giorgi in Geometric Measure Theory.

The result is called a "partial regularity" result because the regularity does not hold everywhere. It should be noted that while the scalar regularity theory was essentially a theory for PDEs, and hence applies to *all* solutions of the Euler–Lagrange equations, this is not the case for the present result: There is *no* regularity theory for critical points of the Euler–Lagrange equation for a quasiconvex or even polyconvex (see next chapter) integral functional. This was shown by Müller & Švérak in 2003 [75] (for the quasiconvex case) and Székelyhidi Jr. in 2004 [92] (for the polyconvex case). We finally remark that sometimes better estimates on the "smallness" of Σ_u than merely $|\Sigma_u| = 0$ are available. In fact, for strongly convex integrands it can be shown that the (Hausdorff-)dimension of the singular set is at most $d-2$, this is actually within reach of our discussion in Section 2.5, because we showed $\mathrm{W}^{2,2}_{\mathrm{loc}}$-regularity, and general properties of such functions imply the conclusion, see Chapter 2 of [44]. In the strongly quasiconvex case, much less is known, but at least for minimizers that happen to be $\mathrm{W}^{1,\infty}$ it was established in 2007 by Kristensen & Mingione that the dimension of the singular set is strictly less than d, see [56]. Many other questions are open.

3.9 Notes and bibliographical remarks

The notion of quasiconvexity was first introduced in Morrey's fundamental paper [69] and later refined by Meyers in [65]. The results about null-Lagrangians, in particular Lemma 3.8 and Lemma 3.10, go back to Morrey [70], Reshetnyak [82] and Ball [5]. Our proof is based on the idea that certain combinations of derivatives might have good convergence properties even if the individual derivatives do not. This is also pivotal for so-called *compensated compactness*, which originated in work of Tartar [94, 95] and Murat [76, 77], in particular in their celebrated *Div–Curl Lemma*, see for example the discussion in [35]. It is quite remarkable that one can prove Brouwer's fixed point theorem using the fact that minors are null-Lagrangians, see Theorem 8.1.3 in [36]. Proposition 3.6 is originally due to Morrey [70], we follow the presentation in [12]. We also remark that if only $r = p$, then a weaker version of Lemma 3.10 is true where we only have convergence of the minors in the sense of distributions, see Theorem 8.20 in [25].

The convexity properties of the quadratic function $A \mapsto \mathbf{M}A : A$, where $\mathbf{M}$ is a fourth-order tensor (or, equivalently, a matrix in $\mathbb{R}^{md\times md}$) has received considerable attention because it corresponds to a *linear* Euler–Lagrange equation. In this case, quasi-convexity and rank-one convexity are the same, and polyconvexity (see next chapter) is also equivalent to quasiconvexity if $d = 2$ or $m = 2$, but this does not hold anymore for $d, m \geq 3$. These results together with pointers to the vast literature can be found in Section 5.3.2 in [25].

A more general result on why convexity is inadmissible for realistic problems in nonlinear elasticity can be found in Section 4.8 of [20].

The result that rank-one convex functions are locally Lipschitz continuous is well-known for convex functions, see for example Corollary 2.4 in [34] and an adapted version for rank-one convex (even separately convex) functions is in Theorem 2.31 in [25]. Our proof with a quantitative bound is from Lemma 2.2 in [12].

The Fundamental Theorem of Young Measure Theory 3.11 has seen considerable evolution since its original proof by Young in the 1930s and 1940s [97–99]. It was popularized in the Calculus of Variations by Tartar [95] and Ball [7]. Further, much more general, versions have been contributed by Balder [4], also see [19] for probabilistic applications.

The Ball–James Rigidity Theorem 3.23 is from [10], see [73] for further results in the same spirit.

Lemma 3.21 is an expression of the well-known *decomposition lemma* from [38], another version is in [53].

Many results in this section that are formulated for Carathéodory integrands continue to hold for $f\colon \Omega \times \mathbb{R}^N \to \mathbb{R}$ that are Borel measurable and lower semicontinuous in the second argument, so called **normal integrands**, see [16] and [37].

The linear growth case $p = 1$ is very special and requires more sophisticated tools, so-called *generalized* Young measures, see [1, 32, 57, 58].

One issue in the linear-growth case is demonstrated in the following counterpart of Lemma 3.18 for $p = 1$:

Lemma 3.32. *Let $(V_j) \subset \mathrm{L}^1(\Omega;\mathbb{R}^N)$ be a bounded sequence generating the Young measure $(\nu_x)_{x\in\Omega} \subset \mathbf{M}^1(\mathbb{R}^N)$. Define $v(x) := [\nu_x] = \langle \mathrm{id}, \nu_x\rangle$. Then there exists an increasing sequence $\Omega_k \subset \Omega$ with $|\Omega_k| \uparrow |\Omega|$ such that $V_j \rightharpoonup v$ in $\mathrm{L}^1(\Omega_k;\mathbb{R}^N)$ for all k; this is called **biting convergence**.*

Once one knows the so-called Chacon Biting Lemma, which asserts that every L^1-bounded sequence has a subsequence converging in the biting sense, the preceding lemma is easy to see using (III) from the Fundamental Theorem of Young Measure Theory 3.11. More information on this topic can be found in [14] and some of the references cited therein.

It is possible to prove the Lower Semicontinuity Theorem 3.25 for quasiconvex integrands without the use of Young measures, see for instance Chapter 8 in [25] for such an approach. However, many of the ideas are essentially the same, they are just carried out directly without the intermediate steps of Young measures. However, Young measures allow one to very clearly organize and compartmentalize these ideas, which aids the understanding of the material. We will also use Young measures later for the relaxation of non-lower semicontinuous functionals. More on lower semicontinuity and Young measures can be found in the book [81]. The first use of Young measures to investigate questions of lower semicontinuity for quasiconvex integral functionals seems to be [50].

The blow-up technique is related to the use of tangent measures in Geometric Measure Theory, see for example [61].

Chapter 4

Polyconvexity

At the beginning of Chapter 3 we saw that convexity cannot hold concurrently with frame-indifference (and a mild positivity condition). Thus, we were lead to consider quasiconvex integrands. However, while quasiconvexity is of huge importance in the theory of the Calculus of Variations, our Lower Semicontinuity Theorem 3.25 has one major drawback: we needed to require the p-growth bound

$$|f(x,A)| \leq M(1+|A|^p) \qquad \text{for all } x \in \Omega, A \in \mathbb{R}^{d\times d} \text{ and some } M > 0,\ 1 < p < \infty.$$

Unfortunately, this is not a realistic assumption for nonlinear elasticity because it ignores the fact that infinite compressions should cost infinite energy. Indeed, realistic integrands have the property that

$$f(A) \to +\infty \quad \text{as} \quad \det A \downarrow 0$$

and

$$f(A) = +\infty \quad \text{if} \quad \det A \leq 0.$$

For example, the family of matrices

$$A_\alpha := \begin{pmatrix} 1 & 0 \\ 0 & \alpha \end{pmatrix}, \qquad \alpha > 0,$$

satisfies $\det A_\alpha \downarrow 0$ as $\alpha \downarrow 0$, but $|A_\alpha|$ remains uniformly bounded, so the above p-growth bound cannot hold.

The question whether the Lower Semicontinuity Theorem 3.25 for quasiconvex integrands can be extended to integrands with the above growth is currently a major unsolved problem, see [8]. For the time being, we have to confine ourself to a more restrictive notion of convexity if we want to allow for the above "elastic" growth. This type of convexity is called *polyconvexity* by John Ball in [5], who proved the corresponding existence theorem for minimization problems and enabled a mature mathematical treatment of nonlinear elasticity theory, including the realistic Ogden materials, see Section 1.4 and Example 4.3.

We will only look at the three-dimensional theory because it is by far the most physically important case and because this restriction eases the notational burden; however, other dimensions can be treated as well, we refer to the comprehensive treatment in [25].

4.1 Polyconvexity

A continuous integrand $h\colon \mathbb{R}^{3\times 3} \to \mathbb{R} \cup \{+\infty\}$ (now we allow the value $+\infty$) is called **polyconvex** if it can be written in the form

$$h(A) = H(A, \operatorname{cof} A, \det A), \qquad A \in \mathbb{R}^{3\times 3},$$

where $H\colon \mathbb{R}^{3\times 3} \times \mathbb{R}^{3\times 3} \times \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ is *convex*. Here, $\operatorname{cof} A$ denotes the cofactor matrix as defined in Appendix A.1.

While convexity obviously implies polyconvexity, the converse is clearly wrong. However, we have:

Proposition 4.1. *A polyconvex $h\colon \mathbb{R}^{3\times 3} \to \mathbb{R}$ (not taking the value $+\infty$) is quasiconvex.*

Proof. Let h be as stated and let $\psi \in \mathrm{W}^{1,\infty}(B(0,1);\mathbb{R}^3)$ with $\psi(x) = Ax$ on $\partial B(0,1)$. Then, using Jensen's inequality

$$\begin{aligned}
\fint_{B(0,1)} h(\nabla\psi)\,\mathrm{d}x &= \fint_{B(0,1)} H(\nabla\psi, \operatorname{cof}\nabla\psi, \det\nabla\psi)\,\mathrm{d}x \\
&\geq H\biggl(\fint_{B(0,1)} \nabla\psi\,\mathrm{d}x, \fint_{B(0,1)} \operatorname{cof}\nabla\psi\,\mathrm{d}x, \fint_{B(0,1)} \det\nabla\psi\,\mathrm{d}x\biggr) \\
&= H(A, \operatorname{cof} A, \det A) = h(A),
\end{aligned}$$

where for the last line we used the fact that minors are null-Lagrangians as proved in Lemma 3.8. □

At this point we have seen all four major convexity conditions that play a role in the modern Calculus of Variations. They can be put into a linear order by implications:

$$\text{convexity} \;\Rightarrow\; \text{polyconvexity} \;\Rightarrow\; \text{quasiconvexity} \;\Rightarrow\; \text{rank-one convexity}$$

While in the scalar case ($d = 1$ or $m = 1$) all these notions are equivalent, in higher dimensions the reverse implications do not hold in general. Clearly, the determinant function is polyconvex but not convex. However, the fact that the notions of polyconvexity, quasiconvexity, and rank-one convexity are all different in higher dimensions, are non-trivial. The fact that rank-one convexity does not imply quasiconvexity, at least for $d \geq 2$, $m \geq 3$, is the subject of Šverák's famous counterexample, see Example 5.10. The case $d = m = 2$ is still open and one of the major unsolved problems in the field. Here we discuss an example of a quasiconvex, but not polyconvex function:

Example 4.2 (Alibert–Dacorogna–Marcellini). Recall from Example 3.5 the Alibert–Dacorogna–Marcellini function

$$h_\gamma(A) := |A|^2\bigl(|A|^2 - 2\gamma \det A\bigr), \qquad A \in \mathbb{R}^{2\times 2}.$$

It can be shown that h_γ is polyconvex if and only if $|\gamma| \leq 1$. Since it is quasiconvex if and only if $|\gamma| \leq \gamma_{\mathrm{QC}}$, where $\gamma_{\mathrm{QC}} > 1$, we see that there are quasiconvex functions that are not polyconvex. See again Section 5.3.8 in [25] for details.

Example 4.3 (Ogden materials). Functions W of the form

$$W(F) := \sum_{i=1}^{M} a_i \operatorname{tr}\big[(F^T F)^{\gamma_i/2}\big] + \sum_{j=1}^{N} b_j \operatorname{tr}\operatorname{cof}\big[(F^T F)^{\delta_j/2}\big] + \Gamma(\det F), \qquad F \in \mathbb{R}^{3\times 3},$$

where $a_i > 0$, $\gamma_i \geq 1$, $b_j > 0$, $\delta_j \geq 1$, and $\Gamma\colon \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ is a convex function with $\Gamma(d) \to +\infty$ as $d \downarrow 0$, $\Gamma(d) = +\infty$ for $s \leq 0$ and good enough growth from below, can be shown to be polyconvex. These stored energy functionals correspond to so-called Ogden materials and occur in a wide range of elasticity applications, see [20] for details.

It can also be shown that in three dimensions convex functions of certain combinations of the singular values of a matrix (the singular values of A are the square roots of the eigenvalues of $A^T A$) are polyconvex, see Section 5.3.3 in [25] for details.

4.2 Existence Theorems

Let $\Omega \subset \mathbb{R}^3$ be a *connected* bounded Lipschitz domain. In this section we will prove existence of a minimizer of the variational problem

$$\begin{cases} \mathscr{F}[u] := \displaystyle\int_\Omega W(x, \nabla u(x)) - b(x) \cdot u(x) \, \mathrm{d}x \quad \to \quad \min, \\ \text{over all } u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3) \text{ with } \det \nabla u > 0 \text{ a.e. and } u|_{\partial\Omega} = g, \end{cases} \tag{4.1}$$

where $W\colon \Omega \times \mathbb{R}^{3\times 3} \to \mathbb{R} \cup \{+\infty\}$ is Carathéodory and $W(x, \cdot)$ is polyconvex for almost every $x \in \Omega$. Thus,

$$W(x,A) = \mathbb{W}(x, A, \operatorname{cof} A, \det A), \qquad (x,A) \in \Omega \times \mathbb{R}^{3\times 3},$$

for $\mathbb{W}\colon \Omega \times \mathbb{R}^{3\times 3} \times \mathbb{R}^{3\times 3} \times \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ with $\mathbb{W}(x, \cdot, \cdot, \cdot)$ jointly convex and continuous for almost every $x \in \Omega$. As the only upper growth assumptions on W we impose

$$\begin{cases} W(x,A) \to +\infty \quad \text{as} \quad \det A \downarrow 0, \\ W(x,A) = +\infty \quad \text{if} \quad \det A \leq 0. \end{cases} \tag{4.2}$$

Furthermore, we assume as usual that $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^3)$ and that $b \in \mathrm{L}^q(\Omega;\mathbb{R}^3)$ where $1/p + 1/q = 1$. The exponent $p > 1$ will remain unspecified for now. Later, when we impose conditions on the *coercivity* of W, we will also specify p.

The first lower semicontinuity result is relatively straightforward:

Theorem 4.4. *If in addition to the above assumptions it holds that*

$$\mu|A|^p - \mu^{-1} \leq W(x,A) \qquad \text{for some } \mu > 0 \text{ and } p > 3, \tag{4.3}$$

then the minimization problem (4.1) *has at least one solution in the space*

$$\mathscr{A} := \big\{ u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3) \ : \ \det \nabla u > 0 \text{ a.e. and } u|_{\partial\Omega} = g \big\},$$

whenever this set is non-empty.

Proof. We apply the usual Direct Method. For a minimizing sequence $(u_j) \subset \mathscr{A}$ with $\mathscr{F}[u_j] \to \min$, we get from the coercivity estimate (4.3) in conjunction with the Poincaré–Friedrichs inequality (and the fixed boundary values) that for any $\delta > 0$,

$$\begin{aligned}
&\int_\Omega W(x, \nabla u_j) - b(x) \cdot u_j(x) \,\mathrm{d}x \\
&\qquad \geq \mu \|\nabla u_j\|_{\mathrm{L}^p}^p - \mu^{-1}|\Omega| - \|b\|_{\mathrm{L}^q} \cdot \|u_j\|_{\mathrm{L}^p} \\
&\qquad \geq \frac{\mu}{C}\|u_j\|_{\mathrm{W}^{1,p}}^p - \frac{\mu}{C}\|g\|_{\mathrm{W}^{1-1/p,p}}^p - \frac{|\Omega|}{\mu} - \frac{1}{\delta q}\|b\|_{\mathrm{L}^q}^q - \frac{\delta}{p}\|u_j\|_{\mathrm{W}^{1,p}}^p.
\end{aligned}$$

Choosing $\delta = p\mu/(2C)$, we arrive at

$$\sup_{j\in\mathbb{N}} \|u_j\|_{\mathrm{W}^{1,p}} \leq C\Big(\sup_{j\in\mathbb{N}} \mathscr{F}[u_j] + 1\Big).$$

Thus, we may select a subsequence (not renumbered) such that $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$.

Now, by Lemma 3.10 and $p > 3$,

$$\begin{cases} \det \nabla u_j \rightharpoonup \det \nabla u & \text{in } \mathrm{L}^{p/3} \qquad \text{and} \\ \operatorname{cof} \nabla u_j \rightharpoonup \operatorname{cof} \nabla u & \text{in } \mathrm{L}^{p/2}. \end{cases}$$

Thus, an argument entirely analogous to the proof of Theorem 2.7 (note that in that theorem we did not need any upper growth condition) yields that the hyperelastic part of $\mathscr{F}$,

$$u \mapsto \int_\Omega W(x, \nabla u_j) \,\mathrm{d}x = \int_\Omega \mathbb{W}(x, \nabla u_j, \operatorname{cof} \nabla u_j, \det \nabla u_j) \,\mathrm{d}x$$

is weakly lower semicontinuous in $\mathrm{W}^{1,p}$. The second part of $\mathscr{F}$ is weakly continuous by Lemma 2.37. Thus,

$$\mathscr{F}[u] \leq \liminf_{j\to\infty} \mathscr{F}[u_j] = \inf_{\mathscr{A}} \mathscr{F}.$$

In particular, $\det \nabla u > 0$ almost everywhere and $u|_{\partial\Omega} = g$ by the weak continuity of the trace. Hence $u \in \mathscr{A}$ and the proof is finished. □

Unfortunately, the preceding theorem has a major drawback in that $p > 3$ has to be assumed. In applications in elasticity theory, however, a more realistic form of W is

$$W(A) = \frac{\lambda}{2}(\operatorname{tr} E)^2 + \mu \operatorname{tr} E^2 + \mathrm{O}(|E|^2), \qquad E = \frac{1}{2}(A^T A - I), \tag{4.4}$$

where $\lambda, \mu > 0$ are the **Lamé constants**. Except for the last term $\mathrm{O}(|E|^2)$, which vanishes as $|E| \downarrow 0$, this energy corresponds to a so-called **St. Venant–Kirchhoff material**. It was shown by Ciarlet & Geymonat [21] (also see Theorem 4.10-2 in [20]) that for any such Lamé constants, there exists a polyconvex function W of the form

$$W(A) = a|A|^2 + b|\operatorname{cof} A|^2 + \gamma(\det A) + c,$$

Comment...

where $a,b,c > 0$ and $\gamma\colon \mathbb{R} \to [0,\infty]$ is convex, such that the above expansion (4.4) holds. Clearly, such W has 2-growth in $|A|$. Thus, we need an existence theorem for functions with these growth properties.

The core of the refined argument will be an improvement of Lemma 3.10:

Lemma 4.5. *Let $1 \le p,q,r \le \infty$ with*

$$p \ge 2, \qquad \frac{1}{p} + \frac{1}{q} \le 1, \qquad r \ge 1,$$

and assume that the sequence $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ satisfies for some $H \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3})$, $d \in \mathrm{L}^r(\Omega)$ that

$$\begin{cases} u_j \rightharpoonup u & \text{in } \mathrm{W}^{1,p}, \\ \operatorname{cof}\nabla u_j \rightharpoonup H & \text{in } \mathrm{L}^q, \\ \det\nabla u_j \rightharpoonup d & \text{in } \mathrm{L}^r. \end{cases}$$

Then, $H = \operatorname{cof}\nabla u$ and $d = \det\nabla u$.

Proof. The idea is to use *distributional* versions of the cofactor matrix and Jacobian determinant and to show that they agree with the usual definitions for sufficiently regular functions. To this aim we will use the representation of minors as divergences already employed in Lemmas 3.8, 3.10.

Step 1. From (3.7) we get that for all $\varphi = (\varphi^1,\varphi^2,\varphi^3) \in \mathrm{C}^1(\Omega;\mathbb{R}^3)$,

$$(\operatorname{cof}\nabla\varphi)^k_l = \partial_{l+1}(\varphi^{k+1}\partial_{l+2}\varphi^{k+2}) - \partial_{l+2}(\varphi^{k+1}\partial_{l+1}\varphi^{k+2}),$$

where $k,l \in \{1,2,3\}$ are cyclic indices. Thus, for all $\psi \in \mathrm{C}^\infty_c(\Omega)$,

$$\begin{aligned} \int_\Omega (\operatorname{cof}\nabla\varphi)^k_l \psi \,\mathrm{d}x &= -\int_\Omega (\varphi^{k+1}\partial_{l+2}\varphi^{k+2})\partial_{l+1}\psi - (\varphi^{k+1}\partial_{l+1}\varphi^{k+2})\partial_{l+2}\psi \,\mathrm{d}x \\ &=: \big\langle (\operatorname{Cof}\nabla\varphi)^k_l, \psi \big\rangle \end{aligned} \tag{4.5}$$

and we call the quantity "$\operatorname{Cof}\nabla\varphi$" the **distributional cofactors**.

To investigate the continuity properties of the distributional cofactors, we consider

$$G(\varphi) := \int_\Omega (\varphi^k \partial_l \varphi^m)\partial_n\psi \,\mathrm{d}x, \qquad \varphi \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3),$$

for some $k,l,m,n \in \{1,2,3\}$ and fixed $\psi \in \mathrm{C}^\infty_c(\Omega)$. We can estimate this using the Hölder inequality as follows:

$$|G(\varphi)| \le \|\varphi\|_{\mathrm{L}^s} \|\nabla\varphi\|_{\mathrm{L}^p} \|\nabla\psi\|_\infty$$

whenever

$$\frac{1}{s} + \frac{1}{p} \le 1. \tag{4.6}$$

Comment...

In particular this is true for $s = p \geq 2$. Thus, since $\mathrm{C}^1(\Omega;\mathbb{R}^3)$ is dense in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$, we have that the equality (4.5) also holds for $\varphi \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ whenever $p \geq 2$.

Moreover, the above arguments yield for a sequence $(\varphi_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ that

$$G(\varphi_j) \to G(\varphi) \qquad \text{if} \qquad \begin{cases} \varphi_j \to \varphi & \text{in } \mathrm{L}^s, \\ \nabla\varphi_j \rightharpoonup \nabla\varphi & \text{in } \mathrm{L}^p. \end{cases}$$

The first convergence is automatically satisfied by the compact Rellich–Kondrachov Theorem A.18 if

$$s < \begin{cases} \frac{3p}{3-p} & \text{if } p < 3, \\ \infty & \text{if } p \geq 3. \end{cases} \tag{4.7}$$

We can always choose s such that it *simultaneously* satisfies (4.6) and (4.7), as a quick calculation shows. Thus,

$$\big\langle \operatorname{Cof}\nabla\varphi_j, \psi \big\rangle \to \big\langle \operatorname{Cof}\nabla\varphi, \psi \big\rangle \qquad \text{if} \qquad \varphi_j \rightharpoonup \varphi \quad \text{in } \mathrm{W}^{1,p},\ p \geq 2. \tag{4.8}$$

Step 2. First, if $\varphi = (\varphi^1, \varphi^2, \varphi^3) \in \mathrm{C}^1(\Omega;\mathbb{R}^3)$, then we know from (3.8) that

$$\det\nabla\varphi = \sum_{l=1}^{3} \partial_l \varphi^1 (\operatorname{cof}\nabla\varphi)_l^1 = \sum_{l=1}^{3} \partial_l \big(\varphi^1 (\operatorname{cof}\nabla\varphi)_l^1\big).$$

Thus, in this case, we have for all $\psi \in \mathrm{C}_c^\infty(\Omega)$ that

$$\int_\Omega (\det\nabla\varphi)\psi \,\mathrm{d}x = \sum_{l=1}^{3} \int_\Omega \partial_l \varphi^1 (\operatorname{cof}\nabla\varphi)_l^1 \psi \,\mathrm{d}x = -\sum_{l=1}^{3} \int_\Omega \big(\varphi^1 (\operatorname{cof}\nabla\varphi)_l^1\big) \partial_l \psi \,\mathrm{d}x. \tag{4.9}$$

The key idea now is that by Hölder's inequality this quantity makes sense if only $\varphi \in \mathrm{L}^p(\Omega;\mathbb{R}^3)$ and $\operatorname{cof}\nabla\varphi \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3})$ with $\frac{1}{p} + \frac{1}{q} \leq 1$. Analogously to the argument for the cofactor matrix, this motivates us to define for

$$\varphi \in \mathrm{L}^p(\Omega;\mathbb{R}^3) \quad \text{with} \quad \operatorname{cof}\nabla\varphi \in \mathrm{L}^q(\Omega;\mathbb{R}^3), \quad \text{where} \quad \frac{1}{p} + \frac{1}{q} \leq 1,$$

the **distributional determinant**

$$\big\langle \operatorname{Det}\nabla\varphi, \psi \big\rangle := -\sum_{l=1}^{3} \int_\Omega \big(\varphi^1 (\operatorname{cof}\nabla\varphi)_l^1\big) \partial_l \psi \,\mathrm{d}x, \qquad \psi \in \mathrm{C}_c^\infty(\Omega).$$

From (4.9) we see that if $\varphi \in \mathrm{C}^1(\Omega;\mathbb{R}^3)$, then "det = Det", i.e.

$$\int_\Omega (\det\nabla\varphi)\psi \,\mathrm{d}x = \big\langle \operatorname{Det}\nabla\varphi, \psi \big\rangle \tag{4.10}$$

for all $\psi \in \mathrm{C}_c^\infty(\Omega)$. We want to show that this equality remains to hold if merely $\varphi \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ and $\operatorname{cof}\nabla\varphi \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3})$ with p, q as in the statement of the lemma. For the moment fix $\psi \in \mathrm{C}_c^\infty(\Omega)$ and define for $\varphi \in \mathrm{C}^1(\Omega;\mathbb{R}^3)$ and $F \in \mathrm{C}^1(\Omega;\mathbb{R}^{3\times 3})$,

$$Z(\varphi, F) := \sum_{l=1}^{3} \int_\Omega \partial_l \varphi^1 F_l^1 \psi + \varphi^1 F_l^1 \partial_l \psi \,\mathrm{d}x = \sum_{l=1}^{3} \int_\Omega F_l^1 \partial_l (\varphi^1 \psi) \,\mathrm{d}x.$$

To establish (4.10), we need to show $Z(\varphi, \operatorname{cof}\nabla\varphi) = 0$.

Recall the Piola identity (3.9),

$$\operatorname{div}\operatorname{cof}\nabla v = 0 \qquad \text{if } v \in \mathrm{C}^1(\Omega;\mathbb{R}^3).$$

As a consequence,

$$Z(\varphi, \operatorname{cof}\nabla v) = 0 \qquad \text{for } \varphi, v \in \mathrm{C}^1(\Omega;\mathbb{R}^3).$$

Furthermore, we have the estimate

$$|Z(\varphi, F)| \leq \|\nabla\varphi\|_{\mathrm{L}^p} \|F\|_{\mathrm{L}^q} \|\psi\|_\infty \qquad \text{whenever} \qquad \frac{1}{p} + \frac{1}{q} \leq 1.$$

Thus, we get, via two approximation procedures (first in $F = \operatorname{cof}\nabla v$, then in φ)

$$Z(\varphi, \operatorname{cof}\nabla v) = 0 \qquad \text{for } \varphi \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3),\ \operatorname{cof}\nabla v \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3}).$$

In particular,

$$Z(\varphi, \operatorname{cof}\nabla\varphi) = 0 \qquad \text{for } \varphi \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3),\ \operatorname{cof}\nabla\varphi \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3}).$$

Therefore, (4.10) holds also for $\varphi = u$ as in the statement of the lemma.

Moreover, we see from the definition of the distributional determinant that for a sequence $(\varphi_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ we have

$$\big\langle \operatorname{Det}\nabla\varphi_j, \psi \big\rangle \to \big\langle \operatorname{Det}\nabla\varphi, \psi \big\rangle \qquad \text{if} \qquad \begin{cases} \varphi_j \to \varphi & \text{in } \mathrm{L}^s, \\ \operatorname{cof}\nabla\varphi_j \rightharpoonup \operatorname{cof}\nabla\varphi & \text{in } \mathrm{L}^q. \end{cases}$$

whenever

$$\frac{1}{s} + \frac{1}{q} \leq 1. \tag{4.11}$$

The first convergence follows, as before, from the Rellich–Kondrachov Theorem A.18 if

$$s < \begin{cases} \frac{3p}{3-p} & \text{if } p < 3, \\ \infty & \text{if } p \geq 3. \end{cases} \tag{4.12}$$

However, since we assumed

$$\frac{1}{p} + \frac{1}{q} \leq 1,$$

we can always choose s such that it satisfies (4.11) and (4.12) *simultaneously*, as can be seen by some elementary algebra. Thus,

$$\big\langle \operatorname{Det}\nabla\varphi_j, \psi \big\rangle \to \big\langle \operatorname{Det}\nabla\varphi, \psi \big\rangle \qquad \text{if} \qquad \begin{cases} \varphi_j \rightharpoonup \varphi & \text{in } \mathrm{W}^{1,p}, \\ \operatorname{cof}\nabla\varphi_j \rightharpoonup \operatorname{cof}\nabla\varphi & \text{in } \mathrm{L}^q, \end{cases} \tag{4.13}$$

where p,q satisfy the assumptions of the lemma.

Step 3. Assume that, as in the statement of the lemma, we are given a sequence $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ such that for $H \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3})$, $d \in \mathrm{L}^r(\Omega)$ it holds that

$$\begin{cases} u_j \rightharpoonup u & \text{in } \mathrm{W}^{1,p}, \\ \operatorname{cof}\nabla u_j \rightharpoonup H & \text{in } \mathrm{L}^q, \\ \det\nabla u_j \rightharpoonup d & \text{in } \mathrm{L}^r. \end{cases}$$

Then, (4.5), (4.10) imply that for all $\psi \in \mathrm{C}_c^\infty(\Omega)$,

$$\big\langle \operatorname{Cof}\nabla u_j, \psi \big\rangle \to \big\langle H, \psi \big\rangle \qquad \text{and} \qquad \big\langle \operatorname{Det}\nabla u_j, \psi \big\rangle \to \big\langle d, \psi \big\rangle.$$

On the other hand, from (4.8), (4.13),

$$\big\langle \operatorname{Cof}\nabla u_j, \psi \big\rangle \to \big\langle \operatorname{Cof}\nabla u, \psi \big\rangle \qquad \text{and} \qquad \big\langle \operatorname{Det}\nabla u_j, \psi \big\rangle \to \big\langle \operatorname{Det}\nabla u, \psi \big\rangle.$$

Thus,

$$\int_\Omega (\operatorname{Cof}\nabla u - H)\psi \,\mathrm{d}x = 0 \qquad \text{and} \qquad \int_\Omega (\operatorname{Det}\nabla u - d)\psi \,\mathrm{d}x = 0$$

for all $\psi \in \mathrm{C}_c^\infty(\Omega)$. The Fundamental Lemma of the Calculus of Variations 2.21 then gives immediately

$$H = \operatorname{cof}\nabla u \qquad \text{and} \qquad d = \det\nabla u.$$

This finishes the proof. □

With this tool at hand, we can now prove the improved existence result:

Theorem 4.6 (Ball). *Let* $1 \le p,q,r < \infty$

$$p \ge 2, \qquad \frac{1}{p} + \frac{1}{q} \le 1, \qquad r > 1,$$

such that in addition to the assumptions at the beginning of this section, it holds that

$$W(x,A) \ge \mu\big(|A|^p + |\operatorname{cof} A|^q + |\det A|^r\big) - \mu^{-1} \qquad \textit{for some } \mu > 0. \tag{4.14}$$

Then, the minimization problem (4.1) *has at least one solution in the space*

$$\begin{aligned} \mathscr{A} := \big\{ u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3) \ : \ & \operatorname{cof}\nabla u \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3}), \det\nabla u \in \mathrm{L}^r(\Omega), \\ & \det\nabla u > 0 \textit{ a.e. and } u|_{\partial\Omega} = g \big\}, \end{aligned}$$

whenever this set is non-empty.

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Proof. This follows in a completely analogous way to the proof of Theorem 4.4, but now we select a subsequence of a minimizing sequence $(u_j) \subset \mathscr{A}$ such that for some $H \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3})$, $d \in \mathrm{L}^r(\Omega)$ we have

$$\begin{cases} u_j \rightharpoonup u & \text{in } \mathrm{W}^{1,p}, \\ \operatorname{cof}\nabla u_j \rightharpoonup H & \text{in } \mathrm{L}^q, \\ \det\nabla u_j \rightharpoonup d & \text{in } \mathrm{L}^r, \end{cases}$$

which is possible by the usual weak compactness results in conjunction with the coercivity assumption (4.14). Thus, Lemma 4.5 yields

$$\begin{cases} u_j \rightharpoonup u & \text{in } \mathrm{W}^{1,p}, \\ \operatorname{cof}\nabla u_j \rightharpoonup \operatorname{cof}\nabla u & \text{in } \mathrm{L}^q, \\ \det\nabla u_j \rightharpoonup \det\nabla u & \text{in } \mathrm{L}^r, \end{cases}$$

and we may argue as in Theorem 4.4 to conclude that u is indeed a minimizer. The proof that $u \in \mathscr{A}$ is the same as in Theorem 4.4. □

4.3 Global injectivity

For reasons of *physical admissibility*, we need to additionally prove that we can find a minimizer that is globally injective, at least almost everywhere (since we work with Sobolev functions). There are several approaches to this issue, for example via the *topological degree*. We here present a classical argument by Ciarlet & Nečas [22]. It is important to notice that on physical grounds it is only realistic to expect this injectivity for $p > d$. For lower exponents, complex effects such as cavitation and (microscopic) fracture have to be considered. This is already expressed in the fact that Sobolev functions in $\mathrm{W}^{1,p}$ for $p \le d$ are not necessarily continuous anymore.

We will prove the following theorem:

Theorem 4.7. *In the situation of Theorem 4.4, in particular $p > 3$, the minimization problem* (4.1) *has at least one solution in the space*

$$\mathscr{A} := \bigl\{ u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^d) \ \textbf{:}\ \det\nabla u > 0 \textit{ a.e., } u \textit{ is injective a.e., } u|_{\partial\Omega} = g \bigr\},$$

whenever this set is non-empty.

Proof. Let $(u_j) \subset \mathscr{A}$ be a minimizing sequence. The proof is completely analogous to the one of Theorem 4.4, we only need to show in addition that u is injective almost everywhere.

For our choice $p > 3$, the space $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ embeds continuously into $\mathrm{C}(\Omega;\mathbb{R}^3)$, so we have that

$$u_j \to u \quad \text{uniformly.}$$

Let $U \subset \mathbb{R}^3$ be any relatively compact open set with $u(\Omega) \subset\subset U$ (note that $u(\Omega)$ is relatively compact). Then, $u_j(\Omega) \subset U$ for j sufficiently large by the uniform convergence. Hence, for such j,

$$|u_j(\Omega)| \le |U|.$$

Furthermore, since $\det \nabla u_j$ converges weakly to $\det \nabla u$ in $\mathrm{L}^{p/3}$ by Lemma 3.10 and all u_j are injective almost everywhere by definition of the space $\mathscr{A}$,

$$\int_\Omega \det \nabla u \,\mathrm{d}x = \lim_{j\to\infty} \int_\Omega \det \nabla u_j \,\mathrm{d}x = \lim_{j\to\infty} |u_j(\Omega)| \le |U|.$$

Letting $|U| \downarrow |u(\Omega)|$, we get the **Ciarlet–Nečas non-interpenetration condition**

$$\int_\Omega \det \nabla u \,\mathrm{d}x \le |u(\Omega)|. \tag{4.15}$$

It is known (see for example [60] or [17]) that even if only $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ it holds that

$$\int_\Omega |\det \nabla u| \,\mathrm{d}x = \int_{u(\Omega)} \mathscr{H}^0(u^{-1}(x'))\,\mathrm{d}x',$$

where we denote by $\mathscr{H}^0$ the counting measure. We estimate

$$|u(\Omega)| \le \int_{u(\Omega)} \mathscr{H}^0(u^{-1}(x'))\,\mathrm{d}x' = \int_\Omega \det \nabla u \,\mathrm{d}x \le |u(\Omega)|,$$

where we used that $\det \nabla u > 0$ almost everywhere and (4.15). Thus,

$$\mathscr{H}^0(u^{-1}(x')) = 1 \qquad \text{for a.e. } x' \in u(\Omega),$$

which is nothing else than the almost everywhere injectivity of u. □

4.4 Notes and bibliographical remarks

Theorem 4.6 is a refined version by Ball, Currie & Olver [9] of Ball's original result in [5]. Also see [10, 11] for further reading.

Many questions about polyconvex integral functionals remain open to this day: In particular, the regularity of solutions and the validity of the Euler–Lagrange equations are largely open in the general case. Note that the regularity theory from the previous chapter is not in general applicable, at least if we do not assume the upper p-growth. These questions are even open for the more restricted situation of nonlinear elasticity theory. See [8] for a survey on the current state of the art and a collection of challenging open problems.

As for the almost injectivity (for $p > 3$), this is in fact sometimes automatic, as shown by Ball [6], but the arguments do not apply to all situations. Tang [93] extended this to $p > 2$, but since then we have to deal with non-continuous functions, it is not even clear how to define $u(\Omega)$. For fracture, there is a huge literature, see [74] and the recent [46], which also contains a large bibliography. A study on the surprising (mis)behavior of the determinant for $p < d$ can be found in [52].

Chapter 5

Relaxation

When trying to minimize an integral functional over a Sobolev space, it it a very real possibility that there is no minimizer if the integrand is not quasiconvex; of course, even functionals with non-quasiconvex integrands can have minimizers, but, generally speaking, we should expect non-quasiconvexity to be correlated with the non-existence of minimizers.

For instance, consider the functional

$$\mathscr{F}[u] := \int_0^1 |u(x)|^2 + \bigl(|u'(x)|^2 - 1\bigr)^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}_0^{1,4}(0,1).$$

The gradient part of the integrand, $a \mapsto (a^2-1)^2$, is called a **double-well potential**, see Figure 5.1. Approximate minimizers of $\mathscr{F}$ try to satisfy $u' \in \{-1,+1\}$ as closely as possible, while at the same time staying close to zero because of the first term under the integral. These contradicting requirements lead to minimizing sequences that develop faster and faster oscillations similar to the ones shown in Figure 3.2 on page 48. It should be intuitively clear that no classical function can be a minimizer of $\mathscr{F}$.

In this situation, we have essentially two options: First, if we only care about the infimal *value* of $\mathscr{F}$, we can compute the *relaxation* $\mathscr{F}_*$ of $\mathscr{F}$, which (by definition) will be lower semicontinuous and then find its minimizer. This will give us the infimum of $\mathscr{F}$ over all admissible functions, but the minimizer of $\mathscr{F}_*$ says only very little about the minimizing sequence of our original $\mathscr{F}$ since all oscillations (and concentrations in some cases) have been "averaged out".

Second, we can focus on the minimizing sequences themselves and try to find a generalized limit object to a minimizing sequence encapsulating "interesting" information about the minimization. This limit object will be a Young measure and in fact this was the original motivation for introducing them. Effectively, Young measure theory allows one to replace a minimization problem over a Sobolev space by a generalized minimization problem over (gradient) Young measures that *always* has a solution. In applications, the emerging oscillations correspond to **microstructure**, which is very important for example in material science.

In this chapter we will consider both approaches in turn.

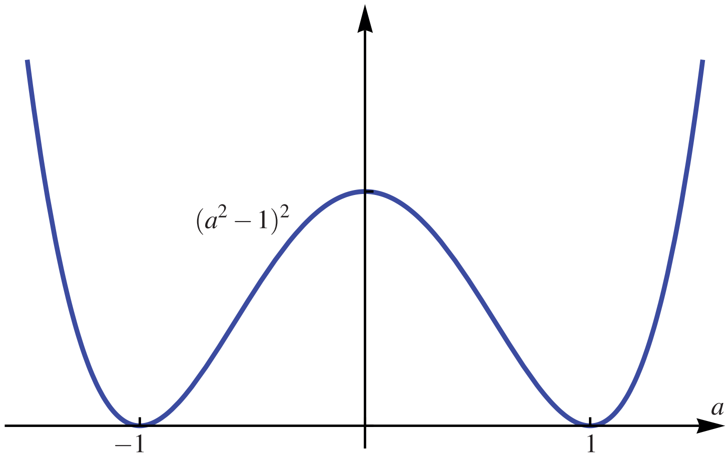

Figure 5.1: A double-well potential.

5.1 Quasiconvex envelopes

We have seen in Chapter 3 that integral functionals with quasiconvex integrands are weakly lower semicontinuous in $\mathrm{W}^{1,p}$, where the exponent $1 < p < \infty$ is determined by growth properties of the integrand. If the integrand is *not* quasiconvex, then we would like to compute the functional's **relaxation**, i.e. the largest (weakly) lower semicontinuous function below it. We can expect that when passing from the original functional to the relaxed one, we should also pass from the original integrand to a related but quasiconvex integrand. For this, we define the **quasiconvex envelope** $Qh\colon \mathbb{R}^{m\times d} \to \mathbb{R}\cup\{-\infty\}$ of a locally bounded Borel-function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ as

$$Qh(A) := \inf\left\{ \fint_\omega h(A+\nabla\psi(z))\,\mathrm{d}z \;:\; \psi \in \mathrm{W}_0^{1,\infty}(\omega;\mathbb{R}^m) \right\}, \qquad A \in \mathbb{R}^{m\times d}, \tag{5.1}$$

where $\omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain. Clearly, $Qh \le h$. By a similar argument as in Lemma 3.2, one can see that this formula is independent of the choice of ω. If h has p-growth at infinity, we may replace the space $\mathrm{W}_0^{1,\infty}(\omega;\mathbb{R}^m)$ in the above formula by $\mathrm{W}_0^{1,p}(\omega;\mathbb{R}^m)$. Finally, by a covering argument as for instance in Proposition 3.27, we may furthermore restrict the class of ψ in the above infimum to those satisfying $\|\psi\|_{\mathrm{L}^\infty} \le \varepsilon$ for any $\varepsilon > 0$.

It is well-known that for continuous h the quasiconvex envelope Qh is equal to

$$Qh = \sup\left\{ g \;:\; g \text{ quasiconvex and } g \le h \right\}, \tag{5.2}$$

see Section 6.3 in [25]. Note that the equality also holds if one (hence both) of the two expressions is $-\infty$.

We first need the following lemma:

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Lemma 5.1. *For continuous $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth, $|h(A)| \le M(1+|A|^p)$, the quasiconvex envelope Qh as defined in* (5.1) *is either identically $-\infty$ or real-valued with p-growth, and quasiconvex.*

Proof. Step 1. If Qh takes the value $-\infty$, say $Qh(B_0) = -\infty$, then for all $N \in \mathbb{N}$ there exists $\hat{\psi}_N \in \mathrm{W}_0^{1,\infty}(B(0,1/2);\mathbb{R}^m)$ such that

$$\int_{B(0,1/2)} h(A_0 + \nabla \hat{\psi}_N(z))\,\mathrm{d}z \le -N.$$

For any $A_0 \in \mathbb{R}^{m\times d}$ pick $\psi_* \in \mathrm{W}_{A_0x}^{1,\infty}(B(0,1);\mathbb{R}^m)$ that has trace B_0x on $\partial B(0,1/2)$ and set

$$\psi_N(z) := \begin{cases} \hat{\psi}_N(z) & \text{if } z \in B(0,1/2), \\ \psi_*(z & \text{otherwise,} \end{cases} \qquad z \in B(0,1).$$

Hence,

$$Qh(A_0) \le \fint_{B(0,1)} h(\nabla \psi_N(z))\,\mathrm{d}z \le \frac{-N+C}{B(0,1)}$$

for some fixed $C > 0$ (depending on our choice of ψ_*). Letting $N \to \infty$, we get $Qh(A_0) = -\infty$. Thus, Qh is *identically* $-\infty$.

Step 2. For any $\psi \in \mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^m)$ and any $A_0 \in \mathbb{R}^{m\times d}$ we need to show

$$\fint_{B(0,1)} Qh(A_0 + \nabla\psi(z))\,\mathrm{d}z \ge Qh(A_0). \tag{5.3}$$

It can be shown that also Qh has p-growth, see for instance Lemma 2.5 in [54]. Thus, we can use an approximation argument in conjunction with Theorem 2.30 to prove that it suffices to show the inequality (5.3) for piecewise affine $\psi \in \mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^m)$, say $\psi(x) = v_k + A_kx$ ($v_k \in \mathbb{R}^m, A_k \in \mathbb{R}^{m\times d}$) for $x \in G_k$ from a disjoint collection of Lipschitz subdomains $G_k \subset \Omega$ ($k = 1,\dots,N$) with $|\Omega \setminus \bigcup_k G_k| = 0$. Here we use the $\mathrm{W}^{1,p}$-density of piecewise affine functions in $\mathrm{W}_0^{1,\infty}$, which can be proved by mollification and an elementary approximation of smooth functions by piecewise affine ones (e.g. employing finite elements).

Fix $\varepsilon > 0$. By the definition of Qh, for every k we can find $\varphi_k \in \mathrm{W}_0^{1,\infty}(G_k;\mathbb{R}^m)$ such that

$$Qh(A_0 + A_k) \ge \fint_{G_k} f(A_0 + A_k + \nabla\varphi_k(z))\,\mathrm{d}z - \varepsilon.$$

Then,

$$\begin{aligned}
\int_{B(0,1)} Qh(A_0 + \nabla\psi(z))\,\mathrm{d}z &= \sum_{k=1}^{N} |G_k| Qh(A_0 + A_k) \\
&\ge \sum_{k=1}^{N} \int_{G_k} h(A_0 + A_k + \nabla\varphi_k(z))\,\mathrm{d}z - \varepsilon|G_k| \\
&= \int_{B(0,1)} h(A_0 + \nabla\varphi(z))\,\mathrm{d}z - \varepsilon|B(0,1)| \\
&\ge |B(0,1)|\big(Qh(A_0) - \varepsilon\big),
\end{aligned}$$

where $\varphi \in \mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^m)$ is defined as

$$\varphi(x) := \begin{cases} v_k + A_k x + \varphi_k(x) & \text{if } x \in G_k, \\ 0 & \text{otherwise,} \end{cases} \qquad x \in B(0,1),$$

and the last step follows from the definition of Qh. Letting $\varepsilon \downarrow 0$, we have shown (5.3). $\square$

We can also use the notion of the quasiconvex envelope to introduce a class of non-trivial quasiconvex functions with p-growth, $1 < p < \infty$ (the linear growth case is also possible using a refined technique).

Lemma 5.2. *Let $F \in \mathbb{R}^{2\times 2}$ with* $\operatorname{rank} F = 2$ *and let $1 < p < \infty$. Define*

$$h(A) := \operatorname{dist}(A, \{-F, F\})^p, \qquad A \in \mathbb{R}^{2\times 2}.$$

Then, $Qh(0) > 0$ and the quasiconvex envelope Qh is not convex (at zero).

Proof. We first show $Qh(0) > 0$. Assume to the contrary that $Qh(0) = 0$. Then, by (5.1) there exists a sequence $(\psi_j) \subset \mathrm{W}_0^{1,\infty}(B(0,1);\mathbb{R}^2)$ with

$$\fint_{B(0,1)} h(\nabla\psi_j)\,\mathrm{d}z \to 0. \tag{5.4}$$

Let $P\colon \mathbb{R}^{2\times 2} \to \mathbb{R}^{2\times 2}$ be the projection onto the orthogonal complement of $\operatorname{span}\{F\}$. Some linear algebra shows $|P(A)|^p \le h(A)$ for all $A \in \mathbb{R}^{2\times 2}$. Therefore,

$$P(\nabla\psi_j) \to 0 \quad \text{in } \mathrm{L}^p(B(0,1);\mathbb{R}^{2\times 2}). \tag{5.5}$$

With the definition

$$\hat{g}(\xi) := \int_{B(0,1)} g(x)\mathrm{e}^{-2\pi\mathrm{i}\xi\cdot x}\,\mathrm{d}x, \qquad \xi \in \mathbb{R}^d,$$

for the Fourier transform $\hat{g}$ of a function $g \in \mathrm{L}^1(B(0,1);\mathbb{R}^N)$, we get the rule $\widehat{\nabla\psi_j}(\xi) = (2\pi\mathrm{i})\,\hat{\psi}_j(\xi) \otimes \xi$. The fact that $a \otimes \xi \notin \operatorname{span}\{F\}$ for all $a, \xi \in \mathbb{R}^2 \setminus \{0\}$ implies $P(a \otimes \xi) \neq 0$. Some algebra implies that we may write

$$\widehat{\nabla\psi_j}(\xi) = M(\xi)P(\hat{\psi}_j(\xi) \otimes \xi), \qquad \xi \in \mathbb{R}^d,$$

where $M(\xi)\colon \mathbb{R}^{2\times 2} \to \mathbb{R}^{2\times 2}$ is linear and furthermore depends smoothly and positively 0-homogeneously on ξ. We omit the details of this construction.

For $p = 2$, Parseval's formula $\|g\|_{\mathrm{L}^2} = \|\hat{g}\|_{\mathrm{L}^2}$ together with (5.5) then implies

$$\begin{aligned} \|\nabla\psi_j\|_{\mathrm{L}^2} &= \|\widehat{\nabla\psi_j}\|_{\mathrm{L}^2} = \|M(\xi)P(\hat{\psi}_j(\xi) \otimes \xi)\|_{\mathrm{L}^2} \\ &\le \|M\|_\infty \|P(\hat{\psi}_j(\xi) \otimes \xi)\|_{\mathrm{L}^2} = \|M\|_\infty \|P(\nabla\psi_j)\|_{\mathrm{L}^2} \to 0. \end{aligned}$$

But then $h(\nabla\psi_j) \to |F|$ in L^1, contradicting (5.4).

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For $1 < p < \infty$, we may apply the Mihlin Multiplier Theorem (see for example Theorem 5.2.7 in [45] and Theorem 6.1.6 in [15]) to get analogously to the case $p = 2$ that

$$\|\nabla \psi_j\|_{\mathrm{L}^p} \leq \|M\|_{\mathrm{C}^d} \|P(\nabla \psi_j)\|_{\mathrm{L}^p} \to 0,$$

which is again at odds with (5.4).

Finally, if Qh was convex at zero, we would have

$$Qh(0) \leq \frac{1}{2}\big(Qh(F) + Qh(-F)\big) \leq \frac{1}{2}\big(h(F) + h(-F)\big) = 0,$$

a contradiction. □

5.2 Relaxation of integral functionals

We first consider the abstract principles of relaxation before moving on to more concrete integral functionals. Suppose we have a functional $\mathscr{F} : X \to \mathbb{R} \cup \{+\infty\}$, where X is a reflexive Banach space. Assume furthermore that our $\mathscr{F}$ is *not* weakly lower semicontinuous[1], i.e. there exists at least one sequence $x_j \rightharpoonup x$ in X with $\mathscr{F}[x] > \liminf_{j\to\infty} \mathscr{F}[x_j]$. Then, the **relaxation** $\mathscr{F}_* : X \to \mathbb{R} \cup \{-\infty, +\infty\}$ of F is defined as

$$\mathscr{F}_*[x] := \inf\left\{ \liminf_{j\to\infty} \mathscr{F}[x_j] \;:\; x_j \rightharpoonup x \text{ in } X \right\}.$$

Theorem 5.3 (Abstract relaxation theorem). *Let X be a separable, reflexive Banach space and let $\mathscr{F} : X \to \mathbb{R} \cup \{+\infty\}$ be a functional. Assume furthermore:*

(WH1) ***Weak Coercivity:*** *For all $\Lambda > 0$, the sublevel set*

$$\{x \in X \;:\; \mathscr{F}[x] \leq \Lambda\} \qquad \textit{is sequentially weakly relatively compact.}$$

Then, the relaxation $\mathscr{F}_*$ of $\mathscr{F}$ is weakly lower semicontinuous and $\min_X \mathscr{F}_* = \inf_X \mathscr{F}$.

Proof. Since X^* is also separable, we can find a countable set $\{f_l\}_{l\in\mathbb{N}} \subset X^*$ such that

$$x_j \rightharpoonup x \quad \text{in } X \qquad \iff \qquad \sup_j \|x_j\| < \infty \text{ and } \langle x_j, f_l\rangle \to \langle x, f_l\rangle \text{ for all } l \in \mathbb{N}.$$

Step 1. We first show weak lower semicontinuity of $\mathscr{F}_*$: Let $x_j \rightharpoonup x$ in X such that $\liminf_{j\to\infty} \mathscr{F}_*[x_j] < \infty$ (if this is not satisfied, there is nothing to show). Selecting a (not renumbered) subsequence if necessary, we may further assume that $\alpha := \sup_j \mathscr{F}_*[x_j] < \infty$.

For every $j \in \mathbb{N}$, there exists a sequence $(x_{j,k})_k \subset X$ such that $x_{j,k} \rightharpoonup x_j$ in X as $k \to \infty$ and

$$\liminf_{k\to\infty} \mathscr{F}[x_{j,k}] \leq \mathscr{F}_*[x_j] + \frac{1}{j}.$$

[1] In principle, if $\mathscr{F}$ nevertheless is weakly lower semicontinuous for a *minimizing* sequence, then no relaxation is necessary. It is, however, very difficult to establish such a property without establishing general weak lower semicontinuity.

Now select a $y_j := x_{j,k(j)}$ such that

$$\big|\langle y_j, f_l\rangle - \langle x_j, f_l\rangle\big| \le \frac{1}{j} \qquad \text{for all } l = 1,\dots,j,$$

and

$$\mathscr{F}[y_j] \le \mathscr{F}_*[x_j] + \frac{2}{j} \le \alpha + \frac{2}{j}.$$

Then, from the weak coercivity (WH1) we get $\sup_j \|y_j\| < \infty$. Thus, it follows that the so-selected diagonal sequence (y_j) satisfies $y_j \rightharpoonup x$ and

$$\mathscr{F}_*[x] \le \liminf_{j\to\infty} \mathscr{F}[y_j] \le \liminf_{j\to\infty} \mathscr{F}_*[x_j]$$

and $\mathscr{F}_*$ is shown to be weakly lower semicontinuous.

Step 2. Next take a minimizing sequence $(x_j) \subset X$ such that

$$\lim_{j\to\infty} \mathscr{F}_*[x_j] = \inf_X \mathscr{F}_*.$$

Now, for every x_j select a $y_j \in X$ such that

$$\mathscr{F}[y_j] \le \mathscr{F}_*[x_j] + \frac{1}{j}.$$

From the coercivity assumption (WH1) we get that we may select a weakly converging subsequence (not renumbered) $y_j \rightharpoonup y$, where $y \in X$. Then, since $\mathscr{F}_*$ is weakly lower semicontinuous and $\mathscr{F}_* \le \mathscr{F}$,

$$\mathscr{F}_*[y] \le \liminf_{j\to\infty} \mathscr{F}_*[y_j] \le \liminf_{j\to\infty} \mathscr{F}[y_j] \le \lim_{j\to\infty} \mathscr{F}_*[x_j] = \inf_X \mathscr{F}_*,$$

so $\mathscr{F}_*[y] = \min_X \mathscr{F}_*$. Furthermore,

$$\inf_X \mathscr{F} \le \liminf_{j\to\infty} \mathscr{F}[y_j] \le \min_X \mathscr{F}_* \le \inf_X \mathscr{F}.$$

Thus, $\min_X \mathscr{F}_* = \inf_X \mathscr{F}$, completing the proof. □

As usual, we here are most interested in the concrete case of integral functionals

$$\mathscr{F}[u] := \int_\Omega f(x, \nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m),$$

where, as usual, $f\colon \Omega \times \mathbb{R}^{m\times d} \to \mathbb{R}$ is a Carathéodory integrand satisfying the p-growth and coercivity assumption

$$\mu|A|^p - \mu^{-1} \le f(x,A) \le M(1+|A|^p), \qquad (x,A) \in \Omega \times \mathbb{R}^{m\times d}, \tag{5.6}$$

for some $\mu, M > 0$, $1 < p < \infty$, and $\Omega \subset \mathbb{R}^d$ a bounded Lipschitz domain. The main result of this section is:

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Theorem 5.4 (Relaxation). *Let $\mathscr{F}$ be as above and assume furthermore that there exists a* ***modulus of continuity*** *ω, that is, $\omega\colon [0,\infty) \to [0,\infty)$ is continuous, increasing, and $\omega(0) = 0$, such that*

$$|f(x,A) - f(y,A)| \le \omega(|x-y|)(1+|A|^p), \qquad x,y \in \Omega,\ A \in \mathbb{R}^{m\times d}. \tag{5.7}$$

Then, the relaxation $\mathscr{F}_*$ of $\mathscr{F}$ is

$$\mathscr{F}_*[u] = \int_\Omega Qf(x,\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m),$$

where $Qf(x,\cdot)$ is the quasiconvex envelope of $f(x,\cdot)$ for all $x \in \Omega$.

Proof. We define

$$\mathscr{G}[u] := \int_\Omega Qf(x,\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m).$$

As $Qf(x,\cdot)$ is quasiconvex for all $x \in \Omega$ by Lemma 5.1, it is continuous by Proposition 3.6. We will see below that Qf is continuous in its first argument, so Qf is Carathéodory and $\mathscr{G}$ is thus well-defined.

We will show in the following that (I) $\mathscr{G} \le \mathscr{F}_*$ and (II) $\mathscr{F}_* \le \mathscr{G}$.

To see (I), it suffices to observe that $\mathscr{G}$ is weakly lower semicontinuous by Theorem 3.25 and that $\mathscr{G} \le \mathscr{F}$. Thus, from the definition of $\mathscr{F}_*$ we immediately get $\mathscr{G} \le \mathscr{F}_*$.

We will show (II) in several steps.

Step 1. Fix $A \in \mathbb{R}^{m\times d}$. For $x \in \Omega$ let $\psi_x \in \mathrm{W}^{1,p}_0(B(0,1);\mathbb{R}^m)$ be such that

$$\fint_{B(0,1)} f(x,A+\nabla\psi_x(z))\,\mathrm{d}z \le Qf(x,A)+1.$$

Then we use (5.6) to observe

$$\mu \fint_{B(0,1)} |A+\nabla\psi_x(z)|^p\,\mathrm{d}z - \mu^{-1} \le Qf(x,A)+1 \le M(1+|A|^p)+1.$$

Let now $x,y \in \Omega$ and estimate using (5.7) and the definition of $Qf(x,\cdot)$,

$$\begin{aligned} Qf(y,A) - Qf(x,A) &\le \fint_{B(0,1)} \bigl|f(y,A+\nabla\psi_x(z)) - f(x,A+\nabla\psi_x(z))\bigr|\,\mathrm{d}z + 1 \\ &\le \omega(|x-y|) \fint_{B(0,1)} 1+|A+\nabla\psi_x(z)|^p\,\mathrm{d}z + 1 \\ &\le C\omega(|x-y|)(1+|A|^p), \end{aligned}$$

where $C = C(\mu,M)$ is a constant. Exchanging the roles of x and y, we see

$$\bigl|Qf(x,A) - Qf(y,A)\bigr| \le C\omega(|x-y|)(1+|A|^p) \tag{5.8}$$

for all $x,y \in \Omega$ and $A \in \mathbb{R}^{m\times d}$. In particular, Qf is continuous in x.

Step 2. Next we show that it suffices to consider piecewise affine u. If $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, then there exists a sequence $(v_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ of piecewise affine functions such that $v_j \to u$ in $\mathrm{W}^{1,p}$. Since Qf is Carathéodory and has p-growth (as $Qf \le f$), Theorem 2.30 shows that $\mathscr{G}[v_j] \to \mathscr{G}[u]$ and, by lower semicontinuity, $\mathscr{F}_*[u] \le \liminf_{j\to\infty} \mathscr{F}_*[v_j]$. Thus, if (II) holds for all piecewise affine u, it also holds for all $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$.

Step 3. Fix $\varepsilon > 0$ and let $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ be piecewise affine, say $u(x) = v_k + A_k x$ (where $v_k \in \mathbb{R}^m$, $A_k \in \mathbb{R}^{m\times d}$) on the set G_k from a disjoint collection of open sets $G_k \subset \Omega$ ($k = 1,\ldots,N$) such that $|\Omega \setminus \bigcup_{k=1}^N G_k| = 0$. Fix such k and cover G_k up to a nullset with countably many balls $B_{k,l} := B(x_{k,l}, r_{k,l}) \subset G_k$, where $x_{k,l} \in G_k$, $0 < r_{k,l} < \varepsilon$ ($l \in \mathbb{N}$) such that

$$\left| \fint_{B_{k,l}} Qf(x,A_k)\,\mathrm{d}x - Qf(x_{k,l},A_k) \right| \le C\omega(\varepsilon)(1+|A_k|^p).$$

This is possible by the Vitali Covering Theorem A.8 in conjunction with (5.8).

Now, from the definition of Qf and the independence of this definition from the domain, we can find a function $\psi_{k,l} \in \mathrm{W}^{1,\infty}_0(B_{k,l};\mathbb{R}^m)$ with $\|\psi_{k,l}\|_{\mathrm{L}^\infty} \le \varepsilon$ and

$$\left| Qf(x_{k,l},A_k) - \fint_{B_{k,l}} f\bigl(x_{k,l}, A_k + \nabla\psi_{k,l}(z)\bigr)\,\mathrm{d}z \right| \le \varepsilon.$$

Set

$$v_\varepsilon(x) := u(x) + \begin{cases} \psi_{k,l}(x) & \text{if } x \in B_{k,l}, \\ 0 & \text{otherwise,} \end{cases} \qquad x \in \Omega.$$

In a similar way to Step 1 we can show that

$$\mu \fint_{B_{k,l}} |A_k + \nabla\psi_{k,l}(z)|^p\,\mathrm{d}z - \mu^{-1} \le M(1+|A_k|^p) + \varepsilon.$$

Thus,

$$\begin{aligned} \int_\Omega |\nabla v_\varepsilon|^p\,\mathrm{d}x &= \sum_{k,l} \int_{B_{k,l}} |A_k + \nabla\psi_{k,l}(x)|^p\,\mathrm{d}x \\ &\le \frac{M}{\mu} \int_\Omega 1 + |\nabla u(x)|^p\,\mathrm{d}x + \frac{|\Omega|}{\mu^2} + \frac{\varepsilon|\Omega|}{\mu} < \infty, \end{aligned}$$

so, by the Poincaré–Friedrichs inequality, $(v_\varepsilon)_{\varepsilon>0} \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ is uniformly bounded in $\mathrm{W}^{1,p}$.

By the continuity assumption (5.7),

$$\left| \fint_{B_{k,l}} f(x_{k,l}, \nabla v_\varepsilon(x))\,\mathrm{d}x - \fint_{B_{k,l}} f(x, \nabla v_\varepsilon(x))\,\mathrm{d}x \right| \le \omega(\varepsilon) \fint_{B_{k,l}} 1 + |\nabla v_\varepsilon(x)|^p\,\mathrm{d}x.$$

Then, we can combine all the above arguments to get

$$\begin{aligned}\left| \fint_{B_{k,l}} Qf(x, \nabla u(x)) \,\mathrm{d}x - \fint_{B_{k,l}} f(x, \nabla v_\varepsilon(x)) \,\mathrm{d}x \right| \\ \le C\omega(\varepsilon) \fint_{B_{k,l}} 2 + |A_k|^p + |\nabla v_\varepsilon(x)|^p \,\mathrm{d}x + \varepsilon.\end{aligned}$$

Multiplying this by $|B_{k,l}|$ and summing over all k, l, we get

$$\big|\mathscr{G}[u] - \mathscr{F}[v_\varepsilon]\big| \le C\omega(\varepsilon) \int_\Omega 2 + |\nabla u(x)|^p + |\nabla v_\varepsilon(x)|^p \,\mathrm{d}x + \varepsilon|\Omega|$$

and this vanishes as $\varepsilon \downarrow 0$. For $u_j := v_{1/j}$ we can convince ourselves that $u_j \rightharpoonup u$ since (u_j) is weakly relatively compact in $\mathrm{W}^{1,p}$ and $\|u_j - u\|_{\mathrm{L}^\infty} \le 1/j \to 0$. Hence,

$$\mathscr{G}[u] = \lim_{j\to\infty} \mathscr{F}[u_j] \ge \mathscr{F}_*[u],$$

which is (II) for piecewise affine u. □

5.3 Young measure relaxation

As discussed at the beginning of this chapter, the relaxation strategy as outlined in the preceding section has one serious drawback: While it allows to find the infimal value, the relaxed functional does not tell us anything about the structure (e.g. oscillations) of minimizing sequences. Often, however, in applications this structure of minimizing sequences is decisive as it represents the development of **microstructure**. For example, in material science, this microstructure can often be observed and greatly influences material properties. Therefore, in this section we implement the second strategy mentioned at the beginning of the chapter, that is, we extend the minimization problem to a larger space and look for solutions there.

Let us first, in an abstract fashion, collect a few properties that our extension should satisfy. Assume we are given a space X together with a notion of convergence "$\to$" and a functional $\mathscr{F}: X \to \mathbb{R} \cup \{+\infty\}$, which is not lower semicontinuous with respect to the convergence "$\to$" in X. Then, we need to extend X to a larger space $\overline{X}$ with a convergence "$\rightsquigarrow$" and $\mathscr{F}$ to a functional $\overline{\mathscr{F}}: \overline{X} \to \mathbb{R} \cup \{+\infty\}$, called the **extension–relaxation** such that the following conditions are satisfied:

(i) **Extension property:** $X \subset \overline{X}$ (in the sense of an embedding) and, using this embedding, $\overline{\mathscr{F}}|_X = \mathscr{F}$.

(ii) **Lower bound:** If $x_j \to x$ in X, then, up to a subsequence, there exists $\overline{x} \in \overline{X}$ such that $x_j \rightsquigarrow \overline{x}$ in $\overline{X}$ and

$$\liminf_{j\to\infty} \mathscr{F}[x_j] \ge \overline{\mathscr{F}}[\overline{x}].$$

(iii) **Recovery sequence:** For all $\overline{x} \in \overline{X}$ there exists a **recovery sequence** $(x_j) \subset X$ with $x_j \rightsquigarrow \overline{x}$ in $\overline{X}$ and such that

$$\lim_{j\to\infty} \mathscr{F}[x_j] = \overline{\mathscr{F}}[\overline{x}].$$

Intuitively, (ii) entails that we can solve our minimization problem for $\mathscr{F}$ by passing to the extended space; in particular, if $(x_j) \subset X$ is minimizing and relatively compact (which, up to a subsequence, implies $x_j \to x$ in X), then (ii) tells us that $\overline{x}$ should be considered the "minimizer". On the other hand, (iii) ensures that the minimization problems for $\mathscr{F}$ and for $\overline{\mathscr{F}}$ are sufficiently related. In particular, it is easy to see that the infima agree and indeed $\overline{\mathscr{F}}$ attains its minimum (for this, we need to assume some coercivity of $\mathscr{F}$):

$$\min_{\overline{X}} \overline{\mathscr{F}} = \inf_{X} \mathscr{F}.$$

If we consider integral functionals defined on $X = \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, we have already seen a very convenient extended space $\overline{X}$, namely the space of gradient p-Young measures. In fact, these generalized variational problems are the original *raison d'être* for Young measures. In the following we will show how this fits into the above abstract framework. This, we return to our prototypical integral functional

$$\mathscr{F}[u] := \int_\Omega f(x,\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m),$$

with $f\colon \Omega \times \mathbb{R}^{m\times d} \to \mathbb{R}$ Carathéodory and

$$\mu|A|^p - \mu^{-1} \le f(x,A) \le M(1+|A|^p), \qquad (x,A) \in \Omega \times \mathbb{R}^{m\times d}, \tag{5.9}$$

for some $\mu, M > 0$, $1 < p < \infty$, and $\Omega \subset \mathbb{R}^d$ a bounded Lipschitz domain.

Motivated by the considerations in Sections 3.3 and 3.4, we let for a gradient Young measure $\nu = (\nu_x) \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ the **extension–relaxation** $\overline{\mathscr{F}}\colon \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d}) \to \mathbb{R}$ be defined as

$$\overline{\mathscr{F}}[\nu] := \langle\!\langle f, \nu \rangle\!\rangle = \int_\Omega \int f(x,A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x.$$

Then, our relaxation result takes the following form:

Theorem 5.5 (Relaxation with Young measures). *Let $\mathscr{F}, \overline{\mathscr{F}}$ be as above.*

(I) ***Extension property:*** *For every $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ define the* elementary Young measure *$(\delta_{\nabla u}) \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ as*

$$(\delta_{\nabla u})_x := \delta_{\nabla u(x)} \qquad \text{for a.e. } x \in \Omega.$$

Then, $\mathscr{F}[u] = \overline{\mathscr{F}}[\delta_{\nabla u}]$.

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(II) ***Lower bound:*** *If* $u_j \rightharpoonup u$ *in* $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$*, then, up to a subsequence, there exists a Young measure* $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ *such that* $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$ *(i.e.* (∇u_j) *generates* ν*) with* $[\nu] = \nabla u$ *a.e., and*

$$\liminf_{j\to\infty} \mathscr{F}[u_j] \geq \overline{\mathscr{F}}[\nu]. \tag{5.10}$$

(III) ***Recovery sequence:*** *For all* $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ *there exists a recovery generating sequence* $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ *with* $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$ *and such that*

$$\lim_{j\to\infty} \mathscr{F}[u_j] = \overline{\mathscr{F}}[\nu]. \tag{5.11}$$

(IV) ***Equality of minima:*** $\overline{\mathscr{F}}$ *attains its minimum and*

$$\min_{\mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})} \overline{\mathscr{F}} = \inf_{\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)} \mathscr{F}.$$

Furthermore, all these statements remain true if we prescribe boundary values. For a Young measure we here prescribe the boundary values for an underlying deformation (which is only determined up to a translation, of course).

Proof. *Ad (I).* This follows directly from the definition of $\overline{\mathscr{F}}$.

Ad (II). The existence of ν follows from the Fundamental Theorem 3.11. The lower bound (5.10) is a consequence of Proposition 3.19.

Ad (III). By virtue of Lemma 3.21 we may construct an (improved) sequence $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ such that

(i) $u_j|_{\partial\Omega} = u$, where $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ is an underlying deformation of ν, that is, $\nabla u = [\nu]$,

(ii) the family $\{\nabla u_j\}_j$ is p-equiintegrable, and

(iii) $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$.

For this improved generating sequence we can now use the statement about representation of limits of integral functionals from the Fundamental Theorem 3.11 (here we need the p-equiintegrability) to get

$$\mathscr{F}[u_j] \to \langle\!\langle f, \nu \rangle\!\rangle = \overline{\mathscr{F}}[\nu],$$

which is nothing else than (5.11).

Ad (IV). This is not hard to see using (II), (III) and the coercivity assumption, that is, the lower bound in (5.9). □

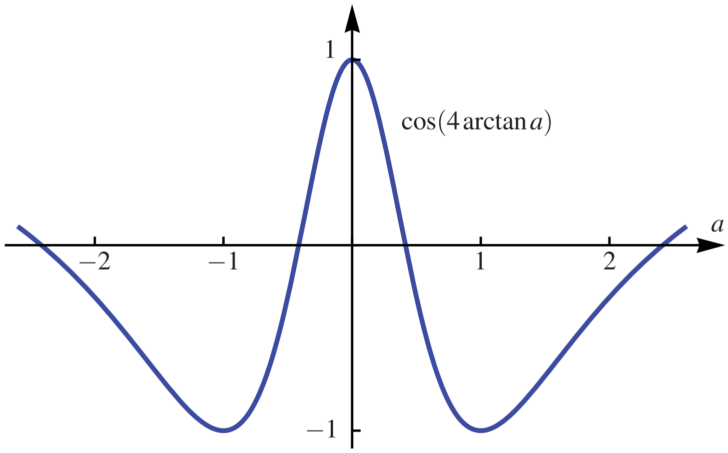

Figure 5.2: The double-well potential in the sailing example.

Example 5.6. In our sailing example from Section 1.6, we were tasked with solving

$$\begin{cases} \mathscr{F}[r] := \displaystyle\int_0^T \cos(4 \arctan r'(t)) - v_{\max}\left(1 - \frac{r^2}{R^2}\right) \mathrm{d}t \quad \to \quad \min, \\ r(0) = r(T) = 0, \qquad |r(t)| \leq R. \end{cases}$$

Here, because of the additional constraints we can work in any L^p-space that we want, even L^∞. Clearly, the integrand

$$W(r,a) := \cos(4 \arctan a) - v_{\max}\left(1 - \frac{r^2}{R^2}\right), \qquad |r(t)| \leq R,$$

is not convex in a, see Figure 5.2. Technically, the preceding theorem is not applicable, since W depends on r and a and $p = \infty$, but it is obvious that we can simply extend it to consider Young measures $(\delta_{r(x)} \otimes \nu_x)_x \in \mathbf{Y}^\infty(\Omega; \mathbb{R} \times \mathbb{R})$ with $\nu \in \mathbf{GY}^\infty(\Omega; \mathbb{R})$ and $[\nu] = r'$ as in Section 3.7 (we omit the details for $p = \infty$). We collect all such product Young measures in the set $\mathbf{GY}^\infty(\Omega; \mathbb{R} \times \mathbb{R})$.

Then, we consider the extended–relaxed variational problem

$$\begin{cases} \overline{\mathscr{F}}[\delta_r \otimes \nu] := \langle\!\langle W, \delta_r \otimes \nu \rangle\!\rangle = \displaystyle\int_0^T W(r(t), a) \, \mathrm{d}\nu_x(a) \, \mathrm{d}x \quad \to \quad \min, \\ \delta_r \otimes \nu = (\delta_{r(x)} \otimes \nu_x)_x \in \mathbf{GY}^\infty(\Omega; \mathbb{R} \times \mathbb{R}), \\ r(0) = r(T) = 0, \qquad |r(t)| \leq R. \end{cases}$$

Let us also construct a sequence of solutions that generate the optimal Young measure solution. Some elementary analysis yields that $g(a) := \cos(4 \arctan a)$ has two minima in $a = \pm 1$, where $g(a) = -1$ (see Figure 5.2). Moreover, $h(r) := -v_{\max}(1 - r^2/R^2)$ attains its minimum $-v_{\max}$ for $r = 0$. Thus,

$$W \geq -\big(1 + v_{\max}\big) =: W_{\min}.$$

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We let

$$h(s) := \begin{cases} s-1 & \text{if } s \in [0,2], \\ 3-s & \text{if } s \in (2,4], \end{cases} \qquad s \in [0,4],$$

and consider h to be extended to all $s \in \mathbb{R}$ by periodicity. Then set

$$r_j(t) := \frac{T}{4j} h\left(\frac{4j}{T}t + 1\right).$$

It is easy to see that $r_j' \in \{-1,+1\}$ and $r_j \to 0$ uniformly. Thus,

$$\mathscr{F}[r_j] \to T \cdot W_{\min} = \inf \mathscr{F}.$$

By the Fundamental Theorem 3.11 on Young measures, we know that we may select a subsequence of j's (not renumbered) such that (see Lemma 3.28)

$$(r_j, r_j') \overset{\mathbf{Y}}{\to} \delta_r \otimes \nu = (\delta_{r(x)} \otimes \nu_x)_x \in \mathbf{GY}^\infty(\Omega; \mathbb{R} \times \mathbb{R}).$$

From the construction of r_j we see

$$\delta_r \otimes \nu = \delta_0 \otimes \left(\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_{+1}\right).$$

Clearly, this $\delta_r \otimes \nu$ is minimizing for $\overline{\mathscr{F}}$.

The following lemma creates a connection to the other relaxation approach from the last section:

Lemma 5.7. *Let $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be continuous and satisfy*

$$\mu|A|^p - \mu^{-1} \le h(A) \le M(1+|A|^p) \qquad A \in \mathbb{R}^{m\times d},$$

for some $\mu, M > 0$. Then, for all $A \in \mathbb{R}^{m\times d}$ there exists a homogeneous gradient Young measure $\nu_A \in \mathbf{GY}^p(B(0,1); \mathbb{R}^{m\times d})$ such that

$$Qh(A) = \int h \,\mathrm{d}\nu_A.$$

Proof. According to (5.1) and the remarks following it,

$$Qh(A) := \inf\left\{ \fint_{B(0,1)} h(A + \nabla\psi(z)) \,\mathrm{d}z \ \textbf{:}\ \psi \in \mathrm{W}_0^{1,p}(B(0,1); \mathbb{R}^m) \right\}.$$

Let now $(\psi_j) \subset \mathrm{W}^{1,p}(B(0,1); \mathbb{R}^m)$ be a minimizing sequence in the above formula. This minimization problem is admissible in Theorem 5.5, which yields a Young measure minimizer $\nu \in \mathbf{GY}^p(B(0,1); \mathbb{R}^m)$ such that

$$Qh(A) = \fint_{B(0,1)} \int h \,\mathrm{d}\nu_x \,\mathrm{d}x.$$

It only remains to show that we can replace (ν_x) by a *homogeneous* Young measure ν_A. This follows from the following lemma. $\square$

Lemma 5.8 (Homogenization). *Let $\nu = (\nu_x) \in \mathbf{GY}^p(\Omega;\mathbb{R}^m)$ such that $[\nu] = \nabla u$ for some $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with linear boundary values. Then, for any Lipschitz domain $D \subset \mathbb{R}^d$ there exists a homogeneous $\overline{\nu} = (\overline{\nu}_y) \in \mathbf{GY}^p(D;\mathbb{R}^m)$ such that*

$$\int h \,\mathrm{d}\overline{\nu} = \frac{1}{|D|} \int_\Omega \int h \,\mathrm{d}\nu_x \,\mathrm{d}x \tag{5.12}$$

for all continuous $h \colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with $|h(A)| \le M(1+|A|^p)$. Furthermore, this result remains valid if $\Omega = Q^d = (-1/2,1/2)^d$, the d-dimensional unit cube and u has periodic boundary values.

Proof. Let $(u_j) \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $u_j(x) = Mx$ on $\partial\Omega$ (in the sense of trace) for a fixed matrix $M \in \mathbb{R}^{m\times d}$ such that $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$, in particular $\sup_j \|\nabla u_j\|_{\mathrm{L}^p} < \infty$. Now choose for every $j \in \mathbb{N}$ a Vitali cover of D, see Theorem A.8,

$$D = Z \uplus \biguplus_{k=1}^{N(j)} \Omega(a_k^{(j)}, r_k^{(j)}), \qquad |Z| = 0,$$

with $a_k^{(j)} \in D$, $r_k^{(j)} \le 1/j$ ($k = 1,\ldots,N(j)$), and $\Omega(a,r) := a + r\Omega$. Then define

$$v_j(y) := \begin{cases} r_k^{(j)} u_j\left(\dfrac{y - a_k^{(j)}}{r_k^{(j)}}\right) + M a_k^{(j)} & \text{if } y \in \Omega(a_k^{(j)}, r_k^{(j)}), \\ 0 & \text{otherwise,} \end{cases} \qquad y \in D.$$

We have $v_j \in \mathrm{W}^{1,p}(D;\mathbb{R}^m)$ (it is easy to see that there are no jumps over the gluing boundaries) and

$$\nabla v_j(y) = \nabla u_j\left(\frac{y - a_k^{(j)}}{r_k^{(j)}}\right) \qquad \text{if } y \in \Omega(a_k^{(j)}, r_k^{(j)}).$$

We can then use a change of variables to compute for all $\varphi \in \mathrm{C}_0(D)$ and $h \in \mathrm{C}(\mathbb{R}^{m\times d})$ with p-growth,

$$\begin{aligned} \int_D \varphi(y) h(\nabla v_j(y)) \,\mathrm{d}y &= \sum_{k=1}^{N(j)} \int_{\Omega(a_k^{(j)}, r_k^{(j)})} \varphi(y)\, h\left(\nabla u_j\left(\frac{y - a_k^{(j)}}{r_k^{(j)}}\right)\right) \mathrm{d}y \\ &= \sum_{k=1}^{N(j)} (r_k^{(j)})^d \varphi(a_k^{(j)}) \int_\Omega h(\nabla u_j(x)) \,\mathrm{d}x + \mathrm{O}\left(\frac{1}{j}\right)|D| \end{aligned}$$

where we also used that φ is uniformly continuous. Letting $j \to \infty$ and using that (Riemann sums)

$$\lim_{j\to\infty} \sum_{k=1}^{N(j)} (r_k^{(j)})^d \varphi(a_k^{(j)}) = \fint_D \varphi(x) \,\mathrm{d}x,$$

we arrive at

$$\lim_{j\to\infty} \int_D \varphi(y)h(\nabla v_j(y))\,\mathrm{d}y = \fint_D \varphi(x)\,\mathrm{d}x \cdot \int_\Omega \int h(A)\,\mathrm{d}\nu_x\,\mathrm{d}x. \tag{5.13}$$

For $\varphi = \mathbb{1}$ and $h(A) := |A|^p$, this gives

$$\sup_j \|\nabla v_j\|_{\mathrm{L}^p}^p = \sup_j \|\nabla u_j\|_{\mathrm{L}^p}^p < \infty.$$

Thus, there exists $\overline{\nu} \in \mathbf{GY}^p(D;\mathbb{R}^m)$ such that $\nabla v_j \xrightarrow{\mathbf{Y}} \overline{\nu}$.

Using Lemma 3.21 we may moreover assume that (∇u_j), and hence also (∇v_j), is p-equiintegrable. Then, (5.13) implies

$$\int_D \int \varphi(y)h(A)\,\mathrm{d}\overline{\nu}_y\,\mathrm{d}y = \fint_D \varphi(x)\,\mathrm{d}x \cdot \int_\Omega \int h(A)\,\mathrm{d}\nu_x\,\mathrm{d}x$$

for all φ, h as above. This implies in particular that $(\overline{\nu}_y) = \overline{\nu}$ is homogeneous. For $\varphi = \mathbb{1}$, we get (5.12).

The additional claim about $\Omega = Q^d$ and an underlying deformation u with periodic boundary values follows analogously, since also in this case we can "glue" generating functions on a (special) covering. □

5.4 Characterization of gradient Young measures

In the previous section we replaced a minimization problem over a Sobolev space by its extension–relaxation, defined on the space of gradient Young measures. Unfortunately, this subset of the space of Young measures is so far defined only "extrinsically", i.e. through the existence of a generating sequence of gradients. The question arises whether there is also an "intrinsic" characterization of gradient Young measures. Intuitively, trying to understand gradient Young measures amounts to understanding the "structure" of sequences of gradients, which is a useful endeavor.

Recall that in Lemma 3.22 we showed that a homogeneous gradient Young measure $\nu \in \mathbf{GY}^p(B(0,1);\mathbb{R}^{m\times d})$ with barycenter $B := [\nu] \in \mathbb{R}^{m\times d}$ satisfies the Jensen-type inequality

$$h(B) \le \int h\,\mathrm{d}\nu$$

for all *quasiconvex* functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with $|h(A)| \le M(1+|A|^p)$ ($M > 0$). There, we interpreted this as an expression of the (generalized) convexity of h, in analogy with the classical Jensen-inequality.

However, we can also dualize the situation and see the validity of the above Jensen-type inequality for quasiconvex functions as a property of gradient Young measures. The following (perhaps surprising) result shows that this dual point of view is indeed valid and the Jensen-type inequality (essentially) characterizes gradient Young measures, a result due to David Kinderlehrer and Pablo Pedregal from the early 1990s:

Theorem 5.9 (Kinderlehrer–Pedregal). *Let $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^{m\times d})$ be a Young measure such that $[\nu] = \nabla u$ for some $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ and such that for almost every $x \in \Omega$ the Jensen-type inequality*

$$h(\nabla u(x)) \le \int h \,\mathrm{d}\nu_x$$

holds for all quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with $|h(A)| \le M(1+|A|^p)$ $(M > 0)$. Then $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, i.e. ν is a gradient p-Young measure.

The idea of the proof is to show that the set of gradient Young measures is *convex* and (weakly*) closed in the set of Young measures and the Jensen-type inequalities entail that ν cannot be separated from this set by a hyperplane (which turn out to be in one-to-one correspondence with integrands). Then, the Hahn–Banach Theorem implies that ν actually lies in this set and is hence a gradient Young measure. One first shows this for homogeneous Young measures and then uses a gluing procedure to extend the result to x-dependent $\nu = (\nu_x)$. The details, however, are quite technical, see [81] or the original works [49,51] for a full proof.

This theorem is conceptually very important. It entails that if we could understand the class of quasiconvex functions, then we also could understand gradient Young measures and thus sequences of gradients. Unfortunately, our knowledge of quasiconvex functions (and hence of gradient Young measures) is very limited at present. Most results are for specific situations only and much of it has an "ad-hoc" flavor.

We close this chapter with a result showing that quasiconvex functions are not easy to understand. In Proposition 3.3 we proved that quasiconvexity implies rank-one convexity and Proposition 4.1 showed that polyconvexity implies quasiconvexity. Hence, in a sense, rank-one convexity is a lower bound and polyconvexity an upper bound on quasiconvexity. Moreover, the Alibert–Dacorogna–Marcellini function (see Examples 3.5, 4.2)

$$h_\gamma(A) := |A|^2\big(|A|^2 - 2\gamma \det A\big), \qquad A \in \mathbb{R}^{2\times 2},\ \gamma \in \mathbb{R}.$$

has the following convexity properties:

- h_γ is convex if and only if $|\gamma| \le \dfrac{2\sqrt{2}}{3} \approx 0.94$,
- h_γ is rank-one convex if and only if $|\gamma| \le \dfrac{2}{\sqrt{3}} \approx 1.15$,
- h_γ is quasiconvex if and only if $|\gamma| \le \gamma_{\mathrm{QC}}$ for some $\gamma_{\mathrm{QC}} \in \left(1, \dfrac{2}{\sqrt{3}}\right]$,
- h_γ is polyconvex if and only if $|\gamma| \le 1$.

However, it is open whether $\gamma_{\mathrm{QC}} = \frac{2}{\sqrt{3}}$ and so it could be that quasiconvexity and rank-one convexity are actually equivalent. This was open for a long time until Vladimir Šverák's 1992 counterexample [89], which settled the question in the negative, at least for $d \ge 2$,

$m \geq 3$. The case $d = m = 2$ is still open and considered to be very important, also because of its connection to other branches of mathematics.

Example 5.10 (Šverák). We will show the non-equivalence of quasiconvexity and rank-one convexity for $m = 3$, $d = 2$ only. Higher dimensions can be treated using an embedding of $\mathbb{R}^{3\times 2}$ into $\mathbb{R}^{m\times d}$. To accomplish this, we construct $h\colon \mathbb{R}^{3\times 2} \to \mathbb{R}$ that is quasiconvex, but not rank-one convex. Define a linear subspace L of $\mathbb{R}^{3\times 2}$ as follows:

$$L := \left\{ \begin{pmatrix} x & 0 \\ 0 & y \\ z & z \end{pmatrix} \;:\; x,y,z \in \mathbb{R} \right\}.$$

We denote by $P\colon \mathbb{R}^{3\times 2} \to L$ a linear projection onto L given as

$$P(A) := \begin{pmatrix} a & 0 \\ 0 & d \\ (e+f)/2 & (e+f)/2 \end{pmatrix} \qquad \text{for} \qquad A = \begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix} \in \mathbb{R}^{3\times 2}.$$

Also define $g\colon L \to \mathbb{R}$ to be

$$g\begin{pmatrix} x & 0 \\ 0 & y \\ z & z \end{pmatrix} := -xyz.$$

For $\alpha, \beta > 0$, we let $h_{\alpha,\beta}\colon \mathbb{R}^{3\times 2} \to \mathbb{R}$ be given as

$$h_{\alpha,\beta}(A) := g(P(A)) + \alpha\big(|A|^2 + |A|^4\big) + \beta|A - P(A)|^2.$$

We show the following two properties of $h_{\alpha,\beta}$:

(I) For every $\alpha > 0$ sufficiently small and all $\beta > 0$, the function $h_{\alpha,\beta}$ is *not* quasiconvex.

(II) For every $\alpha > 0$, there exists $\beta = \beta(\alpha) > 0$ such that $h_{\alpha,\beta}$ is rank-one convex.

Ad (I): For $A = 0$, $D = (0,1)^2$, and a periodic $\psi \in \mathrm{W}^{1,\infty}(D;\mathbb{R}^3)$ given as

$$\psi(x_1,x_2) := \frac{1}{2\pi}\begin{pmatrix} \sin 2\pi x_1 \\ \sin 2\pi x_2 \\ \sin 2\pi(x_1 + x_2) \end{pmatrix},$$

we have $\nabla\psi \in L$ and so $P(\nabla\psi) = \nabla\psi$. Thus, we may compute (using the identity for the cosine of a sum and the fact that the sine is an odd function)

$$\int_D g(\nabla\psi)\,\mathrm{d}x = -\int_0^1\int_0^1 (\cos 2\pi x_1)^2(\cos 2\pi x_2)^2\,\mathrm{d}x_1\,\mathrm{d}x_2 = -\frac{1}{4} < 0.$$

Thus, for $\alpha > 0$ sufficiently small,

$$\int_D h_{\alpha,\beta}(\nabla\psi)\,\mathrm{d}x < 0 = h_{\alpha,\beta}(0).$$

Comment...

It turns out that in the definition of quasiconvexity we may alternatively test with functions that have merely periodic boundary values on D (see Proposition 5.13 in [25]). Hence, (I) follows.

Ad (II): Rank-one convexity is equivalent to the **Legendre–Hadamard condition**

$$\mathrm{D}^2 h_{\alpha,\beta}(A)[B,B] := \frac{\mathrm{d}^2}{\mathrm{d}t^2} h_{\alpha,\beta}(A+tB)\bigg|_{t=0} \geq 0 \qquad \text{for all } A,B \in \mathbb{R}^{3\times 2},\ \operatorname{rank} B \leq 1.$$

Here, $\mathrm{D}^2 h_{\alpha,\beta}(A)[B,B]$ is the second (Gâteaux-)derivative of $h_{\alpha,\beta}$ at A in direction B.

The function g is a homogeneous polynomial of degree three, whereby we can find $c > 0$ such that for all $A,B \in \mathbb{R}^{3\times 2}$ with $\operatorname{rank} B \leq 1$,

$$G(A,B) := \mathrm{D}^2(g \circ P)(A)[B,B] := \frac{\mathrm{d}^2}{\mathrm{d}t^2} g(P(A+tB))\bigg|_{t=0} \geq -c|A||B|^2.$$

A computation then shows that

$$\begin{aligned}\mathrm{D}^2 h_{\alpha,\beta}(A)[B,B] &= G(A,B) + 2\alpha|B|^2 + 4\alpha|A|^2|B|^2 + 8\alpha(A:B)^2 + 2\beta|B-P(B)|^2 \\ &\geq (-c + 4\alpha|A|)|A||B|^2.\end{aligned}$$

where we used the Frobenius product $A : B := \sum_{i,j} A^i_j B^i_j$. Thus, for $|A| \geq c/(4\alpha)$, the Legendre–Hadamard condition (and hence rank-one convexity) holds.

We still need to prove the Legendre–Hadamard condition for $|A| < c/(4\alpha)$ and any fixed $\alpha > 0$. Since $\mathrm{D}^2 h_{\alpha,\beta}(A)[B,B]$ is homogeneous of degree two in B, we only need to consider A,B from the compact set

$$K := \left\{ (A,B) \in \mathbb{R}^{3\times 2} \times \mathbb{R}^{3\times 2} \ \mathbf{:}\ |A| \leq \frac{c}{4\alpha},\ |B| = 1,\ \operatorname{rank} B = 1 \right\}.$$

From the estimate

$$\mathrm{D}^2 h_{\alpha,\beta}(A)[B,B] \geq G(A,B) + 2\alpha|B|^2 + 2\beta|B-P(B)|^2 =: \kappa(A,B,\beta)$$

it suffices to show that there exists $\beta = \beta(\alpha) > 0$ such that $\kappa(A,B,\beta) \geq 0$ for all $(A,B) \in K$. Assume that this is not the case. Then there exists $\beta_j \to \infty$ and $(A_j,B_j) \in K$ with

$$0 > \kappa(A_j,B_j,\beta_j) = G(A_j,B_j) + 2\alpha + 2\beta|B_j - P(B_j)|^2.$$

As K is compact, we may assume $(A_j,B_j) \to (A,B) \in K$, for which it holds that

$$G(A,B) + 2\alpha \leq 0, \qquad P(B) = B, \qquad \text{and} \qquad \operatorname{rank} B = 1.$$

However, since $\operatorname{rank} B = 1$, we have $g(P(A+tB)) = 0$ for all $t \in \mathbb{R}$. This implies in particular $G(A,B) = \mathrm{D}^2(g\circ P)(A)[B,B] = 0$, yielding $2\alpha \leq 0$, which contradicts our assumption $\alpha > 0$. Thus, we have shown that for a suitable choice of $\alpha, \beta > 0$ it indeed holds that $h_{\alpha,\beta}$ is rank-one convex.

So, quasiconvexity and, dually, gradient Young measures, must remain somewhat mysterious concepts. This also shows that the theory of elliptic systems of PDEs, for which traditionally the Legendre–Hadamard condition is the expression of "ellipticity", is incomplete. In particular, non-variational systems (i.e. systems of PDEs which do not arise as the Euler–Lagrange equations to a variational problem) are only poorly understood.

5.5 Notes and bibliographical remarks

Classically, one often defines the quasiconvex envelope through (5.2) and not as we have done through (5.1). In this case, (5.2) is often called the **Dacorogna formula**, see Section 6.3 in [25] for further references.

The construction of Lemma 5.2 is based on a construction of Šverák [88] and some ellipticity arguments similar to those in Lemma 2.7 of [73].

Laurence Chisholm Young originally introduced the objects that are now called Young measures as "generalized curves/surfaces" in the late 1930s and early 1940s [97–99] to treat problems in the Calculus of Variations and Optimal Control Theory that could not be solved using classical methods. At the end of the 1960s he also wrote a book [100] (2nd edition [101]) on the topic, which explained these objects and their applications in great detail (including philosophical discussions on the semantics of the terminology "generalized curve/surface").

More recently, starting from the seminal work of Ball & James in 1987 [10], Young measures have provided a convenient framework to describe fine phase mixtures in the theory of microstructure, also see [33, 73] and the references cited therein. Further, and more related to the present work, relaxation formulas involving Young measures for non-convex integral functionals have been developed and can for example be found in Pedregal's book [81] and also in Chapter 11 of the recent text [3]; historically, Dacorogna's lecture notes [24] were also influential.

A second avenue of development – somewhat different from Young's original intention – is to use Young measures only as a *technical tool* (that is without them appearing in the final statement of a result). This approach is in fact quite old and was probably first adopted in a series of articles by McShane from the 1940s [62–64]. There, the author first finds a generalized Young measure solution to a variational problem, then proves additional properties of the obtained Young measures, and finally concludes that these properties entail that the generalized solution is in fact classical. This idea exhibits some parallels to the hugely successful approach of first proving existence of a weak solution to a PDE and then showing that this weak solution has (in some cases) additional regularity.

The Kinderlehrer–Pedregal Theorem 5.9 also holds for $p = \infty$, which was in fact the first case to be established. See [81] for an overview and proof.

A different proof of Proposition 5.2 can be found in Section 5.3.9 of [25].

The conditions (i)–(iii) at the beginning of Section 5.3 are modeled on the concept of Γ-convergence (introduced by De Giorgi) and are also the basis of the usual treatment of parameter-dependent integral functionals. See [18] for an introduction and [27] for a thorough treatise.

Appendix A

Prerequisites

This appendix recalls some facts that are needed throughout the course.

A.1 General facts and notation

First, recall the following version of the fundamental **Young inequality**

$$ab \leq \frac{\delta}{2}a^2 + \frac{1}{2\delta}b^2,$$

which holds for all $a, b \geq 0$ and $\delta > 0$.

In this text (as in much of the literature), the matrix space $\mathbb{R}^{m\times d}$ of $(m \times d)$-matrices always comes equipped with the the **Frobenius matrix (inner) product**

$$A : B := \sum_{j,k} A^j_k B^j_k, \qquad A, B \in \mathbb{R}^{m\times d},$$

which is just the Euclidean product if we identify such matrices with vectors in $\mathbb{R}^{md}$. This inner product induces the **Frobenius matrix norm**

$$|A| := \sqrt{\sum_{j,k} (A^j_k)^2}, \qquad A \in \mathbb{R}^{m\times d}.$$

Very often we will use the **tensor product** of vectors $a \in \mathbb{R}^m$, $b \in \mathbb{R}^d$,

$$a \otimes b := ab^T \quad \in \mathbb{R}^{m\times d}.$$

The tensor product interacts well with the Frobenius norm,

$$|a \otimes b| = |a| \cdot |b|, \qquad a \in \mathbb{R}^m,\ b \in \mathbb{R}^d.$$

While all norms on the finite-dimensional space $\mathbb{R}^{m\times d}$ are equivalent, some "finer" arguments require to specify a matrix norm; if nothing else is stated, we always use the Frobenius norm.

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It is proved in texts on Matrix Analysis, see for instance [48], that this Frobenius norm can also be expressed as

$$|A| = \sqrt{\sum_i \sigma_i(A)^2}, \qquad A \in \mathbb{R}^{m\times d},$$

where $\sigma_i(A) \geq 0$ is the i'th singular value of A, where $i = 1,\dots,\min(d,m)$. Recall that every matrix $A \in \mathbb{R}^{m\times d}$ has a **(real) singular value decomposition**

$$A = P\Sigma Q^T$$

for orthogonal $P \in \mathbb{R}^{m\times m}$, $Q \in \mathbb{R}^{d\times d}$, and a *diagonal* $\Sigma = \operatorname{diag}(\sigma_1,\dots,\sigma_{\min(d,m)}) \in \mathbb{R}^{m\times d}$ with only positive entries

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0 = \sigma_{r+1} = \sigma_{r+2} = \cdots = \sigma_{\min(d,m)},$$

called the **singular values** of A, where r is the rank of A.

For $A \in \mathbb{R}^{d\times d}$ the **cofactor matrix** $\operatorname{cof} A \in \mathbb{R}^{d\times d}$ is the matrix whose (j,k)'th entry is $(-1)^{j+k} M^{\neg j}_{\neg k}(A)$ with $M^{\neg j}_{\neg k}(A)$ being the (j,k)-minor of A, i.e. the determinant of the matrix that originates from A by deleting the j'th row and k'th column. One important formula (and in fact another way to define the cofactor matrix) is

$$\partial_A[\det A] = \operatorname{cof} A.$$

Furthermore, the **Cramer rule** entails that

$$A(\operatorname{cof} A)^T = \det A \cdot I \qquad \text{for all } A \in \mathbb{R}^{d\times d},$$

where I denotes the identity matrix as usual. In particular, if A is invertible,

$$A^{-1} = \frac{(\operatorname{cof} A)^T}{\det A}.$$

Sometimes, the matrix $(\operatorname{cof} A)^T$ is called the **adjugate matrix** to A in the literature (we will not use this terminology here, however).

A fundamental inequality involving the determinant is the **Hadamard inequality**: Let $A \in \mathbb{R}^{d\times d}$ with columns $A_j \in \mathbb{R}^n$ $(j = 1,\dots,d)$. Then,

$$|\det A| \leq \prod_{j=1}^{d} |A_j| \leq |A|^d.$$

Of course, an analogous formula holds with the rows of A.

Comment...

A.2 Measure theory

We assume that the reader is familiar with the notion of Lebesgue- and Borel-measurability, nullsets, and L^p-spaces; a good introduction is [85]. For the d-dimensional Lebesgue measure we write $\mathscr{L}^d$ (or $\mathscr{L}^d_x$ if we want to stress the integration variable), but the Lebesgue measure of a Borel- or Lebesgue-measurable set $A \subset \mathbb{R}^d$ is simply $|A|$. THe **characteristic function**f such an A is

$$\mathbb{1}_A(x) := \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{otherwise,} \end{cases} \qquad x \in \mathbb{R}^d.$$

In the following, we only recall some results that are needed elsewhere.

Lemma A.1 (Fatou). *Let $f_j \colon \mathbb{R}^N \to [0,+\infty]$, $j \in \mathbb{N}$, be a sequence of Lebesgue-measurable functions. Then,*

$$\liminf_{j\to\infty} \int f_j(x)\,\mathrm{d}x \geq \int \liminf_{j\to\infty} f_j(x)\,\mathrm{d}x.$$

Lemma A.2 (Monotone convergence). *Let $f_j \colon \mathbb{R}^N \to [0,+\infty]$, $j \in \mathbb{N}$, be a sequence of Lebesgue-measurable functions with $f_j(x) \uparrow f(x)$ for almost every $x \in \Omega$. Then, $f \colon \mathbb{R}^N \to [0,+\infty]$ is measurable and*

$$\lim_{j\to\infty} \int f_j(x)\,\mathrm{d}x = \int f(x)\,\mathrm{d}x.$$

Lemma A.3. *If $f_j \to f$ **strongly** in $\mathrm{L}^1(\mathbb{R}^d)$, $1 \leq p \leq \infty$, that is,*

$$\|f_j - f\|_{\mathrm{L}^p} \to 0 \quad \textit{as} \quad j \to \infty,$$

then there exists a subsequence (not renumbered[1]) such that $f_j \to f$ almost everywhere.

Theorem A.4 (Lebesgue dominated convergence theorem). *Let $f_j \colon \mathbb{R}^N \to \mathbb{R}^n$, $j \in \mathbb{N}$, be a sequence of Lebesgue-measurable functions such that there exists an L^p-integrable majorant $g \in \mathrm{L}^p(\mathbb{R}^N)$, $1 \leq p < \infty$, that is,*

$$|f_j| \leq g \qquad \textit{for all } j \in \mathbb{N}.$$

If $f_j \to f$ pointwise almost everywhere, then also $f_j \to f$ strongly in L^p.

The following strengthening of Lebesgue's theorem is very convenient:

Theorem A.5 (Pratt). *If $f_j \to f$ pointwise almost everywhere and there exists a sequence $(g_j) \subset \mathrm{L}^1(\mathbb{R}^N)$ with $g_j \to g$ in L^1 such that $|f_j| \leq g_j$, then $f_j \to f$ in L^1.*

The following convergence theorem is of fundamental significance:

[1] We generally do not choose different indices for subsequences if the original sequence is discarded.

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Theorem A.6 (Vitali convergence theorem). *Let $\Omega \subset \mathbb{R}^d$ be bounded and let $(f_j) \subset \mathrm{L}^p(\Omega;\mathbb{R}^m)$, $1 \leq p < \infty$. Assume furthermore that*

(i) ***No oscillations:*** *$f_j \to f$ in measure, that is, for all $\varepsilon > 0$,*

$$\left|\left\{ x \in \Omega \; : \; |f_j - f| > \varepsilon \right\}\right| \to 0 \quad as \quad j \to \infty.$$

(ii) ***No concentrations:*** *$\{f_j\}_j$ is p-equiintegrable, i.e.*

$$\lim_{K\uparrow\infty} \sup_j \int_{\{|f_j|>K\}} |f_j|^p \,\mathrm{d}x = 0.$$

Then, $f_j \to f$ strongly in L^p.

Lebesgue-integrable functions also have a very useful "continuity" property:

Theorem A.7. *Let $f \in \mathrm{L}^1(\mathbb{R}^d)$. Then, almost every $x_0 \in \mathbb{R}^d$ is a* ***Lebesgue point*** *of f, i.e.*

$$\lim_{r\downarrow 0} \fint_{B(x_0,r)} |f(x) - f(x_0)| \,\mathrm{d}x = 0,$$

where $B(x_0,r) \subset \mathbb{R}^d$ is the ball with center x_0 and radius $r > 0$ and $\fint_{B(x_0,r)} := \frac{1}{|B(x_0,r)|} \int_{B(x_0,r)}$.

The following *covering theorem* is a handy tool for several constructions, see Theorems 2.19 and 5.51 in [2] for a proof:

Theorem A.8 (Vitali covering theorem). *Let $B \subset \mathbb{R}^N$ be a bounded Borel set and let $K \subset \mathbb{R}^d$ be compact. Then, there exist $a_k \in B$, $r_k > 0$, where $k \in \mathbb{N}$, such that*

$$B = Z \uplus \biguplus_{k=1}^{N} K(a_k, r_k), \qquad K(a_k, r_k) = a_k + r_k K,$$

with $Z \subset B$ a Lebesgue-nullset, $|Z| = 0$, and "$\biguplus$" denoting that the sets in the union are disjoint.

We also need other measures than Lebesgue measure on subsets of $\mathbb{R}^N$ (this could be either $\mathbb{R}^d$ or a matrix space $\mathbb{R}^{m\times d}$, identified with $\mathbb{R}^{md}$). All of these abstract measures will be **Borel measures**, that is, they are defined on the **Borel σ-algebra** $\mathfrak{B}(\mathbb{R}^N)$ of $\mathbb{R}^N$. In this context, recall that the Borel-σ-algebra is the smallest σ-algebra that contains the open sets. All Borel measures defined on $\mathbb{R}^N$ that do not take the value $+\infty$ are collected in the set $\mathbf{M}(\mathbb{R}^N)$ of **(finite) Radon measures**, its subclass of **probability measures** is $\mathbf{M}^1(\mathbb{R}^N)$. We also use $\mathbf{M}(\Omega), \mathbf{M}^1(\Omega)$ for the subset of measures that only charge $\Omega \subset \mathbb{R}^N$. We define the **duality pairing**

$$\langle h, \mu \rangle := \int h(A) \,\mathrm{d}\mu(A)$$

for $h\colon \mathbb{R}^N \to \mathbb{R}^m$ and $\mu \in \mathbf{M}(\mathbb{R}^N)$, whenever this integral makes sense (we need at least h μ-measurable). The following notation is convenient for the **barycenter** of a finite Borel measure μ:

$$[\mu] := \langle \mathrm{id}, \mu \rangle = \int A \,\mathrm{d}\mu(A).$$

Probability measures and convex functions interact well:

Lemma A.9 (Jensen inequality). *For all probability measures $\mu \in \mathbf{M}^1(\mathbb{R}^N)$ and all convex $h\colon \mathbb{R}^N \to \mathbb{R}$ it holds that*

$$h([\mu]) = h\biggl(\int A \,\mathrm{d}\mu(A)\biggr) \le \int h(A) \,\mathrm{d}\mu(A) = \langle h, \mu \rangle.$$

There is an alternative, "dual", view on finite measures, expressed in the following important theorem:

Theorem A.10 (Riesz representation theorem). *The space of Radon measures $\mathbf{M}(\mathbb{R}^N)$ is isometrically isomorphic to the dual space $\mathrm{C}_0(\mathbb{R}^N)^*$ via the duality pairing*

$$\langle h, \mu \rangle = \int h \,\mathrm{d}\mu.$$

As an easy, often convenient, consequence, we can *define* measures through their **action** on $\mathrm{C}_0(\mathbb{R}^N)$. Note, however, that we then need to check the boundedness $|\langle h, \mu \rangle| \le \|h\|_\infty$.

If the element $\mu \in \mathrm{C}_0(\mathbb{R}^N)^*$ is additionally positive, that is, $\langle h, \mu \rangle \ge 0$ for $h \ge 0$, and normalized, that is, $\langle \mathbb{1}, \mu \rangle = 1$ (here, $\mathbb{1} \equiv 1$ on the whole space), then μ from the Riesz representation theorem is in fact a *probability measure*, $\mu \in \mathbf{M}^1(\mathbb{R}^N)$.

A.3 Functional analysis

We assume that the reader has a solid foundation in the basic notions of Functional Analysis such as Banach spaces and their duals, weak/weak* convergence (and topology[2]), reflexivity, and weak/weak* compactness. The application of $x^* \in X^*$ to $x \in X$ is often written via the **duality pairing** $\langle x, x^* \rangle = \langle x^*, x \rangle := x^*(x)$. We write $x_j \rightharpoonup x$ for weak convergence and $x_j \overset{*}{\rightharpoonup} x$ for weak* convergence. In this context recall that in Banach spaces topological weak compactness is equivalent to sequential weak compactness, this is the Eberlein–Šmulian Theorem. Also recall that the norm in a Banach space is lower semicontinuous with respect to weak convergence: If $x_j \rightharpoonup x$ in X, then $\|x\| \le \liminf_{j\to\infty} \|x_j\|$. One very thorough reference for most of this material is [23]. The following are just reminders:

Theorem A.11 (Weak compactness). *Let X be a reflexive Banach space. Then, norm-bounded sets in X are sequentially weakly relatively compact.*

[2]However, we here really only need convergence of sequences.

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Theorem A.12 (Sequential Banach–Alaoglu theorem). *Let X be a separable Banach space. Then, norm-bounded sets in the dual space X^* are weakly* sequentially relatively compact.*

Theorem A.13 (Dunford–Pettis). *Let $\Omega \subset \mathbb{R}^d$ be bounded. A bounded family $\{f_j\} \subset \mathrm{L}^1(\Omega)$ ($j \in \mathbb{N}$) is equiintegrable if and only if it is weakly relatively (sequentially) compact in $\mathrm{L}^1(\Omega)$.*

Finally, weak convergence can be "improved" to strong convergence in the following way:

Lemma A.14 (Mazur). *Let $x_j \rightharpoonup x$ in a Banach space X. Then, there exists a sequence $(y_j) \subset X$ of convex combinations,*

$$y_j = \sum_{n=j}^{N(j)} \theta_{j,n} x_n, \qquad \theta_{j,n} \in [0,1], \qquad \sum_{n=j}^{N(j)} \theta_{j,n} = 1.$$

such that $y_j \to x$ in X.

A.4 Sobolev and other function spaces spaces

We give a brief introduction to Sobolev spaces, see [59] or [36] for more detailed accounts and proofs.

Let $\Omega \subset \mathbb{R}^d$. As usual we denote by $\mathrm{C}(\Omega) = \mathrm{C}^0(\Omega), \mathrm{C}^k(\Omega)$, $k = 1,2,\ldots$, the spaces of continuous and k-times continuously differentiable functions. As norms in these spaces we have

$$\|u\|_{\mathrm{C}^k} := \sum_{|\alpha| \leq k} \|\partial^\alpha u\|_\infty, \qquad u \in \mathrm{C}^k(\Omega), \qquad k = 0,1,2,\ldots,$$

where $\|\cdot\|_\infty$ is the supremum norm. Here, the sum is over all **multi-indices** $\alpha \in (\mathbb{N} \cup \{0\})^d$ with $|\alpha| := \alpha_1 + \cdots + \alpha_d \leq k$ and

$$\partial^\alpha := \partial_1^{\alpha_1} \partial_2^{\alpha_2} \cdots \partial_d^{\alpha_d}$$

is the α-derivative operator. Similarly, we define $\mathrm{C}^\infty(\Omega)$ of infinitely-often differentiable functions (but this is not a Banach space anymore). A subscript "$_c$" indicates that all functions u in the respective function space (e.g. $\mathrm{C}_c^\infty(\Omega)$) must have their **support**

$$\operatorname{supp} u := \overline{\{x \in \Omega \,:\, u(x) \neq 0\}}$$

compactly contained in Ω (so, $\operatorname{supp} u \subset \Omega$ and $\operatorname{supp} u$ compact). For the compact containment of a A in an open set B we write $A \subset\subset B$, which means that $\overline{A} \subset B$. In this way, the previous condition could be written $\operatorname{supp} u \subset\subset \Omega$.

For $k \in \mathbb{N}$ a positive integer and $1 \leq p \leq \infty$, the **Sobolev space** $\mathrm{W}^{k,p}(\Omega)$ is defined to contain all functions $u \in \mathrm{L}^p(\Omega)$ such that the **weak derivative** $\partial^\alpha u$ exists in $\mathrm{L}^p(\Omega)$ for all

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multi-indices $\alpha \in (\mathbb{N} \cup \{0\})^d$ with $|\alpha| \le k$. This means that for each such α, there is a (unique) function $v_\alpha \in \mathrm{L}^p(\Omega)$ satisfying

$$\int v_\alpha \cdot \psi \, \mathrm{d}x = (-1)^{|\alpha|} \int u \cdot \partial^\alpha \psi \, \mathrm{d}x \qquad \text{for all } \psi \in \mathrm{C}_c^\infty(\Omega),$$

and we write $\partial^\alpha u$ for this v_α. The uniqueness follows from the Fundamental Lemma of the Calculus of Variations 2.21. Clearly, if $u \in \mathrm{C}^k(\Omega)$, then the weak derivative is the classical derivative. For $u \in \mathrm{W}^{1,p}(\Omega)$ we further define the **weak gradient** and **weak divergence**,

$$\nabla u := (\partial_1 u, \partial_2 u, \dots, \partial_d u), \qquad \operatorname{div} u := \partial_1 u + \partial_2 u + \dots + \partial_d u.$$

As norm in $\mathrm{W}^{k,p}(\Omega)$, $1 \le p < \infty$, we use

$$\|u\|_{\mathrm{W}^{k,p}} := \left(\sum_{|\alpha| \le k} \|\partial^\alpha u\|_{\mathrm{L}^p}^p \right)^{1/p}, \qquad u \in \mathrm{W}^{k,p}(\Omega).$$

For $p = \infty$, we set

$$\|u\|_{\mathrm{W}^{k,\infty}} := \max_{|\alpha| \le k} \|\partial^\alpha u\|_{\mathrm{L}^\infty}, \qquad u \in \mathrm{W}^{k,\infty}(\Omega).$$

Under these norms, the $\mathrm{W}^{k,p}(\Omega)$ become Banach spaces.

Concerning the boundary values of Sobolev functions, we have:

Theorem A.15 (Trace theorem). *For every domain $\Omega \subset \mathbb{R}^d$ with Lipschitz boundary and every $1 \le p \le \infty$, there exists a bounded linear* ***trace operator*** $\mathrm{tr}_\Omega \colon \mathrm{W}^{1,p}(\Omega) \to \mathrm{L}^p(\partial\Omega)$ *such that*

$$\operatorname{tr} \varphi = \varphi|_{\partial\Omega} \qquad \text{if } \varphi \in \mathrm{C}(\overline{\Omega}).$$

We write $\mathrm{tr}_\Omega u$ *simply as* $u|_{\partial\Omega}$. *Furthermore,* tr_Ω *is bounded and weakly continuous between* $\mathrm{W}^{1,p}(\Omega)$ *and* $\mathrm{L}^p(\partial\Omega)$.

For $1 < p < \infty$ denote the image of $\mathrm{W}^{1,p}(\Omega)$ under tr_Ω by $\mathrm{W}^{1-1/p,p}(\partial\Omega)$, called the **trace space** of $\mathrm{W}^{1,p}(\Omega)$. As a makeshift norm on $\mathrm{W}^{1-1/p,p}(\partial\Omega)$ one can use the $\mathrm{L}^p(\partial\Omega)$-norm, but $\mathrm{W}^{1-1/p,p}(\partial\Omega)$ is not closed under this norm. This can be remedied by introducing a more complicated norm involving "fractional" derivatives (hence the notation for $\mathrm{W}^{1-1/p,p}(\partial\Omega)$), under which this set becomes a Banach space.

We write $\mathrm{W}_0^{1,p}(\Omega)$ for the linear subspace of $\mathrm{W}^{1,p}(\Omega)$ consisting of all $\mathrm{W}^{1,p}$-functions with zero boundary values (in the sense of trace). More generally, we use $\mathrm{W}_g^{1,p}(\Omega)$ with $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega)$, for the affine subspace of all $\mathrm{W}^{1,p}$-functions with boundary trace g.

The following are some properties of Sobolev spaces, stated for simplicity only for the first-order space $\mathrm{W}^{1,p}(\Omega)$ with $\Omega \subset \mathbb{R}^d$ a bounded Lipschitz domain.

Theorem A.16 (Poincaré–Friedrichs inequality). *Let* $u \in \mathrm{W}^{1,p}(\Omega)$.

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(I) If $u|_{\partial\Omega} = 0$*, then*

$$\|u\|_{\mathrm{L}^p} \leq C\|\nabla u\|_{\mathrm{L}^p},$$

where $C = C(\Omega) > 0$ *is a constant. If* $1 < p < \infty$ *and* $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega)$*, then*

$$\|u\|_{\mathrm{L}^p} \leq C\big(\|\nabla u\|_{\mathrm{L}^p} + \|g\|_{\mathrm{W}^{1-1/p,p}}\big).$$

(II) Setting $[u]_\Omega := \fint_\Omega u \,\mathrm{d}x$*, it holds that*

$$\|u - [u]_\Omega\|_{\mathrm{L}^p} \leq C\|\nabla u\|_{\mathrm{L}^p},$$

where $C = C(\Omega) > 0$ *is a constant.*

Sobolev functions have higher integrability than originally assumed:

Theorem A.17 (Sobolev embedding). *Let* $u \in \mathrm{W}^{1,p}(\Omega)$.

(I) If $p < d$*, then* $u \in \mathrm{L}^{p^*}(\Omega)$*, where*

$$p^* = \frac{dp}{d-p}$$

and there is a constant $C = C(\Omega) > 0$ *such that* $\|u\|_{\mathrm{L}^{p^*}} \leq C\|u\|_{\mathrm{W}^{1,p}}$.

(II) If $p = d$*, then* $u \in \mathrm{L}^q(\Omega)$ *for all* $1 \leq q < \infty$ *and* $\|u\|_{\mathrm{L}^q} \leq C_q\|u\|_{\mathrm{W}^{1,p}}$.

(III) If $p > d$*, then* $u \in \mathrm{C}(\Omega)$ *and* $\|u\|_\infty \leq C\|u\|_{\mathrm{W}^{1,p}}$.

The second part can in fact be made more precise by considering embeddings into Hölder spaces, see Section 5.6.3 in [36] for details.

Theorem A.18 (Rellich–Kondrachov). *Let* $(u_j) \subset \mathrm{W}^{1,p}(\Omega)$ *with* $u_j \rightharpoonup u$ *in* $\mathrm{W}^{1,p}$.

(I) If $p < d$*, then* $u_j \to u$ *in* $\mathrm{L}^q(\Omega;\mathbb{R}^N)$ *(strongly) for any* $q < p^* = dp/(d-p)$.

(II) If $p = d$*, then* $u_j \to u$ *in* $\mathrm{L}^q(\Omega;\mathbb{R}^N)$ *(strongly) for any* $q < \infty$.

(III) If $p > d$*, then* $u_j \to u$ *uniformly.*

The following is a consequence of the *Stone–Weierstrass Theorem*:

Theorem A.19 (Density). *For every* $1 \leq p < \infty$*,* $u \in \mathrm{W}^{1,p}(\Omega)$*, and all* $\varepsilon > 0$ *there exists* $v \in (\mathrm{W}^{1,p} \cap \mathrm{C}^\infty)(\Omega)$ *with* $u - v \in \mathrm{W}^{1,p}_0(\Omega)$ *and* $\|u - v\|_{\mathrm{W}^{1,p}} < \varepsilon$.

Finally, we note that for $0 < \gamma \leq 1$ a function $u\colon \Omega \to \mathbb{R}$ is **γ-Hölder-continuous function**, in symbols $u \in \mathrm{C}^{0,\gamma}(\Omega)$, if

$$\|u\|_{\mathrm{C}^{0,\gamma}} := \|u\|_\infty + \sup_{\substack{x,y\in\Omega\\ x\neq y}} \frac{|u(x)-u(y)|}{|x-y|^\gamma} < \infty.$$

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Functions in $\mathrm{C}^{0,1}(\Omega)$ are called **Lipschitz-continuous**. We construct higher-order spaces $\mathrm{C}^{k,\gamma}(\Omega)$ analogously.

Next, we define a **family of mollifiers** as follows: Let $\rho \in \mathrm{C}_c^\infty(\mathbb{R}^d)$ be radially symmetric and positive. Then, the family $(\rho_\delta)_\delta \subset \mathrm{C}_c^\infty(B(0,\delta))(\mathbb{R}^d)$ consists of the functions

$$\rho_\delta(x) := \frac{1}{\delta^d}\rho\left(\frac{x}{\delta^d}\right), \qquad x \in \mathbb{R}^d.$$

For $u \in \mathrm{W}^{k,p}(\mathbb{R}^d)$, where $k \in \mathbb{N} \cup \{0\}$ and $1 \leq p < \infty$, we define the **mollification** $u_\delta \in \mathrm{W}^{k,p}(\mathbb{R}^d)$ as the **convolution** between u and ρ_δ, i.e.

$$u_\delta(x) := (\rho_\delta \star u)(x) := \int \rho_\delta(x-y)u(y)\,\mathrm{d}y, \qquad x \in \mathbb{R}^d.$$

Lemma A.20. *If $u \in \mathrm{W}^{k,p}(\mathbb{R}^d)$, then $u_\delta \to u$ strongly in $\mathrm{W}^{k,p}$ as $\delta \downarrow 0$.*

Analogous results also hold for continuously differentiable functions.

Finally, all the above notions and theorems continue to hold for vector-valued functions $u = (u^1, \ldots, u^m)^T : \Omega \to \mathbb{R}^m$ and in this case we set

$$\nabla u := \begin{pmatrix} \partial_1 u^1 & \partial_2 u^1 & \cdots & \partial_d u^1 \\ \partial_1 u^2 & \partial_2 u^2 & \cdots & \partial_d u^2 \\ \vdots & \vdots & & \vdots \\ \partial_1 u^m & \partial_2 u^m & \cdots & \partial_d u^m \end{pmatrix} \quad \in \mathbb{R}^{m \times d}.$$

We use the spaces $\mathrm{C}(\Omega;\mathbb{R}^m), \mathrm{C}^k(\Omega;\mathbb{R}^m), \mathrm{W}^{k,p}(\Omega;\mathbb{R}^m), \mathrm{C}^{k,\gamma}(\Omega;\mathbb{R}^m)$ with analogous meanings.

We also use *local* versions of all spaces, namely $\mathrm{C}_{\mathrm{loc}}(\Omega), \mathrm{C}^k_{\mathrm{loc}}(\Omega), \mathrm{W}^{k,p}_{\mathrm{loc}}(\Omega), \mathrm{C}^{k,\gamma}_{\mathrm{loc}}(\Omega)$, where the defining norm is only finite on every compact subset of Ω.

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